Electric Circuit Design (ECD)

Date

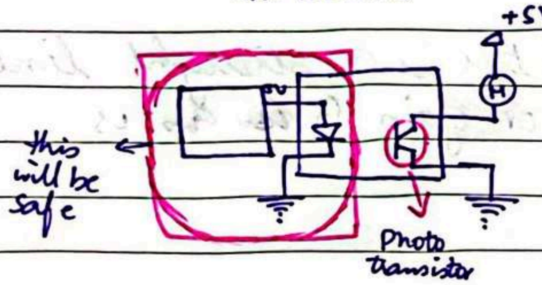
DAY-1

1) Amplification (o/p is greater than i/p)
2) Switching (BJT)
3) Buffering (o/p = i/p)
*4) Isolation
5) Filtering (LP & HP)
6) Math operations

$2 = \frac{twoV}{niV}$

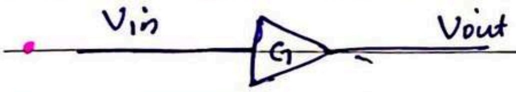
$= twoV$

OPTO-ISOLATOR



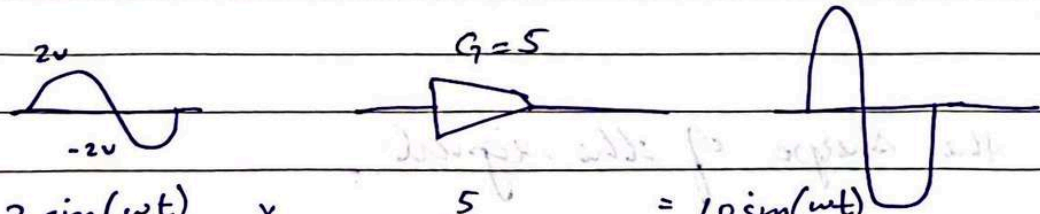
Signal Amplification :-

- Most practical sensors generate o/p of mV, ~~AV~~, μV, mA, μA.
- Since the signal are very weak, they need to be amplified.



$V_{out} = G \times V_{in}$

$\frac{V_{out}}{V_{in}} = G$



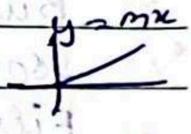
$2\sin(\omega t) \times 5 = 10\sin(\omega t)$

DAY-1

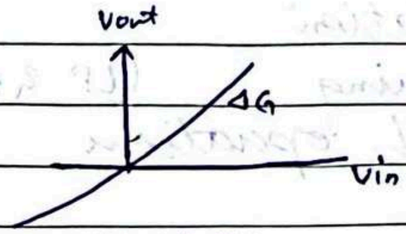
Types of Amplifiers :-

1) Linear
2) Non Linear

Linear :-



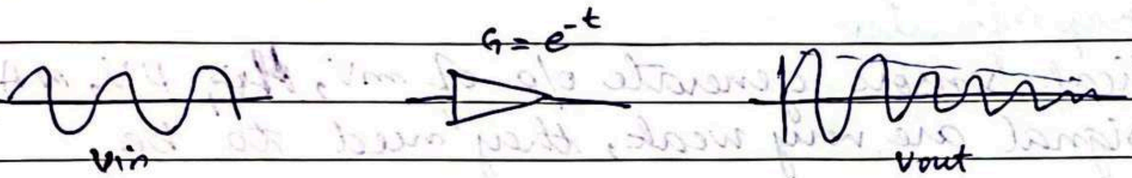
$$\frac{V_{out}}{V_{in}} = 5$$

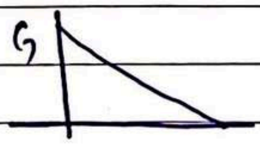


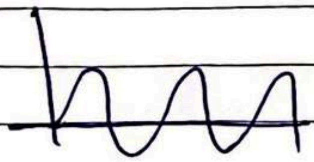
$$V_{out} = 5 V_{in}$$

Vout vs Vin transfer ~~x~~ characteristics is a straight line — hence it is a linear amplifier or gain G ~~is~~ is constant.

Non-linear :-







it changes the shape of the signal.

Date

Signal Amplification :-

Voltage Amplifier $= \frac{V_{out}}{V_{in}} = A_v \quad (V/V)$

Current AMP $= \frac{I_{out}}{I_{in}} = A_i \quad (A/A)$

Power AMP $= \frac{P_{out}}{P_{in}} = \frac{V_{out} I_{out}}{V_{in} I_{in}} = \left(\frac{V_{out}}{V_{in}}\right)\left(\frac{I_{out}}{I_{in}}\right) = A_v A_i$

$= A = \left(\frac{W}{W}\right)$

DAY-2

Amplification Gains (in dB's)

voltage gain $\boxed{A_v = \frac{V_o}{V_i}} \quad (V/V)$

$A_v(dB) = 20 \log |A_v|$

$\boxed{= 20 \log |V_o/V_i| \ dB}$

current gain $\boxed{A_i = \frac{I_o}{I_i}} \quad A/A$

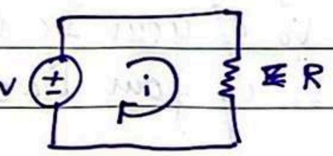
$A_i(dB) = 20 \log |A_i|$

$\boxed{= 20 \log \left|\frac{I_o}{I_i}\right| \ dB}$

Power gain :-

$20 \log |A_v|$

$20 \log \left|\frac{V_o}{V_i}\right|$



Date

$$= 10 \log \left|\frac{V_o}{V_i}\right|^2$$

$$= 10 \log \left(\frac{V_o^2}{V_i^2}\right)$$

$$= 10 \log \left|\frac{V_o^2/R}{V_i^2/R}\right|$$

$$= 10 \log \left|\frac{P_o}{P_i}\right| \ dB$$

$$\boxed{A_p = \frac{P_o}{P_i}}$$ W/W

$$\boxed{A_p(dB) = 10 \log \left|\frac{P_o}{P_i}\right|}$$

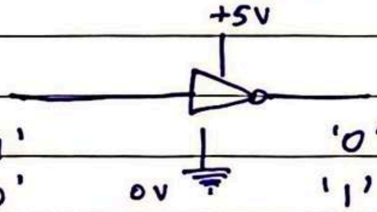
$A_v = \frac{V_o}{V_i} < 0$ (meaning ;) It means that it is inverted.

$A_v(dB) = 20 \log \left|\frac{V_o}{V_i}\right| < 0$ (meaning :) It has been atanuated (lessen the value)

Amplifier Power Supplies & Saturation :-

Max limit of +Vcc

+5V



'1' '0' 0V

'0' → 0V

'1' → 5V

Note :- logic '1' = +Vcc and logic '0' = ground

Max V_o of your IC cct = +Vcc

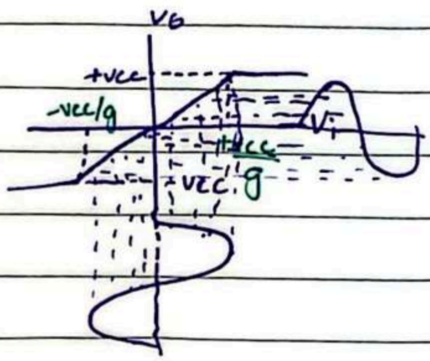
Min V_o of your IC cct = -Vcc

Scanned with CamScanner

Date

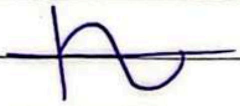
Ideal Transfer characteristics :-

$$G = \frac{V_o}{V_i} = 5 \qquad \left(\begin{matrix} V_o = 5 V_i \\ y = mx \end{matrix} \right)$$

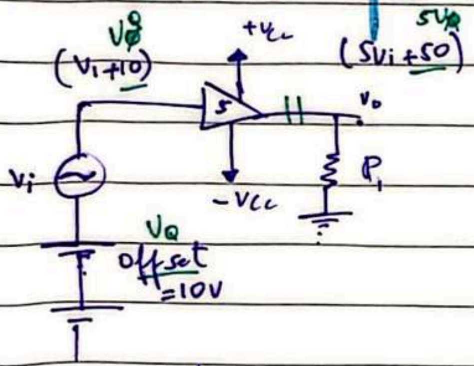


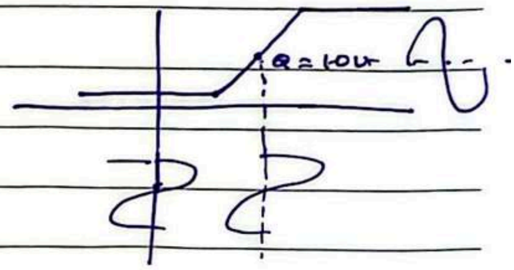
$\pm V_{CC} = \pm 15\,V$

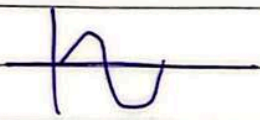
$V_i = 3 \sin(\omega t)$



Practical Transfer characteristics :-



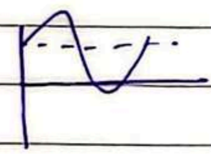




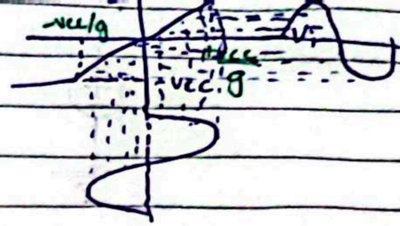
∵ capacitor blocks DC

∵ " allows AC

Q

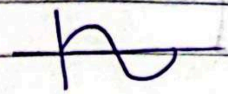


$y = mx$

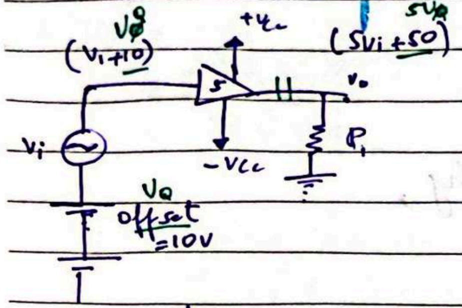


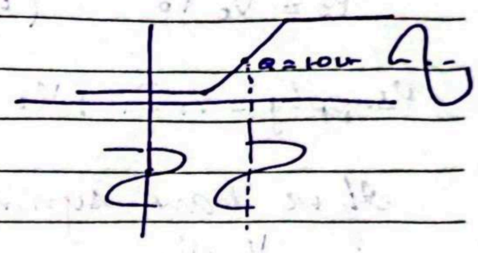
$\_\_ = \pm 15\,V$

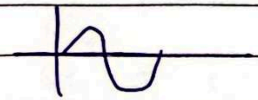
$V_1 = 3 \sin(\omega t)$



Practical Transfer characteristics :-

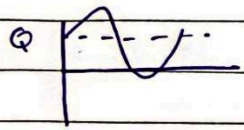






∵ capacitor blocks DC

∵ '' allows AC

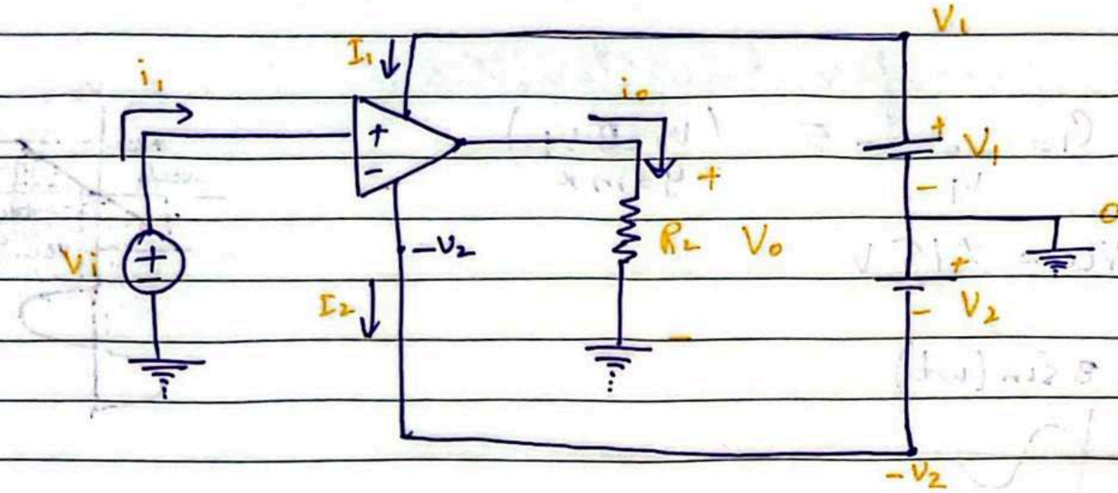


DAY-3

QUESTION -

Consider an amplifier operating at ± 10 V power supplies. It is fed sine voltage ----- Example 1.1.

Date



$P_i = V_i \, i_i$ (input)

$P_o = V_o \, i_o$ (output)

$P_{supply} = V_1 I_1 + V_2 I_2 = 2 V_1 I_1$

If we have symmetrical power supply,

$V_1 = V_2$

$I_1 = I_2$

i/p = o/p

$P_i + P_{supply} = P_o + P_{LOSS}$

$P_{loss} = P_i + P_{supply} - P_o$

$$\eta = \frac{P_o}{P_i + P_{supply}} \times 100\%$$

DAY-3

solution of previous Question:

$V_i = 1 V_p \Rightarrow 1/\sqrt{2}$. $V_o = 9 V_p = 9/\sqrt{2}$

$R_o = 1 k\Omega$

$V = IR$

Scanned with CamScanner

Date

$i_o = \frac{V_o}{R_o} \Rightarrow \frac{9/\sqrt{2}}{1K} \Rightarrow 6.3mA$

$i_i = 0.1m/\sqrt{2} = 0.0707\ mA.$

a) $A_v = \frac{V_o}{V_i} = \frac{9}{1} = \frac{9/\sqrt{2}}{1/\sqrt{2}} = 9\ V/V \Rightarrow 20\log(9)$
$= 19.08\ dB$

b) $A_i = \frac{I_o}{I_i} = \frac{6.3m}{0.07m} = 90\ A/A \Rightarrow 20\log(90)$
$= 39.08\ dB$

c) $P_i = VI$
$= (1/\sqrt{2})(0.1m/\sqrt{2})$
$= 0.05mW$

$P_o = V_{o,rms}\, I_{o,rms}$
$= \left(\frac{9}{\sqrt{2}}\right)(6.3m)$
$= 40.5\ mW$

$A_p = \frac{P_o}{P_i} = \frac{40.5m}{0.05m} = 810\ W/W$ or $A_p = A_v A_i$
$\Rightarrow 10\log(810)$
$= 29.08\ dB$

$P_{supply} = 2 \times 10 \times 9.3m$
$= 190\ mW$

$P_{loss} = P_{supply} + P_i - P_o$
$= 190m + 0.05m - 40.5m$
$= 149.55\ mW$

Scanned with CamScanner

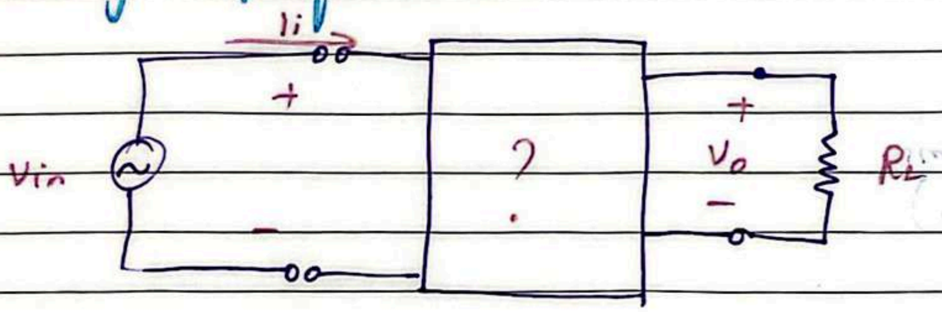
Date

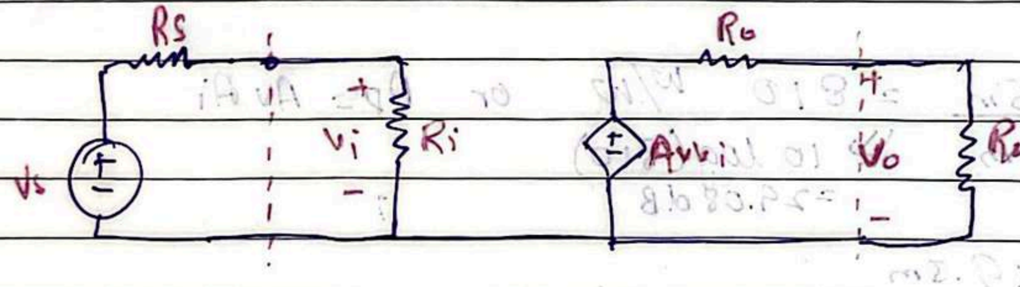
$$\eta = \frac{40.5m}{190 + 0.05} \times 100$$
$$= 21.3\%$$

Note :-
- Always convert units of A_v, A_i & A_p into dB.
- Use the correct method to find A_p and recheck your answer by using the formula $A_p = A_v A_i$
- Convert AC voltages and currents to DC or rms.

Circuit Model For Amplifiers :-

1. Voltage Amplifier :-





Ri accounts for the fact that amplifier draws a finite input current from the input voltage source.

$R_s <<< R_i$

ideally : $R_i \rightarrow \infty$

Date

DAY-4

$R_o <<< R_L$

Ideally $R_o = 0$

Finding expressions :-

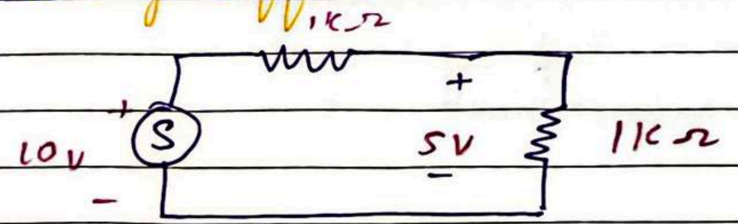
$$V_i = \left(\frac{R_i}{R_i + R_s}\right) V_s \longrightarrow (1)$$

$$V_o = \left(\frac{R_L}{R_o + R_L}\right) A_{v_o} V_i \longrightarrow (2)$$

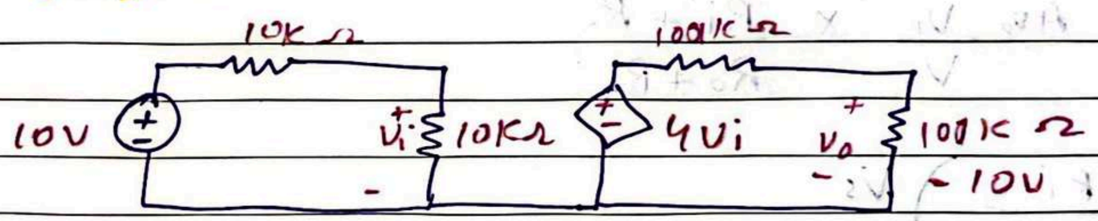
$$V_o = A_{v_o}\left(\frac{R_L}{R_o + R_L}\right)\left(\frac{R_i}{R_i + R_s}\right) V_s$$

$$\boxed{G = \frac{V_o}{V_s} = A_{v_o}\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_s}\right)}$$

Loading Effect :-



$\longleftrightarrow$



$$\frac{V_o}{V_s} = 4\left(\frac{100k}{100k + 100k}\right)\left(\frac{10k}{10k + 10k}\right)$$

$$= 4(0.5)(0.5)$$

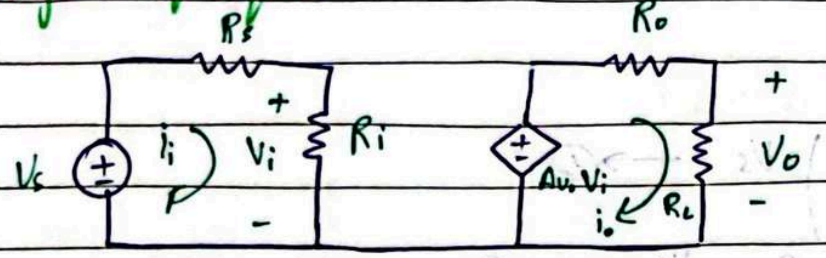
$$\frac{V_o}{V_s} = 1 \Rightarrow V_o = V_s = 10V$$

Scanned with CamScanner

Date

DAY-4

Voltage Amplifier :-



$$G = \frac{V_o}{V_s} = Av_o \left(\frac{R_i}{R_i + R_s}\right)\left(\frac{R_L}{R_o + R_L}\right)$$

Current Gain :-

$$\frac{i_o}{i_i} = ?$$

$$i_o = \frac{Av_o V_i}{R_o + R_L}$$

$$i_i = \frac{V_s}{R_i + R_s}$$

$$\frac{i_o}{i_i} = \frac{Av_o V_i / R_o + R_L}{V_s / R_i + R_s}$$

$$\frac{i_o}{i_i} = Av_o \frac{V_i}{V_s} \times \frac{R_i + R_s}{R_o + R_L} \longrightarrow (1)$$

Now

$$V_i = \left(\frac{R_i}{R_i + R_s}\right) V_s$$

$$\frac{V_i}{V_s} = \frac{R_i}{R_i + R_s}$$

putting in (1)

Date

$$= A_{v_o}\left(\frac{R_i}{\cancel{R_i+R_s}}\right)\left(\frac{\cancel{R_i+R_s}}{R_o+R_L}\right)$$

$$\boxed{\frac{i_o}{i_i} = A_{v_o}\left(\frac{R_i}{R_o+R_L}\right)}$$

(sir does not like this method.)

Preferrable Method :-

$$G = \frac{V_o}{V_s} = A_{v_o}\left(\frac{R_i}{R_i+R_s}\right)\left(\frac{R_L}{R_o+R_L}\right) \quad (V/V)$$

$$\frac{i_o}{i_i} = \frac{V_o/R_L}{V_s/R_s+R_i}$$ (using ohm's law)

$$= \left(\frac{V_o}{V_s}\right)\left(\frac{R_s+R_i}{R_L}\right)$$

$$= A_{v_o}\left(\frac{R_i}{\cancel{R_i+R_s}}\right)\left(\frac{\cancel{R_L}}{R_o+R_L}\right)\left(\frac{\cancel{R_s+R_i}}{\cancel{R_L}}\right)$$

$$\boxed{\frac{i_o}{i_i} = A_{v_o}\left(\frac{R_i}{R_o+R_L}\right)} \quad (A/A)$$

Now finding V_o/i_i ?

$$\frac{V_o}{i_i} = \frac{V_o}{V_s/R_i+R_s} \quad \boxed{= \left(\frac{V_o}{V_s}\right)(R_i+R_s)} \quad (V/A)$$

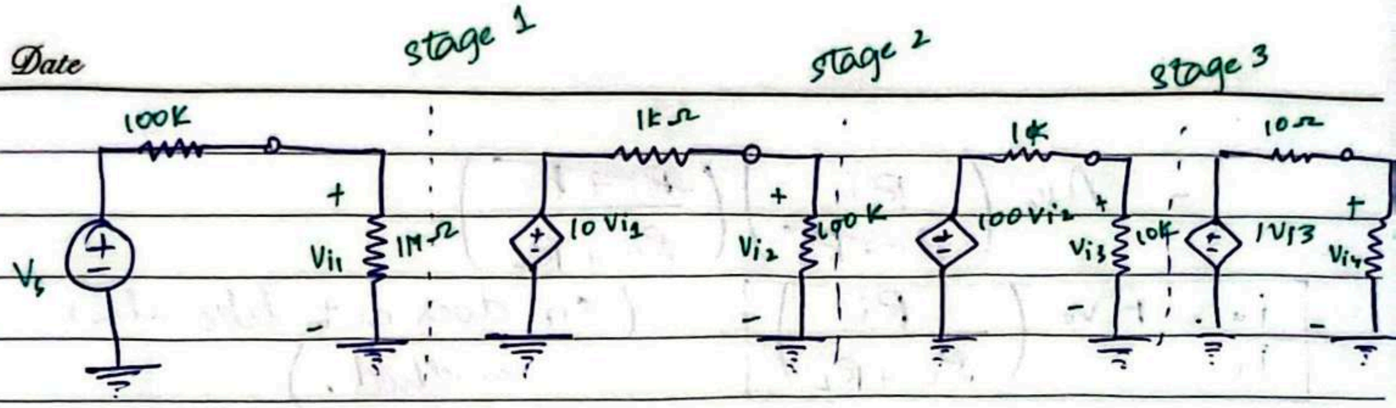
$$\frac{i_o}{V_s} = \frac{V_o/R_L}{V_s} \quad \boxed{= \left(\frac{V_o}{V_s}\right)\left(\frac{1}{R_L}\right)} \quad (A/V)$$

~~Cascade~~

Cascade Amplifier:.

On the next page

Date



$$\frac{V_o}{V_{i3}} = \frac{100}{110}(1) \Rightarrow \boxed{0.909} \quad (V/V)$$

$$\frac{V_{i3}}{V_{i2}} = 100\left(\frac{10K}{1K+10K}\right) \Rightarrow \boxed{90.909} \quad (V/V)$$

$$\frac{V_{i2}}{V_{i1}} = 10\left(\frac{100K}{101K}\right) \Rightarrow \boxed{9.900} \quad (V/V)$$

$$\frac{V_{i1}}{V_s} = \frac{1M}{1.1M} \Rightarrow \boxed{0.9090}$$

$$\frac{V_o}{V_s} = \frac{V_o}{V_{i3}} \times \frac{V_{i3}}{V_{i2}} \times \frac{V_{i2}}{V_{i1}} \times \frac{V_{i1}}{V_s}$$

$$\boxed{\frac{V_o}{V_s} = 743.6} \quad (V/V) \quad \text{or} \quad \boxed{57.4\,dB}$$

$$\frac{V_o}{V_{i1}} = \frac{V_o}{V_{i3}} \times \frac{V_{i3}}{V_{i2}} \times \frac{V_{i2}}{V_{i1}}$$

$$\boxed{\frac{V_o}{V_{i1}} = 818.0 \ (V/V)} \quad \text{or} \quad \boxed{58.25\,dB}$$

$$\frac{i_o}{i_{i1}} = \frac{V_o/R_L}{V_{i1}/R_{i1}}$$

$$= \left(\frac{V_o}{V_{i1}}\right)\left(\frac{R_{i1}}{R_L}\right)$$

Scanned with CamScanner

Date

$$= (818.0)\left(\frac{1M}{100}\right)$$

$$\boxed{= 818 \times 10^4} \quad (A/A) \text{ or } \boxed{138.25\, dB}$$

$$A_P = A_V A_i$$

$$= \left(\frac{V_o}{V_{i_1}}\right)\left(\frac{i_o}{i_{i_1}}\right)$$

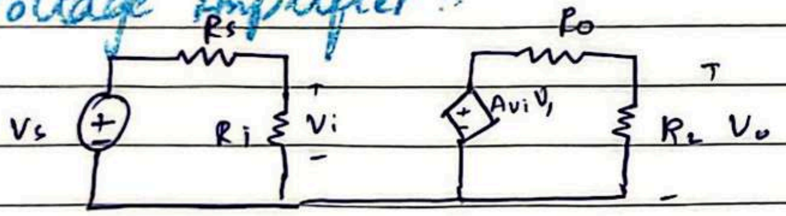
$$= 6.69 \times 10^9$$

$$\boxed{= 6.69\, GW} \quad \text{or} \quad \boxed{98.25\, dB}$$

Four Amplifier Models:-

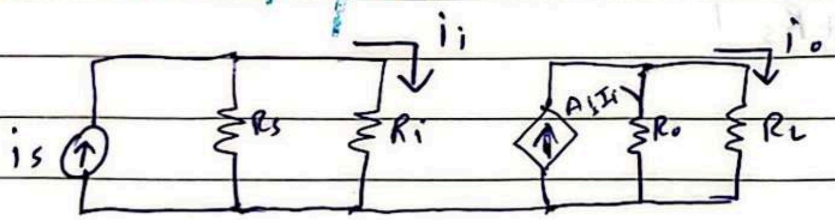
1- Voltage Amplifier
2- Current Amplifier
3- Trans- Resistance
4- Trans - conductance

Voltage Amplifier :-



$$\frac{V_o}{V_s} = A_{vo}\left(\frac{R_i}{R_i + R_s}\right)\left(\frac{R_L}{R_o + R_L}\right)$$

Current Amplifier:



Date

$$i_i = \left(\frac{R_s}{R_s + R_i}\right) i_s$$

$$i_o = \left(\frac{R_o}{R_o + R_L}\right) A_{is} i_i$$

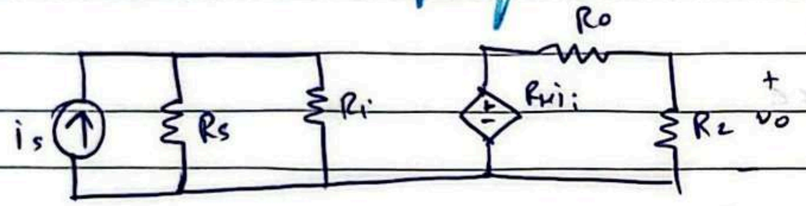
$$\boxed{\frac{i_o}{i_i} = A_{is}\left(\frac{R_o}{R_o + R_L}\right)\left(\frac{R_s}{R_s + R_i}\right)}$$

$$\frac{V_o}{V_i} = \frac{i_o R_L}{i_s (R_i \| R_s)}$$

$$\boxed{\frac{V_o}{V_i} = \left(\frac{i_o}{i_s}\right)\left(\frac{R_L (R_i + R_s)}{R_i R_s}\right)}$$

Trans- Resistance Amplifier :-

V_o / I_s



input current
output voltage

$\cancel{\frac{V_o}{V_i}} =$

$$V_o = \left(\frac{R_L}{R_L + R_o}\right) R_m i_i$$

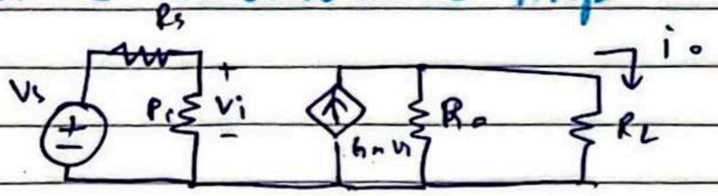
$$i_i = \left(\frac{R_s}{R_i + R_s}\right) i_s$$

$$\frac{V_o}{i_s} = R_m \left(\frac{R_L}{R_o + R_L}\right)\left(\frac{R_s}{R_i + R_s}\right)$$

Scanned with CamScanner

Date

Trans-conductance Amp :- I_o/V_s



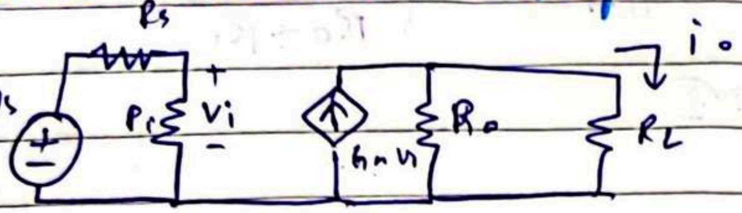
$$i_o = \left(\frac{R_o}{R_o + R_L}\right) G_m V_i$$

$$V_i = \left(\frac{R_i}{R_s + R_i}\right) V_s$$

$$\frac{i_o}{v_i} = G_m \left(\frac{R_o}{R_o + R_L}\right)\left(\frac{R_i}{R_s + R_i}\right)$$

Date

Trans - conductance Amp :- I_o/V_s



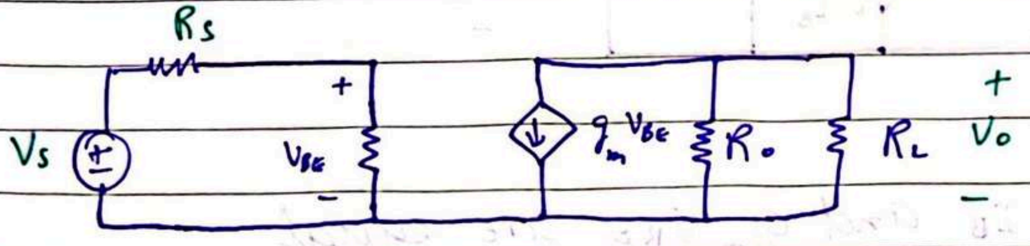
$$i_o = \left(\frac{R_o}{R_o + R_L}\right) G_m V_i$$

$$V_i = \left(\frac{R_i}{R_s + R_i}\right) V_s$$

$$\frac{i_o}{V_i} = G_m \left(\frac{R_o}{R_o + R_L}\right)\left(\frac{R_i}{R_s + R_i}\right)$$

DAY- 5

BJT Model :- (Transconductance Amp)



$R_s = 5K\,\Omega$

$r_\pi = 2.5K\,\Omega$

$g_m = 40m$ (A/V)

$R_o = 100K\,\Omega$

$R_L = 5K\,\Omega$

$\frac{V_o}{V_s} = ?$

Sol :-

Applying ohm's law to second part of circuit.

Scanned with CamScanner

Date

$$V_o = -g_m V_{BE} (R_o || R_L) = -g_m V_{BE} \left(\frac{R_o R_L}{R_o + R_L}\right) \longrightarrow ①$$

$$V_{BE} = \left(\frac{r_\pi}{r_\pi + R_s}\right) V_s \longrightarrow ②$$

Substitute ② in ①

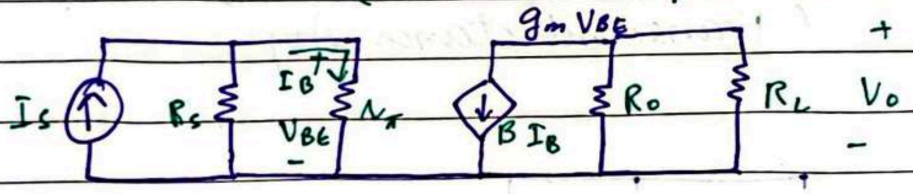
$$V_o = -g_m \left(\frac{r_\pi}{r_\pi + R_s}\right) V_s \left(\frac{R_o R_L}{R_o + R_L}\right)$$

$$\frac{V_o}{V_s} = -g_m \left(\frac{r_\pi}{r_\pi + R_s}\right)\left(\frac{R_o R_L}{R_o + R_L}\right)$$

putting the values

$$\boxed{\frac{V_o}{V_s} = -63.5 \ V/V}$$

BJT Model:- (current Amp)



Sol.

Considering $B I_B$ and $g_m V_{BE}$ are equal

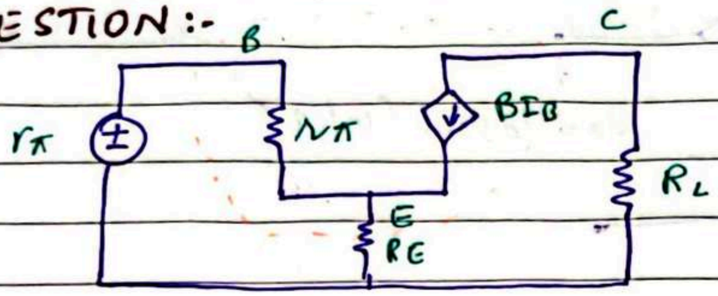
$$g_m V_{BE} = B I_B$$

$$B = g_m \left(\frac{V_{BE}}{I_B}\right) \Rightarrow g_m (r_\pi)$$

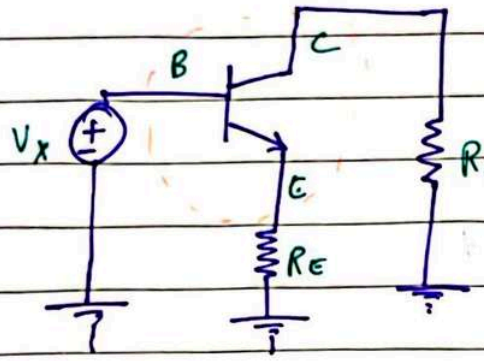
HOME TASK:- Find $\frac{I_o}{V_s}$ for T.C.A

Date

QUESTION :-



(Actual circuit)



(Explained Circuit)

Find internal resistance of this amplifier

Sol :-

$$\frac{V_x}{i_x} = R_{in}$$

$$V_x = V_1 + V_B$$

$$= I_B r_\pi + (I_B + \beta I_B) R_E$$

$$= I_B (r_\pi + (1+\beta) R_E)$$

$$\frac{V_x}{I_B} = r_\pi + (1+\beta) R_E$$

$$i_x = I_B$$

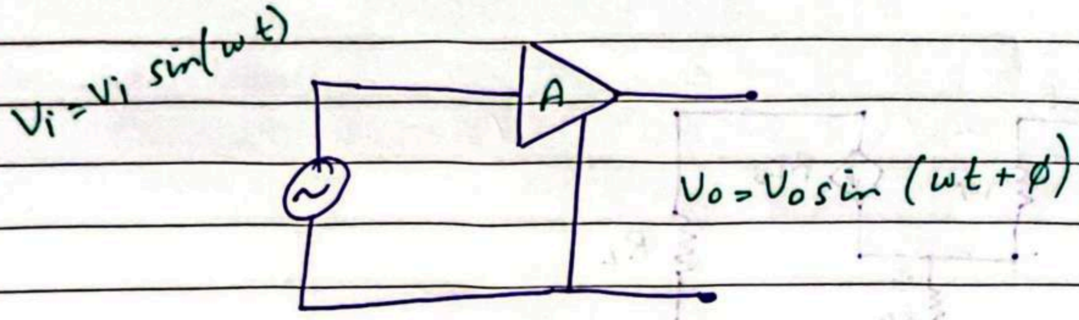
$$\boxed{\frac{V_x}{I_x} = r_\pi + (1+\beta) R_E = R_{in}}$$

Frequency Response of Amplifiers :-

NEXT PAGE

Scanned with CamScanner

Date



Four Axioms:-

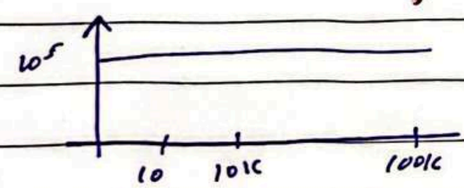
→ Frequency is constant.

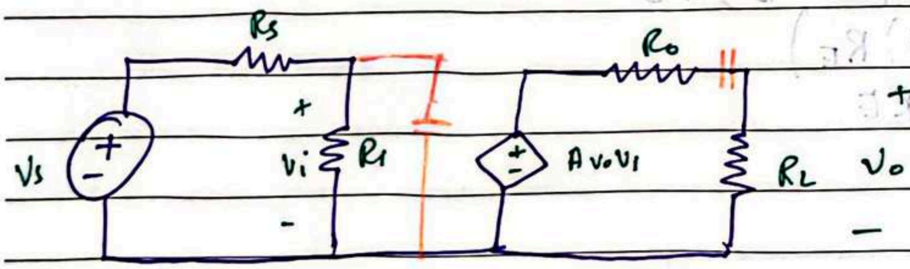
→ Shape of waveform is constant.

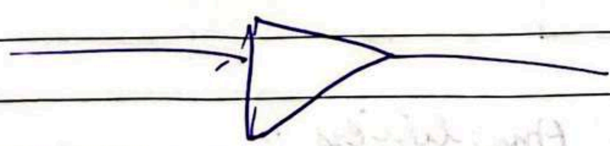
→ Amplitude is changing

→ Phase angle is changing.

Magnitude / gain of an "ideal" voltage Amp.





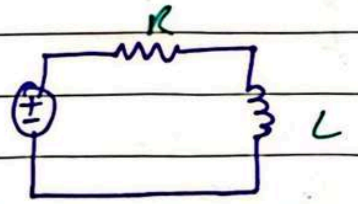


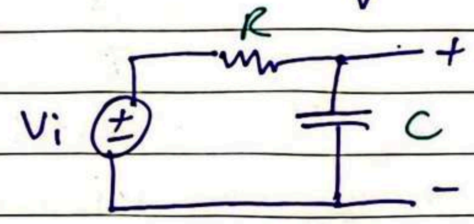
Date

Mag. / gain of a real Voltage Amp :-

Single Time Constant Circuits :-

Those circuits that have only one time constant

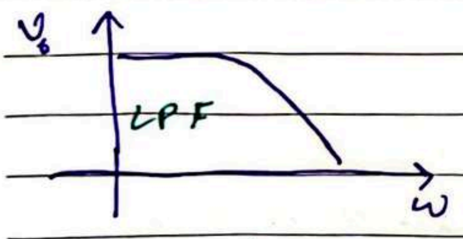




$X_c = \frac{1}{j\omega C}$

$\tau = L/R$ $\tau = RC$

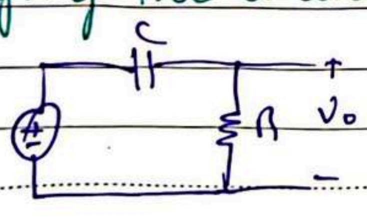
(using this circuit)



$\omega = 0$; $X_c \to \infty$: $V_o = V_i$

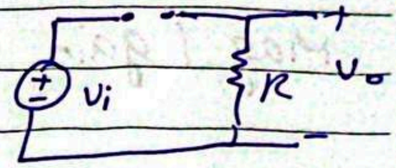
$\omega = \infty$; $X_c \to 0$; $V_o = 0$

Changing the circuit

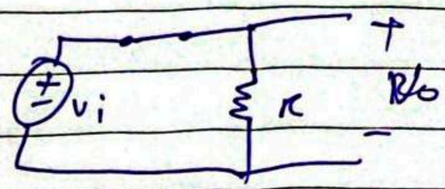


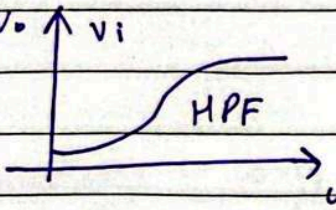
Date

$\omega \rightarrow 0$; $X_C \rightarrow \infty$; $V_0 = 0$ (small)



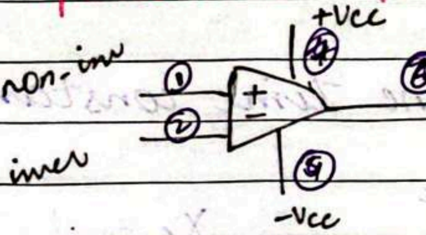
$\omega = \infty$, $X_C \rightarrow 0$; $V_0 = V_i$

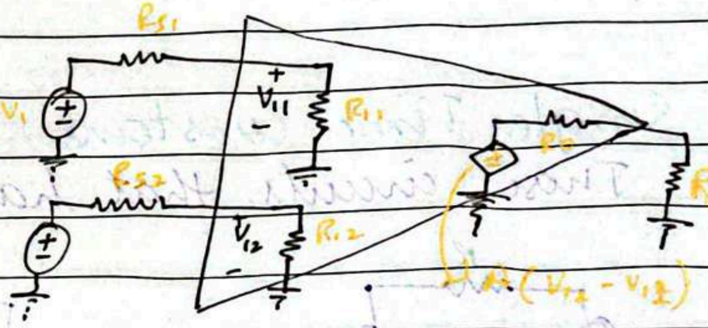




DAY-6 (chapter 5)

Operational Amplifier :-





a) An operational amplifier is used as a voltage amp. and is used to amplify the difference between inv. & non-inv voltage input (V_1 & V_2)

$$V_o = A(\underset{+}{V_2} - \underset{-}{V_1})$$

b) Input current from signal V_1 & V_2 is Zero.
OP-AMP does not draw any current from V_1 & V_2

$R_{i1} \rightarrow \infty$
$R_{i2} \rightarrow \infty$

c) $R_0 = 0$

Scanned with CamScanner

Date

- Common mode rejection is ideally zero but practically it is non-zero.

$V_o = A(V_2 - V_1) \Rightarrow A_D = \frac{V_o}{V_2 - V_1}$

$A_{cm} = \frac{V_o}{V_i} \to 0$

$CMRR = 20\log\left|\frac{A_D}{A_{cm}}\right| \to \infty$

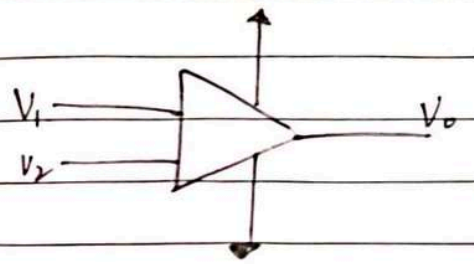
d) C_i & C_o DC coupled

e) $CMRR \to \infty$ ~~(Ideally)~~

f) $\underset{(\text{Ideally})}{A} \to \infty$

$\underset{(\text{Practical})}{A} \to 10^5$ V/V

g) Infinite bandwidth



OP-AMP two way:-

1) Open - loop ✗ Can only be used
2) Closed loop ✓.

Date

If $V_1 = 1$ & $V_2 = 2$

$$V_o = A(V_2 - V_1)$$
$$= 10^5(2-1)$$
$$= \cancel{10^5 V} \Rightarrow 5V$$

If $V_1 = 2$ & $V_2 = 1$

$$V_o = A(V_2 - V_1)$$
$$= A(1-2)$$
$$= \cancel{10^5 V} \Rightarrow 0V$$

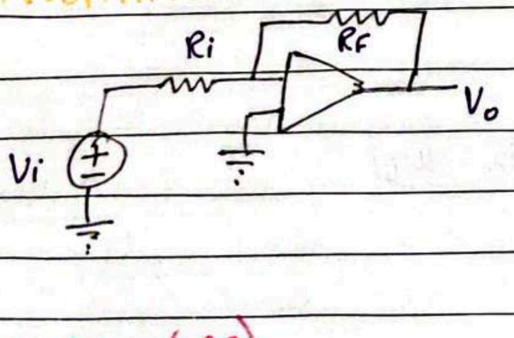
Simply

$V_+ > V_-$: $V_o = +V_{cc}$

$V_- > V_+$: $V_o = -V_{cc}$

- In Open loop configuration an OP-AMP can only be used as comparator and you cannot adjust the gain as per your choice
- In closed loop design becomes more flexible and you can change the gain as per your choice.

INVERTING AMPLIFIER:-



Case 1:- $(A \to \infty)$

$$V_o = A(V_+ - V_-)$$

Scanned with CamScanner

Date

$$V_+ - V_- = \frac{V_o}{A}$$

$$A \rightarrow \infty \;,\; \frac{V_o}{A} \rightarrow 0$$

$$V_+ - V_- = 0$$

$$V_+ = V_-$$ (Virtual Short Circuit)

Note :- If the loop is closed, $V_+ = V_-$

(Ohm's Law)

$V_+ = 0V$

$V_- = 0V$

$$I_i = \frac{V_i - 0}{R_1} = \frac{V_i}{R_i} \longrightarrow (1)$$

∵ Current flows from higher to lower potential

$$0 - V_o = I_i R_F$$

$$V_o = -I_i R_F \longrightarrow (2)$$

Substitute (2) in (1)

$$V_o = -\left(\frac{V_i}{R_i}\right) R_f$$

$$\frac{V_o}{V_i} = -\left(\frac{R_F}{R_i}\right) = G$$

(Node Voltage)

$$\frac{0 - V_i}{R_i} + \frac{0 - V_o}{R_F} = 0$$

$$-\frac{V_i}{R_1} - \frac{V_o}{R_F} = 0$$

$$\frac{V_o}{R_F} = \frac{-V_i}{R_i}$$

Scanned with CamScanner

B

Date

$$\boxed{\frac{V_o}{V_i} = -\frac{R_F}{R_i}}$$

Case 2:- (A → Finite)

$$V_o = A(V_+ - V_-)$$

$$V_+ - V_- = \frac{V_o}{A}$$

$$A \rightarrow \text{finite}$$

$$V_- = V_+ - \frac{V_o}{A}.$$

$$V_+ = 0$$

$$V_- = \frac{-V_o}{A}$$

$$\frac{-V_o/A - V_i}{R_i} + \frac{-V_o/A - V_o}{R_F} = 0$$

$$\frac{-V_o}{AR_1} \quad \frac{-V_i}{R_i} \quad \frac{-V_o}{AR_F} \quad \frac{-V_o}{R_F} = 0$$

$$\frac{-V_o}{AR_i} \quad \frac{-V_o}{AR_F} \quad \frac{-V_o}{R_F} = \frac{V_i}{R_i}$$

$$V_o\left(\frac{1}{AR_i} + \frac{1}{AR_F} + \frac{1}{R_F}\right) = -\frac{V_i}{R_i}$$

$$V_o\left(\frac{1}{A}\left(\frac{1}{R_i} + \frac{1}{R_F}\right) + \frac{1}{R_F}\right) = -\frac{V_i}{R_1}$$

Multiply 'R_F' on both sides of eq

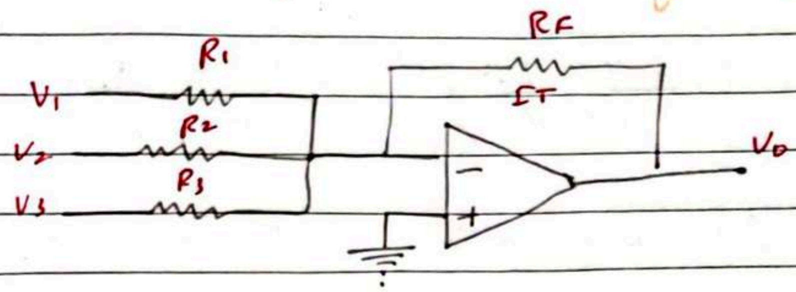
$$V_o\left(\frac{1}{A}\left(\frac{R_F}{R_i} + 1\right) + 1\right) = \left(\frac{-R_F}{R_i}\right)V_i$$

$$\boxed{\frac{V_o}{V_i} = \frac{-R_F/R_i}{1 + 1/A\left(1 + R_F/R_i\right)} = G}$$

Date

DAY-7

Implementing Addition of multiple signals
(Weighted Summing Amplifier)



$I_1 = \frac{V_1}{R_1}$, $I_2 = \frac{V_2}{R_2}$, $I_3 = \frac{V_3}{R_3}$

$I_T = I_1 + I_2 + I_3$

$I_T = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \longrightarrow ①$

$0 - V_o = I_T R_F$

$V_o = -I_T R_F \longrightarrow ②$

Substitute ② in ①

$V_o = -R_F\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)$

$\therefore \boxed{V_o = -\left(\left[\frac{R_F}{R_1}\right]V_1 + \left[\frac{R_F}{R_2}\right]V_2 + \left[\frac{R_F}{R_3}\right]V_3\right)}$

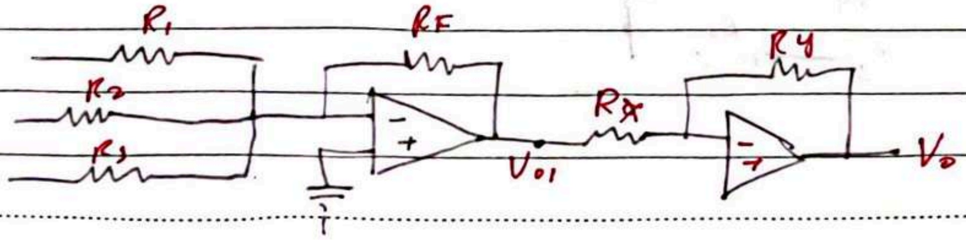
$V_1 = -(2V_1 + 3V_2 + 6V_3)$

$R_F = R_1 = R_2 = R_3 = 10K$

$V_o = -(V_1 + V_2 + V_3)$

$V_o = -V_1 - V_2 - V_3$

Bss samjhaya tha



Scanned with CamScanner

Date

F-VAD

$$V_{o_1} = -\left[\left(\frac{R_F}{R_1}\right)V_1 + \left(\frac{R_F}{R_2}\right)V_2 + \left(\frac{R_F}{R_3}\right)V_3\right]$$

$$V_o = -\left(\frac{R_y}{R_x}\right)V_{o_1}$$

$$= +\left(\frac{R_y}{R_x}\right)\left(\frac{R_F}{R_1}\right)V_1 + \left(\frac{R_y}{R_x}\right)\left(\frac{R_F}{R_2}\right)V_2 + \left(\frac{R_y}{R_x}\right)\left(\frac{R_F}{R_3}\right)V_3$$

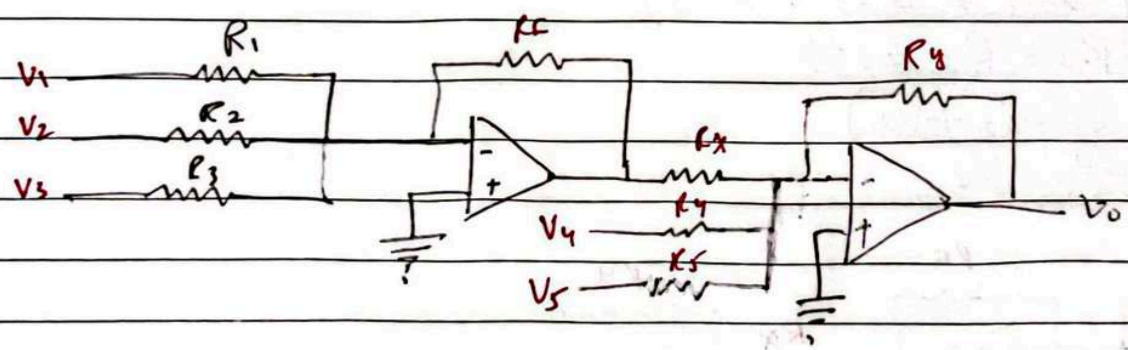
→ If you want to have a +ve gain, add two op-amps.

→ If you want a simpler expression, make $R_y = R_x$.

Implementing Addition / Subtraction circuit :

→ In case of addition & subtraction, we will use a two stage amplifier design. All those voltages that have to be added will be applied at inverting terminal of stage 1. All those inputs that have to be subtracted will be applied at inverting terminal of 2nd op-amp.

$$V_o = \alpha V_1 + \beta V_2 + \gamma V_3 - \varepsilon V_4 - \phi V_5$$



Scanned with CamScanner

Date

$$V_{o1} = -\left(\frac{R_F}{R_1}\right)V_1 - \left(\frac{R_F}{R_2}\right)V_2 - \left(\frac{R_F}{R_3}\right)V_3 \longrightarrow ①$$

$$V_o = -\left(\frac{R_y}{R_x}\right)V_{o1} - \left(\frac{R_y}{R_x}\right)V_4 - \left(\frac{R_y}{R_5}\right)V_5 \longrightarrow ②$$

put ① in ②

$$V_o = -\left(\frac{R_y}{R_x}\right)\left[-\left(\frac{R_F}{R_1}\right)V_1 - \left(\frac{R_F}{R_2}\right)V_2 - \left(\frac{R_F}{R_3}\right)V_3\right] - \left(\frac{R_y}{R_4}\right)V_4 - \left(\frac{R_y}{R_5}\right)$$

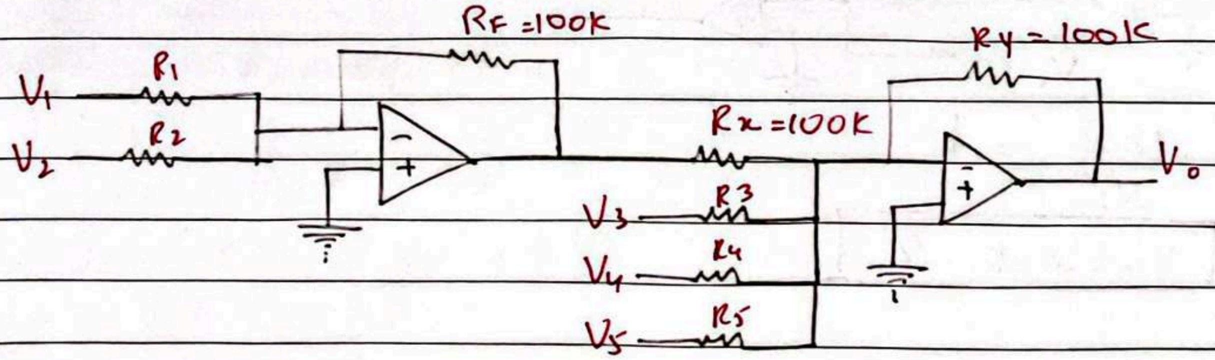
$$= \left(\frac{R_y R_F}{R_x R_1}\right)V_1 + \left(\frac{R_y R_F}{R_x R_2}\right)V_2 + \left(\frac{R_y R_F}{R_x R_3}\right)V_3 - \left(\frac{R_y}{R_4}\right)V_4 - \left(\frac{R_y}{R_5}\right)V_5$$

$$\alpha V_1 + \beta V_2 + \gamma V_3 - \varepsilon V_4 - \phi V_5$$

Design a circuit that performs this equation

$$V_o = 2V_1 + 8V_2 - 5V_3 - 6V_4 - 4V_5$$

You are required to use only 100 kΩ on the feedback.



$$V_o = \left(\frac{R_y R_F}{R_x R_1}\right)V_1 + \left(\frac{R_y R_F}{R_x R_2}\right)V_2 - \left(\frac{R_y}{R_3}\right)V_3 - \left(\frac{R_y}{R_4}\right)V_4 - \left(\frac{R_y}{R_5}\right)V_5$$

$$\frac{R_y}{R_5} = 4 \Rightarrow \frac{100k}{R_5} = 4 \Rightarrow \boxed{R_5 = 25k\,\Omega}$$

Scanned with CamScanner

Date

$\frac{R_y}{R_4} = 6 \Rightarrow \frac{100k}{R_4} = 6 \Rightarrow \boxed{R_4 = 16.66k\,\Omega}$

$\frac{R_y}{R_3} = 5 \Rightarrow \frac{100k}{R_3} = 5 \Rightarrow \boxed{R_3 = 20k\,\Omega}$

$\frac{R_y}{R_x} \cdot \frac{R_F}{R_2} = 8$

$\frac{100k}{100k} \cdot \frac{100k}{R_2} = 8$

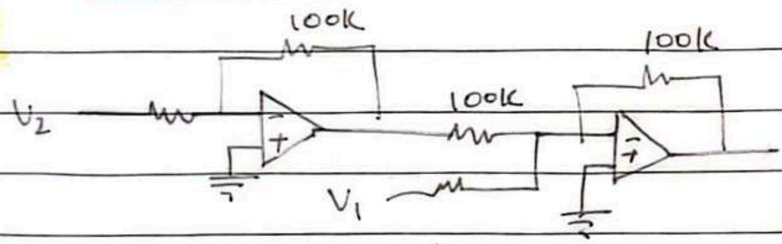
$\boxed{R_2 = 12.5k\,\Omega}$

$\frac{R_y}{R_x} \cdot \frac{R_F}{R_1} = 2$

$\frac{100k}{100k} \cdot \frac{100k}{R_1} = 2$

$\boxed{R_1 = 50k\,\Omega}$

Q



$(V_0 = V_2 - V_1)$

∴ Difference amplifier

$\frac{\cancel{100k}}{\cancel{100k}} \cdot \frac{100k}{V_2} = 1$

$\boxed{V_2 = 100k\,\Omega}$

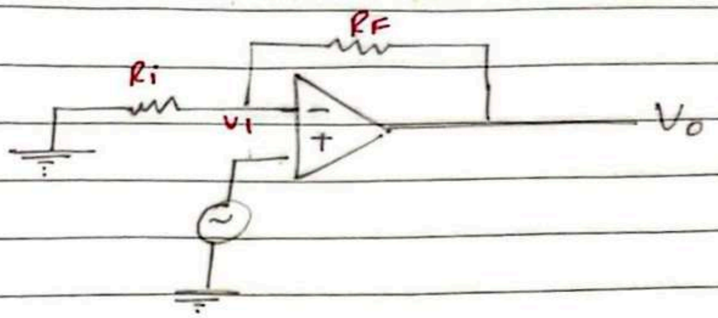
$\frac{100k}{V_1} = 1$

$\boxed{V_1 = 100k}$

Scanned with CamScanner

Date

NON-INVERTING AMP. :-



$(A \rightarrow \infty)$

$$V_o = A(V_+ - V_-)$$

$$\frac{V_o}{A} = V_+ - V_-$$

$$V_+ - V_- = 0$$

$$V_+ = V_-$$

Case 1 :-

$$0 + \frac{V_i - 0}{R_1} + \frac{V_1 - V_o}{R_F} = 0$$

$$\frac{V_i}{R_i} + \frac{V_i}{R_F} = \frac{V_o}{R_F}$$

$$V_i\left(\frac{1}{R_i} + \frac{1}{R_F}\right) = \frac{V_o}{R_F}$$

$$V_o = \left(\frac{R_F}{R_F} + \frac{R_F}{R_i}\right) V_i$$

$$\boxed{\frac{V_o}{V_i} = 1 + \frac{R_F}{R_i}}$$

Case 2 :-

$$I_1 = \frac{V_i - 0}{R_i} = \frac{V_i}{R_1}$$

Scanned with CamScanner

Date

$$\therefore \frac{V_0 - V_i}{R_F} = I_i$$

$$\frac{V_o}{R_F} - \frac{V_i}{R_F} = \frac{V_i}{R_i}$$

$$\frac{V_o}{R_F} = \frac{V_i}{R_i} + \frac{V_i}{R_F}$$

$$\boxed{\frac{V_o}{V_i} = 1 + \frac{R_F}{R_i}}$$

$A \rightarrow$ Finite

$$V_o = A(V_+ - V_-)$$

$$V_+ - V_- = \frac{V_o}{A}$$

$$V_- = V_+ - \frac{V_o}{A}$$

Case 1:-

$$V_+ = V_i$$

$$V_- = V_i - \frac{V_o}{A}$$

$$I_i = \frac{V_i - V_o/A - 0}{R_i}$$

$$I_i = \frac{V_i}{R_i} - \frac{V_o}{AR_i}$$

$$V_o - \left(V_i - \frac{V_o}{A}\right) = I_i R_F$$

$$V_o - V_i + \frac{V_o}{A} = R_F\left(\frac{V_i}{R_i} - \frac{V_o}{AR_i}\right)$$

$$V_o - V_i + \frac{V_o}{A} = \left(\frac{R_F}{R_i}\right) V_i - \left(\frac{R_F}{R_i}\right)\left(\frac{V_o}{A}\right)$$

Scanned with CamScanner

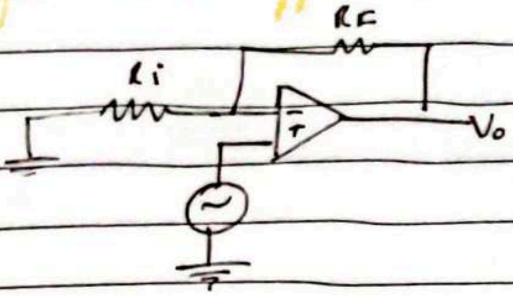
Date

$$V_0 + \frac{V_0}{A} + \left(\frac{R_F}{R_i}\right)\left(\frac{V_0}{A}\right) = V_i + \left(\frac{R_F}{R_i}\right) V_i$$

$$V_0\left(1 + \frac{1}{A}\left(1 + \frac{R_F}{R_i}\right)\right) = V_i\left(1 + \frac{R_F}{R_i}\right)$$

$$\frac{V_0}{V_i} = \frac{1 + R_F/R_i}{1 + 1/A\,(1 + R_F/R_i)}$$

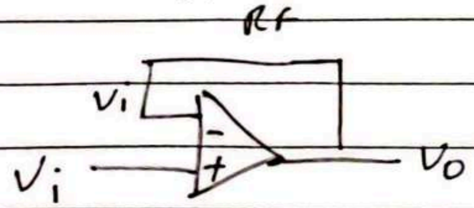
Unity Gain Buffer (voltage follower):-



$$\frac{V_o}{V_i} = 1 + \frac{R_F}{R_i}$$

$R_F = 0 \Rightarrow \frac{V_o}{V_i} = 1$ (short circuit)

$R_i = \infty \Rightarrow \frac{V_o}{V_i} = 1$ (open circuit)



$$V_o = A(V_+ - V_-)$$

$$V_+ - V_- = \frac{V_o}{A}$$

$$A \to \infty$$

$$V_+ - V_- = \frac{V_o}{\infty}$$

Scanned with CamScanner

Date

FYP

$V_+ - V_- = 0$

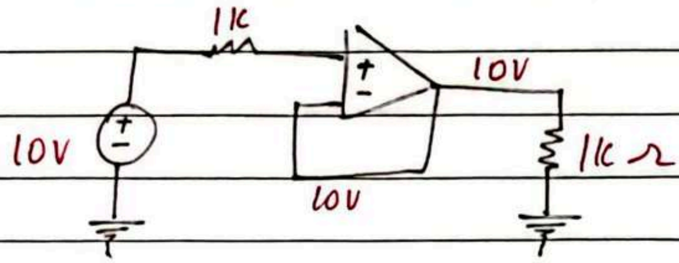
$V_+ = V_-$

$V_+ = V_i$

$V_- = V_+ = V_i$

$\boxed{V_- = V_o = V_i}$

Use VF in FYP



Scanned with CamScanner

Date

$V_+ - V_- = 0$

$V_+ = V_-$

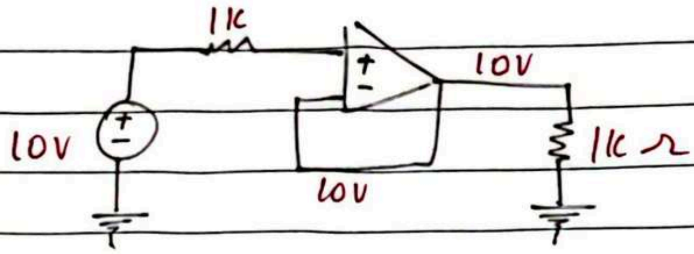
$V_+ = V_i$

$V_- = V_+ = V_i$

$\boxed{V_- = V_0 = V_i}$

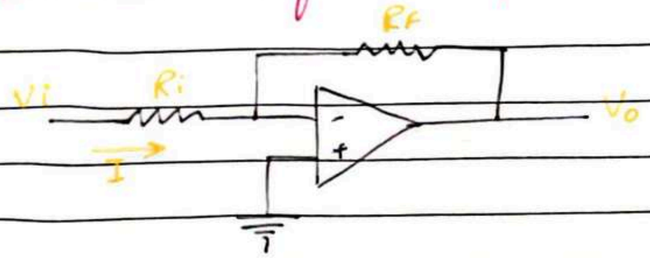
FYP

Use VF in FYP



DAY - 8

A special cas of Inv. Amp. :-



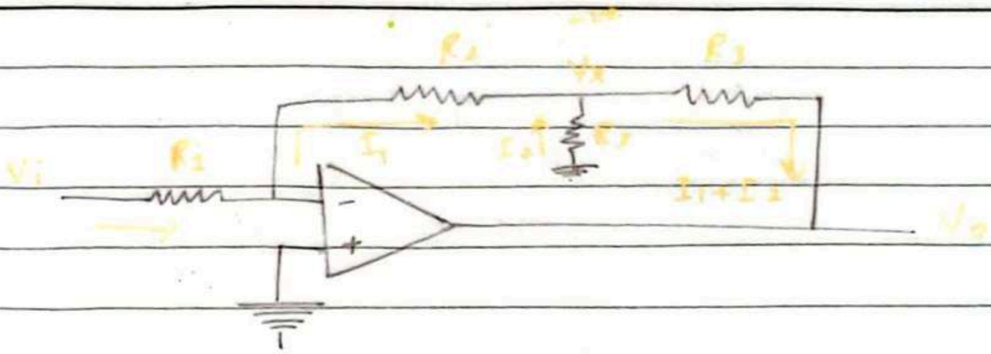
$$G = \frac{V_o}{V_i} = -\frac{R_F}{R_i} = -100$$

Objective :- I want my R_i to be large. I also want my G to be large and I also want my R_F to be practically possible.

sol.

$G = -1000$

Date



$$I_1 = \frac{V_i}{R_1} \longrightarrow (1)$$

$$0 - V_x = I_1 R_2$$

$$V_x = -I_1 R_2 \longrightarrow (2)$$

$$0 - V_x = I_2 R_3 \implies V_x = -I_2 R_3 \longrightarrow (3)$$

$$V_x - V_o = (I_1 + I_2) R_4 \longrightarrow (4)$$

From (1) & (2)

$$V_x = -I_1 R_2 = -\left(\frac{R_2}{R_1}\right) V_1$$

From (3)

$$V_x = -I_2 R_3$$

$$I_2 = -\frac{V_x}{R_3} = \left(\frac{R_2}{R_1 R_3}\right) V_i$$

$$V_x - V_o = (I_1 + I_2) R_4$$

$$-\left(\frac{R_2}{R_1}\right) V_i - V_o = \left(\frac{V_1}{R_1} + \left(\frac{R_2}{R_1 R_3}\right) V_i\right) R_4$$

$$-V_o = \left(\frac{R_2}{R_1}\right) V_i + \left(\frac{R_4}{R_1}\right) V_i + \left(\frac{R_2 R_4}{R_1 R_3}\right) V_i$$

$$-V_o = \left[\frac{R_2}{R_1} + \frac{R_4}{R_1} + \frac{R_2 R_4}{R_1 R_3}\right] V_i$$

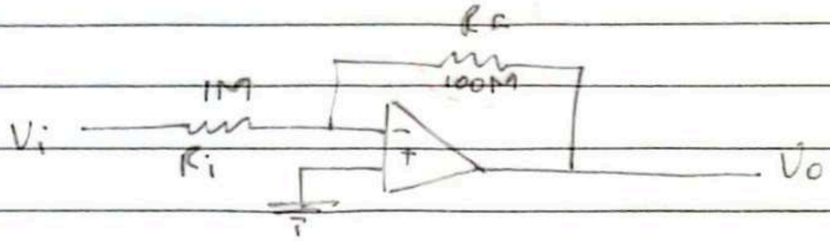
$$\frac{V_o}{V_i} = -\left[\frac{R_2}{R_1} + \frac{R_4}{R_1} + \frac{R_2 R_4}{R_1 R_3}\right]$$

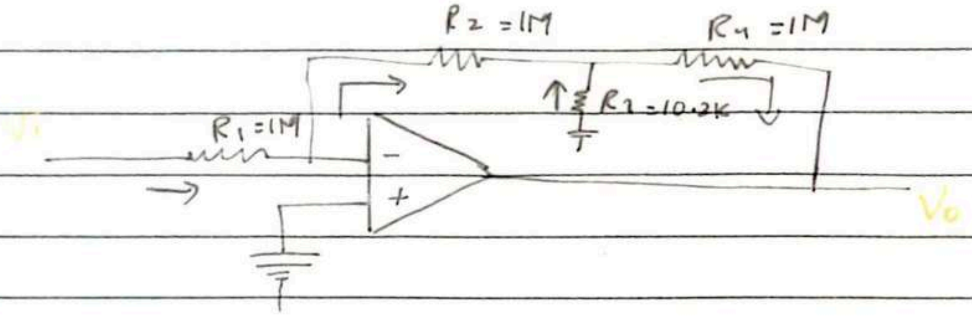
Scanned with CamScanner

Date

QUESTION :-

Design a normal inverting amp. of gain = 1000. The input resistance should be exactly 1 M Ω. Design an improved T network amp. All the resistances should be less than or equal to 1 M Ω.





putting in the formula

$$+1000 = + \left(\frac{1M}{1M} + \frac{1M}{1M} + \frac{1M \times 1M}{1M \times R_3} \right)$$

$$1000 = 1 + 1 + \frac{1M}{R_3}$$

$$\frac{1M}{R_3} = 998$$

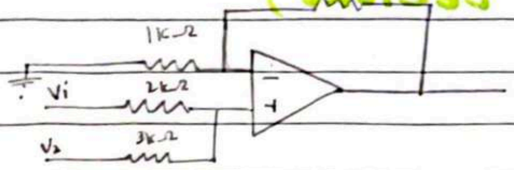
$$R_3 = \frac{1M}{998} \quad \boxed{= 1.02 \, K\Omega}$$

QUESTION :-

$$V_o = +6V_1 + 4V_2 - 3V_3$$

Scanned with CamScanner

Date



Assume $V_+ = V_-$ in every Question

$V_+ = V_- = V_x$

$\frac{V_x - V_1}{2k} + \frac{V_x - V_2}{3k} = 0$ (node voltage on non-inverting terminal)

$3V_x - 3V_1 + 2V_x - 2V_2 = 0$

$5V_x = 3V_1 + 2V_2 \longrightarrow (1)$

$\frac{V_x - 0}{1k} + \frac{V_x - V_0}{9k} = 0$ (N.V on inverting terminal)

$9V_x + V_x - V_0 = V_0$

$V_0 = 10V_x \longrightarrow (2)$

put (1) in (2)

$V_0 = 10\left(\frac{3V_1 + 2V_2}{5}\right)$

$V_0 = 6V_1 + 4V_2$

$\frac{V_x - V_3}{1k} + \frac{V_x - V_0}{9k} = 0$

$9kV_x - 9V_3 + V_x - V_0 = 0$

$V_0 = 10V_x - 9V_3 \longrightarrow (3)$

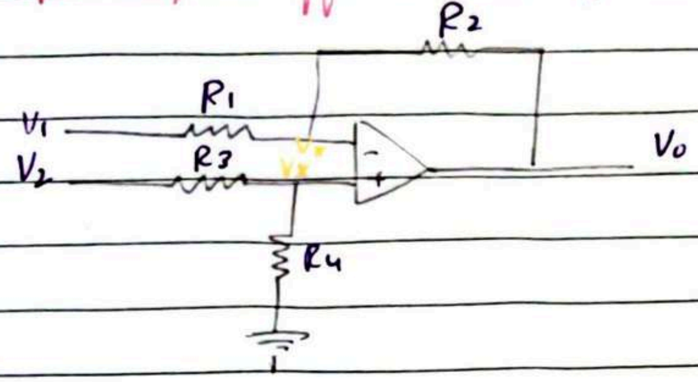
$V_0 = -9V_3 + 10\left(\frac{3V_1 + 2V_2}{5}\right)$

$V_0 = 6V_1 + 4V_2 - 9V_3$

Date

(Make Up LECTURE)

Single op-amp Difference Amp :-



$$V_+ = V_- = V_x$$

$$\frac{V_x - V_2}{R_3} + \frac{V_x - 0}{R_4} = 0$$ (N.V on non-inverting ter)

$$\frac{V_x}{R_3} - \frac{V_2}{R_3} + \frac{V_x}{R_4} = 0$$

$$V_x\left(\frac{1}{R_3} + \frac{1}{R_4}\right) = \frac{V_2}{R_3}$$

$$V_x = \left(\frac{R_4}{R_3 + R_4}\right)V_2 \longrightarrow ①$$

$$\frac{V_x - V_1}{R_1} + \frac{V_x - V_o}{R_2} = 0$$ (N.V on inverting term.)

$$\frac{V_x}{R_1} - \frac{V_1}{R_1} + \frac{V_x}{R_2} - \frac{V_o}{R_2} = 0$$

$$\frac{V_o}{R_2} = V_x\left(\frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{V_1}{R_1}$$

$$\frac{V_o}{R_2} = \left(\frac{R_1 + R_2}{R_1 R_2}\right)V_x - \frac{V_1}{R_1}$$

$$V_o = \left(\frac{R_1 + R_2}{R_1}\right)V_x - \left(\frac{R_2}{R_1}\right)V_1$$

Scanned with CamScanner

Date

$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_n - \left(\frac{R_2}{R_1}\right) V_1 \longrightarrow (2)$$

put (1) in (2)

$$V_o = + \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) V_2 - \left(\frac{R_2}{R_1}\right) V_1$$

Present:-

$$V_o = 3 V_2 - 1 V_2$$

Expect:-

$$\beta (V_2 - V_1)$$

solution:-

$$\gamma = \beta$$

$$\left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) = \frac{R_2}{R_1}$$

$$\frac{R_4}{R_4 + R_3} + \frac{R_2}{R_1}\left(\frac{R_4}{R_4 + R_3}\right) = \frac{R_2}{R_1}$$

$$\frac{R_4}{R_4 + R_3} = \frac{R_2}{R_1} - \frac{R_2}{R_1}\left(\frac{R_4}{R_4 + R_3}\right)$$

$$= \frac{R_2}{R_1}\left(1 - \frac{R_4}{R_4 + R_3}\right)$$

$$\frac{R_4}{\cancel{R_4 + R_3}} = \frac{R_2}{R_1}\left(\frac{R_3}{\cancel{R_4 + R_3}}\right)$$

$$\boxed{\frac{R_4}{R_3} = \frac{R_2}{R_1}}$$

Apply: $\frac{R_4}{R_3} = \frac{R_2}{R_1}$

$$V_o = \frac{R_2}{R_1}(V_2 - V_1)$$

Date

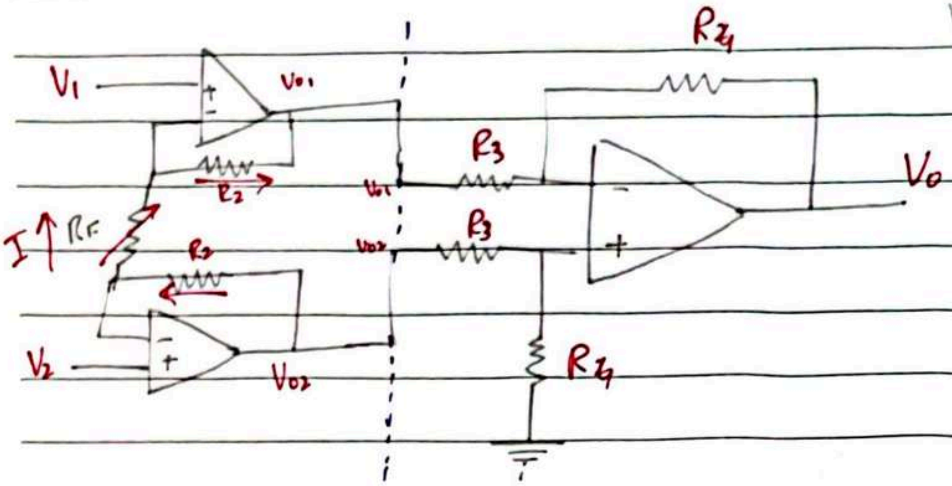
SINGLE OP-AMP DIFFERENCE AMP:

Problems:-

Very difficult to change R_1 and R_2 at the same ratio and so it becomes encuitiating.

And the same problem as before.

1st stage 2nd stage



$$V_o = \frac{R_2}{R_1}(V_2 - V_1)$$

$$V_o = G(V_2 - V_1)$$

(Instumentation Amplifer)

$$V_{o2} = \left(1 + \frac{R_2}{R_1}\right) V_2$$

$$V_{o1} = \left(1 + \frac{R_2}{R_1}\right) V_1$$

$$I = \frac{V_2 - V_1}{2R_1}$$

$$V_{o2} - V_{o1} = I(R_2) + I(2R_1) + I(R_2)$$

$$= I(2R_2 + 2R_1)$$

$$= \left(\frac{V_2 - V_1}{2R_1}\right)(2R_2 + 2R_1)$$

$$= \left(1 + \frac{2R_2}{2R_1}\right)(V_2 - V_1)$$

$$V_o = \frac{R_4}{R_3}(V_{o2} - V_{o1})$$

Scanned with CamScanner

Date

$$\boxed{V_0 = \left(\frac{R_4}{R_3}\right)\left(1 + \frac{2R_2}{R_F}\right)(V_2 - V_1)}$$

Scanned with CamScanner

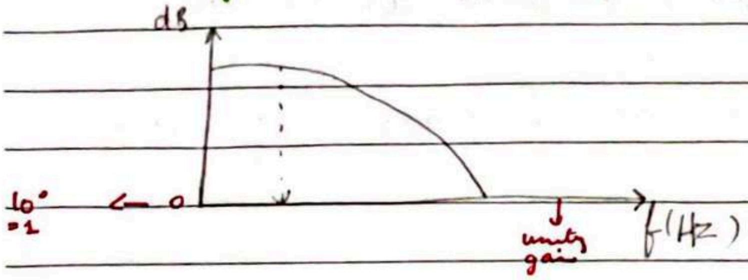
Date

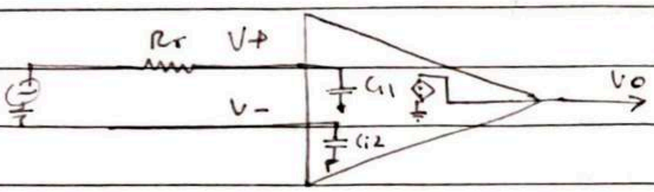
$$\boxed{V_o = \left(\frac{R_4}{R_3}\right)\left(1 + \frac{2R_2}{R_F}\right)(V_2 - V_1)}$$

DAY-9

Frequency Dependance of Open-loop Op-Amp gain :-

(acts like a low pass filter)





ideal = $A(s) = A_o$ (constant gain throughout)

real = $\boxed{A(s) = \left(\frac{1}{1 + s/\omega_b}\right) A_o}$

$$= \frac{A_o}{1 + s/\omega_b}$$

LPF

$$T(s) = \frac{\omega_b}{s + \omega_b}$$

$$= \frac{1}{1 + s/\omega_b}$$

$$s = j\omega$$

$$A(\omega) = \frac{A_o}{1 + j(\omega/\omega_b)}$$

for large frequencies ⇒ at large values of 'ω'

$$1 \ll \frac{\omega}{\omega_b}$$

Date

$$A(\omega) = \frac{A_o}{j(\omega/\omega_b)}$$

$$A(\omega) = \frac{A_o \omega_b}{j\omega}$$

P.YAD

$$\boxed{|A(\omega)| = \frac{A_o \omega_b}{\omega}}$$

magnitude of amplified gain (open loop) at any frequency ω can be evaluated as shown.

Q. What is the amp open loop gain at unity gain frequency?

$$|A(\omega)| = 1$$

Q. What is the unity gain frequency.

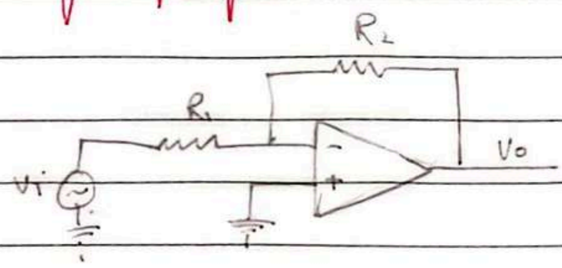
max. gain value

$$1 = \frac{A_o \omega_b}{\omega_T} \Rightarrow \boxed{\omega_T = A_o \omega_b}$$

$$f_T = A_o f_b$$
$$= 10^5 \times 10^1$$
$$= 10^6 \text{ Hz}$$

Frequency Dependance Of Closed loop Op-Amp :-

Inverting Amplifier :-



Inv AM Gain

$A \to \infty$: $-\frac{R_2}{R_1}$

$A \to F$: $\frac{-R_2/R_1}{1 + 1/A(1 + R_2/R_1)}$

$$\frac{-R_2/R_1}{1 + \frac{1}{A_o}(1 + R_2/R_1)} \qquad \frac{-R_2/R_1}{1 + \frac{1}{A(\omega)}(1 + R_2/R_1)}$$

$$G = \frac{V_o}{V_i} = \frac{-R_2/R_1}{1 + 1/A(1 + R_2/R_1)}$$

Scanned with CamScanner

Date

$$\frac{V_o}{V_s} = \frac{-R_2/R_1}{1+\left[\frac{1+R_2/R_1}{A/A+s/\omega_b}\right]}$$

$$\frac{V_o}{V_i} = \frac{-R_2/R_1}{1+(1+R_2/R_1)(1+s/\omega_b/A_0)}$$

$$\frac{V_o}{V_i} = \frac{-R_2/R_1}{1+(1+R_2/R_1)+s/A_0\omega_b(1+R_2/R_1)}$$

$$= \frac{-R_2/R_1}{1+1/A_0(1+R_2/R_1)+s/(A_0\omega_b/1+R_2/R_1)}$$

$A_0 = 10^5$ (very large

$$\frac{V_o(s)}{V_s(s)} = \frac{-R_2/R_1}{1+s/\cancel{A_0}\omega_b/1+R_2/R_1}$$

Open - loop Op - amp:-

$$A(s) = \frac{A_0}{1+s/\omega_b}$$ $\omega_T = A_0\omega_b$ (unity gain freq.)

$$|A(\omega)| = \frac{A_0\omega_b}{\omega}$$ ω_b = cut off freq.

Closed loop Op - amp:-

$$G(s) = \frac{-R_2/R_1}{1+s/A\omega_b/1+R_2/R_1}, \quad \omega_b' = \frac{A_0\omega_b}{1+R_2/R_1}$$

$$|G(\omega)| = \frac{(R_2/R_1)(A_0\omega_b/1+R_2/R_1)}{\omega}$$

Makeup - lecture

$$G(s) = \frac{-R_2/R_1}{1+s/A_0\omega_b/1+R_2/R_1} = \frac{G_0}{1+s/\omega_Q} \Longleftrightarrow \frac{A_0}{1+s/\omega_b}$$

$$-\frac{R_2}{R_1} = G_0$$

Scanned with CamScanner

Date

$$w_Q = \frac{A_0 w_0}{1 + R_2/R_1}$$

$$G(s) = \frac{G_{10}}{1 + S/w_0}$$

$$|G(w)| = G_{10}/w/w_a$$

$$|G(w)| = \frac{G_0 w_\sigma}{w}$$

$$S = jw$$

$$G(w) = \frac{G_0}{1 + j(w/w_\theta)}$$

$$|G(w)| = 1$$

$$\frac{G_0 w_\sigma}{w_2} = 1$$

$$G(w) = \frac{G_0}{j(w/w_\theta)}$$

$$w_2 = G_0 W_\theta$$

Main Formulas:-

$$w_T = A_0 w_b$$

unity gain freq of open loop

$$w_z = \left(\frac{R_2}{R_1}\right)\left(\frac{A_0 w_b}{1 + R_2/R_1}\right)$$

~~cut-off~~ freq.

$$w_\alpha = \frac{A_0 w_b}{1 + R_2/R_1}$$

Closed-loop Op-Amp (Non-inverting) :-

$$G(s) = \frac{1 + R_2/R_1}{1 + S/A_0 w_b/(1 + R_2/R_1)} \Rightarrow \frac{G_0}{1 + S/w_k}$$

$$G_0 = 1 + R_2/R_1$$

$$w_k = \frac{A_0 w_b}{1 + R_2/R_1}$$

$$|G(w)| = \frac{G_{10} w_k}{w} = \frac{(1 + R_2/R_1)(A_0 w_b/1 + R_2/R_1)}{w}$$

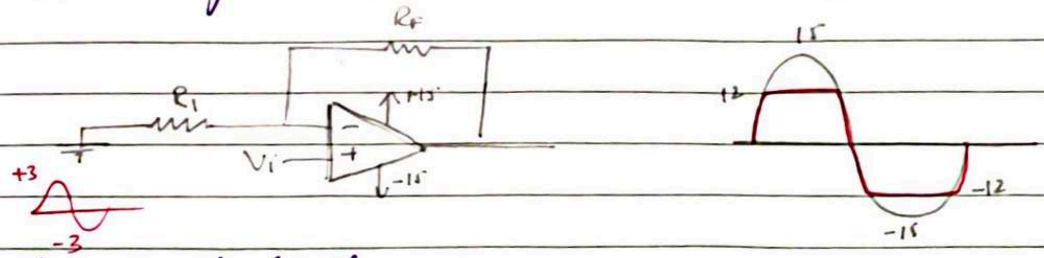
Date

cut off freq. $\omega_k = \dfrac{A_o \omega_b}{1 + R_2/R_1}$

unity gain freq:- $\omega_y = G_o \omega_k = \left(1 + \dfrac{R_2}{R_1}\right)\left(\dfrac{A_o \omega_o}{1 + R_2/R_1}\right)$

DC imperfections Of OP-amps:-

1) O/P voltage saturation:-

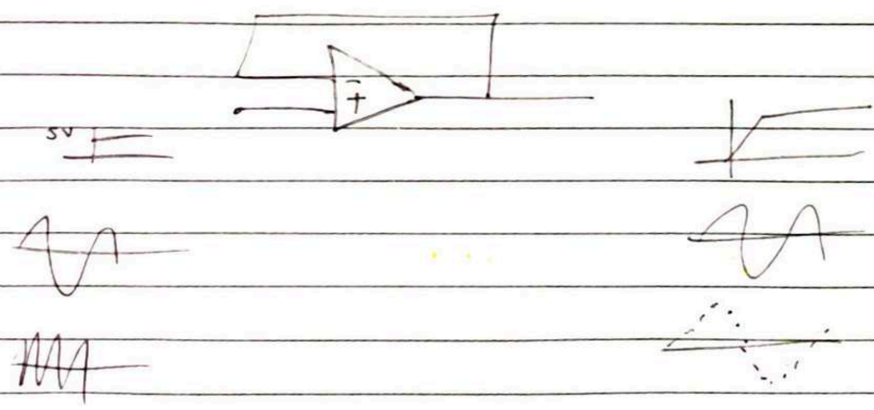


2) O/P current limit:-

→ Max. O/P current limited to ± 20mA

3) SLEW Rate:-

If the input signal applied is changing / varying faster than the value of SR, the OP-AMP will not comply.



Date

$v_i = V_I \sin(\omega t)$

$\frac{dv_i}{dt} = V_I \omega . \cos(\omega t)$

$\left.\frac{dv_i}{dt}\right|_{max} = V_I \omega$

$SR \geq \left.\frac{dV_i}{dt}\right|_{max}$ or $SR \geq \omega V_I$ (safe)

$SR < \omega V_I$ (unsafe)

$SR = \omega_{max} . V_{I.max}$

$SR = 2\pi f_{max} . V_{I.max}$

$f_{max} = \frac{SR}{2\pi (V_{I.max})}$

$V_{I max} = \frac{SR}{2\pi (f_{max})}$

Scanned with CamScanner

Date

$$v_i = V_I \sin(\omega t)$$

$$\frac{dv_i}{dt} = V_I \omega . \cos(\omega t)$$

$$\left.\frac{dv_i}{dt}\right|_{max} = V_I \omega$$

$$SR \geq \left.\frac{dv_i}{dt}\right|_{max} \quad \text{or} \quad SR \geq \omega V_I \text{ (safe)}$$

$$SR < \omega V_I \text{ (unsafe)}$$

$$SR = \omega_{max} . V_{I.max}$$

$$SR = 2\pi f_{max} . V_{I.max}$$

$$f_{max} = \frac{SR}{2\pi(V_{I.max})}$$

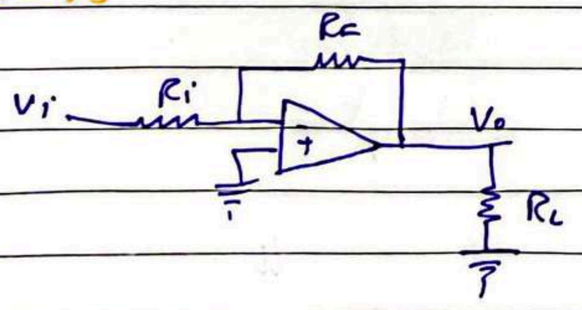
$$V_{I max} = \frac{SR}{2\pi(f_{max})}$$

DAV-10 (After Mid 1)

The output voltage is across R_L and output current is a through R_L.

Dependant Current & Voltage Sources :-

1) VC-VS

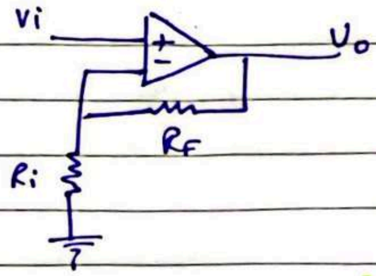


$$V_o = K V_i$$

Shez

Date

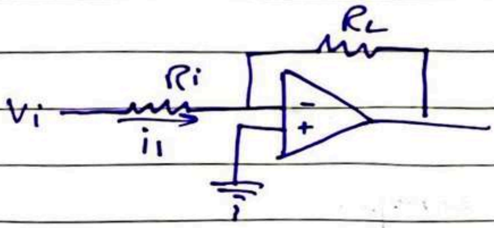
$$V_o = \left(-\frac{R_F}{R_i}\right) V_i$$



$$\boxed{V_o = \left(1 + \frac{R_F}{R_i}\right) V_i}$$

2) VC - CS

$$i_o = K V_i$$



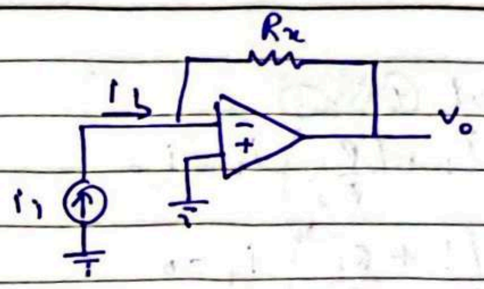
$$i_i = \frac{V_i}{R_i}$$

$$i_o = \frac{V_i}{R_i}$$

$$\boxed{i_o = \left(\frac{1}{R_i}\right) V_i}$$

3) CC - VS

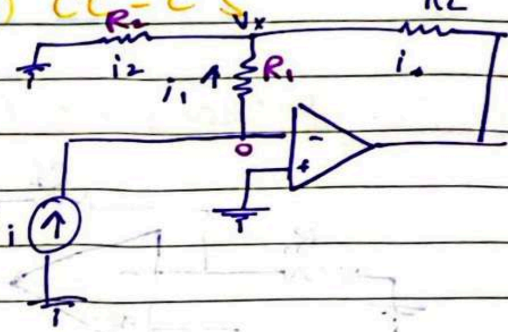
$$V_o = K i_i$$



$$0 - V_o = R_x i_i$$

$$\boxed{V_o = (-R_x) i_i}$$

4) CC - CS



$$i_o = K i_i$$

$$0 - V_x = i_1 R_1$$

$$V_1 = -i_i R_i \rightarrow (1)$$

$$i_1 = -\frac{V_x}{R_i}$$

$$0 - V_x = i_2 R_2$$

$$V_x = -i_2 R_2 \rightarrow (2)$$

$$i_1 + i_2 = i_o \rightarrow (3)$$

put (1) in (2)

$$i_1 R_i = i_2 R_2$$

$$i_2 = \left(\frac{R_1}{R_2}\right) i_1 \rightarrow (4)$$

Scanned with CamScanner

Date

put ④ in ③

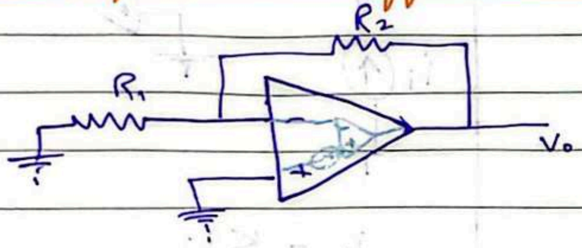
$$i_1 + \left(\frac{R_1}{R_2}\right) i_1 = i_o$$

$$\left(1 + \frac{R_1}{R_2}\right) i_1 = o$$

$$\boxed{i_o = \left(1 + \frac{R_1}{R_2}\right) i_i}$$

DAY-11

DC imperfections (Offset Voltage) :-



($\because$ ideally V_o has to be 0)

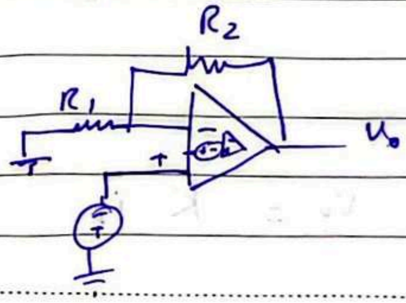
Now, as it is a non inverting op-amp.

$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_i$$

$$V_i = V_{os}$$

$$\boxed{V_o = \left(1 + \frac{R_2}{R_1}\right) V_{os}}$$

Now applying V_i on non-inverting terminal



$$V_o = \left(1 + \frac{R_2}{R_1}\right)(V_i + V_{os})$$

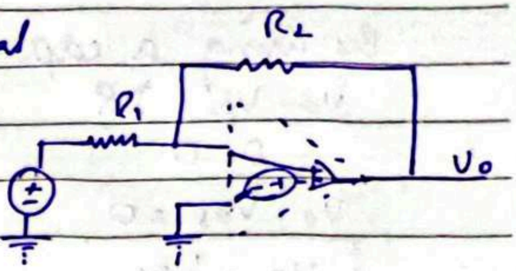
$$V_o = \left(1 + \frac{R_2}{R_1}\right) V_i + \underbrace{\left(1 + \frac{R_2}{R_1}\right) V_{os}}_{\text{parasitic}}$$

Date

Testy

Now applying V_i on inverting terminal

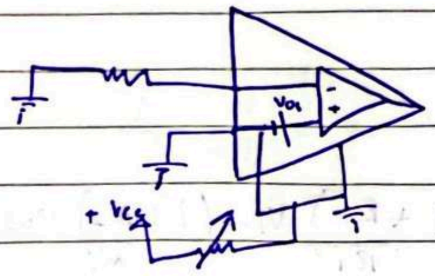
Using principle of superposition

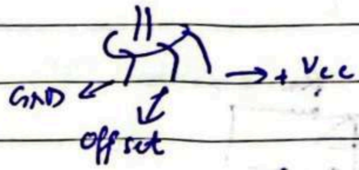


$$\boxed{V_o = \left(1 + \frac{R_2}{R_1}\right) V_{os} - \left(\frac{R_2}{R_1}\right) V_i}$$

∵ in POS, we take one source and ground ~~the other and vice versa~~ and after finding all, we sum all the o/p voltages.

solution to this Imperfection.





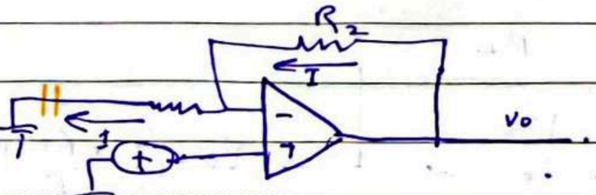
We use variable resistor and check the Vo and keep on rotating until $V_o = 0$. It indicates the V_{os} has been cancelled out.

~~(V_{os} is a DC~~ signal)

1st Remedy :-

Use a potentiometer at 'offset' pin of op-amp to introduce an equal but opposite voltage in series with V_{os}.

2nd Remedy :- (Partial Remedy) (cost effective)



By using a capacitor (non-polar), $I = 0$

$V_0 - V_{os} = IR$

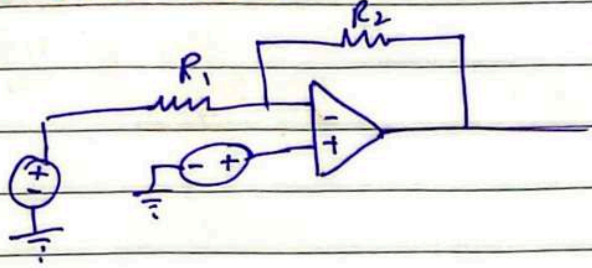
$I = 0$

$V_0 - V_{os} = 0$

$$\boxed{V_0 = V_{os}}$$

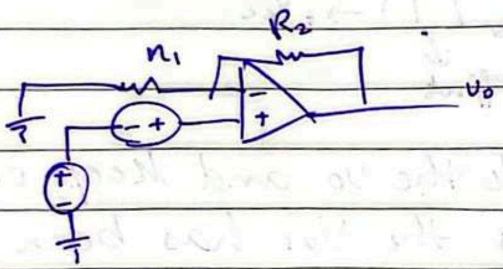
(ONLY WORK IF V_i IS AC)

inv. with Vos :-



$$\boxed{V_0 = -\left(\frac{R_2}{R_1}\right) V_i + \left(1 + \frac{R_2}{R_1}\right) V_{os}}$$

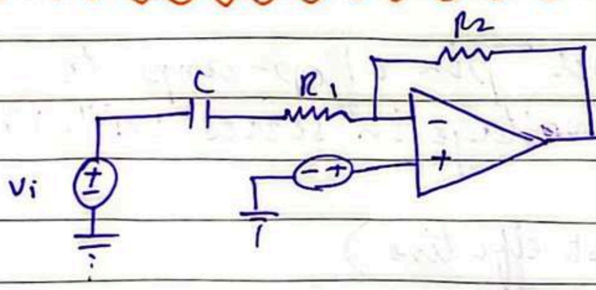
noninv. with Vos :-



$$V_0 = \left(1 + \frac{R_2}{R_1}\right) V_i + \left(1 + \frac{R_2}{R_1}\right) V_{os}$$

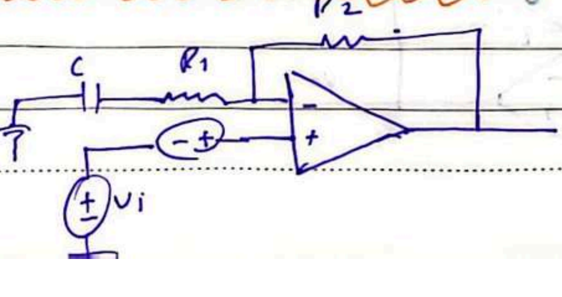
$$\boxed{= \left(1 + \frac{R_2}{R_1}\right) \left(V_1 + V_{os}\right)}$$

inv. with Vos & partial remedy :-



$$\boxed{V_0 = -\left(\frac{R_2}{R_1}\right) V_1 + V_{os}}$$

non-inv. with Vos & partial remedy :-



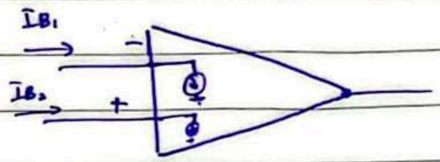
$$\boxed{V_0 = \left(1 + \frac{R_2}{R_1}\right) V_1 + V_{os}}$$

Scanned with CamScanner

Date

DC imperfection (Input Bias Current) :-

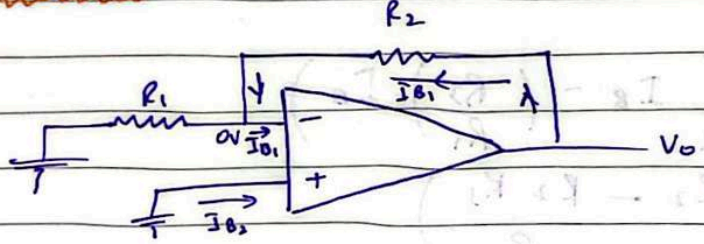
∵ In real op-amp, some non-zero current I_{B_1} & I_{B_2} tends to flow in op-amp terminals.



average input Bias current ⇒ $I_B = \frac{I_{B_1} + I_{B_2}}{2} \approx 100 mA$

Input offset current ⇒ $I_{os} = |I_{B_1} - I_{B_2}| \approx 1 mA\ (\approx 0)$

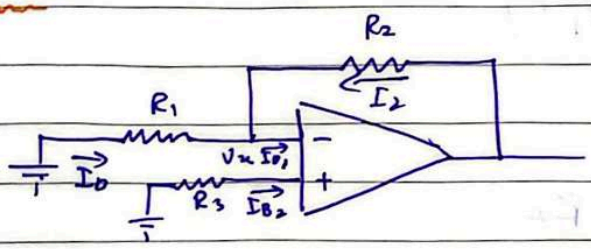
PROBLEM:-



$V_o - 0 = I_{B_1} R_2$

$\boxed{V_o = I_{B_1} R_2} \neq 0$

SOLUTION:-



For simplicity :- $I_{B_1} = I_{B_2} = I_B$

$0 - V_x = I_B R_3$

$V_x = -I_B R_3$

Date

$$I_1 = \frac{0 - V_n}{R_1} = I_B\left(\frac{R_3}{R_1}\right)$$

By applying KCL,

$$I_1 + I_2 = I_B$$
$$I_2 = I_B - I_1$$
$$I_2 = I_B - \left(\frac{R_3}{R_1}\right) I_B$$

$$V_0 - V_n = I_2 R_2$$
$$V_0 = V_n + I_2 R_2$$

putting values

$$= -I_B R_B + R_2\left(I_B - \left(\frac{R_3}{R_1}\right) I_B\right)$$

$$V_0 = I_B\left(-R_3 + R_2 - \frac{R_2 R_3}{R_1}\right)$$

$$V_0 = I_B\left(R_2 - R_3\left(1 + \frac{R_2}{R_1}\right)\right)$$

lets put $V_0 = 0$

$$0 = I_B\left(R_2 - R_3\left(1 + \frac{R_2}{R_1}\right)\right)$$

$$R_2 - R_3\left(1 + \frac{R_2}{R_1}\right) = 0$$

$$R_3\left(1 + \frac{R_2}{R_1}\right) = R_2$$

$$R_3\left(\frac{R_1 + R_2}{R_1}\right) = R_2$$

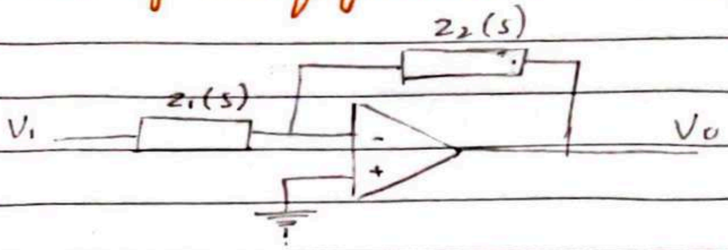
$$R_3 = \frac{R_1 R_2}{R_1 + R_2} \implies \boxed{R_3 = R_1 \parallel R_2}$$

Scanned with CamScanner

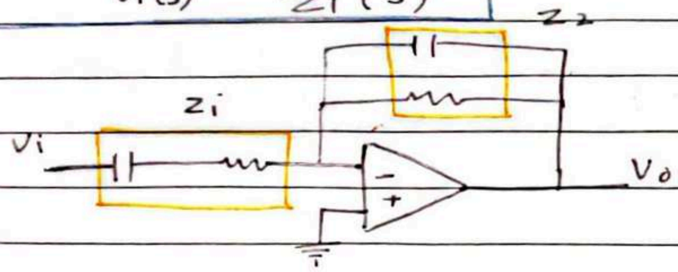
Date

DAY-12

Inverting Configuration with General Inpedences



$$\boxed{G(s) = \frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}}$$

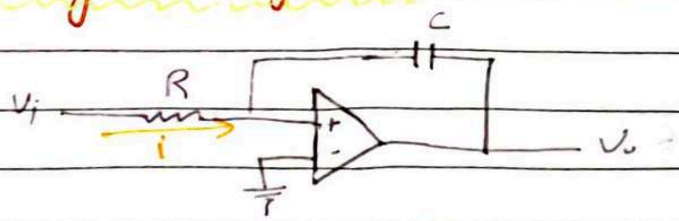


$$\frac{V_o(s)}{V_i(s)} = \frac{Z_2(s)}{Z_1(s)} = -\frac{R_2 \parallel 1/C_2 s}{R_1 + 1/C_1 s}$$

$$= -\frac{R_2(1/C_2 s)/R_2 + 1/C_2 s}{R_1 + 1/C_1 s}$$

$$= -\frac{R_2/C_2 s}{(R_1 + 1/C_1 s)(R_2 + 1/C_2 s)}$$

Inverting "Integrator"



$$i(t) = \frac{V_i(t)}{R}$$

Scanned with CamScanner

Date

$$i(t) = \frac{C\,dv(t)}{dt}$$

$$\int \frac{dv}{dt} = \int \frac{i(t)}{C}$$

$$V_c(t) = \frac{1}{C}\int i(t)\,dt + V_c(0)$$

$$V_c(0) = 0$$

$$V_c(t) = \frac{1}{C}\int i(t)\,dt$$

$$= \frac{1}{C}\int \frac{V_i(t)}{R}\,dt$$

$$V_c(t) = \frac{1}{RC}\int V_i(t)\,dt$$

$$0 - V_o(t) = V_c(t)$$

$$V_o(t) = -V_c(t)$$

$$\boxed{V_o(t) = -\frac{1}{RC}\int V_i(t)\,dt}$$

Another Method :-

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)} = -\frac{1/Cs}{R}$$

$$= -\frac{1}{RCs}$$

$$V_o(s) = -\frac{1}{RCs}\left(\frac{V_i(s)}{s}\right)$$

L^{-1} on both sides

$$\boxed{V_o(t) = \frac{-1}{RC}\int V_i(t)\,dt}$$

Scanned with CamScanner

Date

Acts as a low Pass Filter :-

$$\frac{V_o(s)}{V_i(s)} = \frac{-1}{RCs}$$

$$G(s) = \frac{-1}{RCs}$$

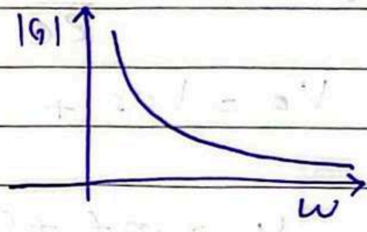
$$s = j\omega$$

$$G(\omega) = -\frac{1}{RCj\omega}$$

$$= \frac{j}{RC\omega}$$

$$G(\omega) = 0 + j\left(\frac{1}{RC\omega}\right)$$

$$|G| = \sqrt{0^2 + (1/RC\omega)^2} = \frac{1}{RC\omega}$$

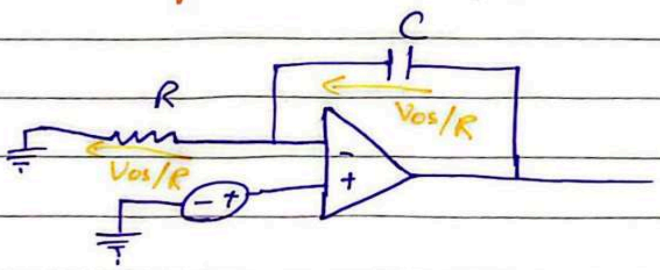


$$\angle G = \tan^{-1}\left(\frac{1/RC\omega}{0}\right)$$

$$= 90°$$

$$G = \frac{1}{RC\omega} \angle 90°$$

Impact Of Vos on Integrator's behaviour :-



$$i = C\frac{dv}{dt}$$

Scanned with CamScanner

$$V_c = \frac{1}{C} \int i(t)\, dt$$

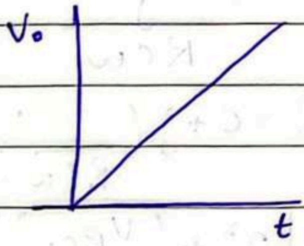
$$V_c = \frac{1}{C} \int \frac{V_{os}}{R}\, dt$$

$$V_c = \frac{1}{RC} \int_0^t V_{os}\, dt$$

$$V_o - V_{os} = V_c$$

$$V_o = V_{os} + V_c$$

$$V_o = V_{os} + \frac{1}{RC} \int_0^t V_{os}\, dt$$

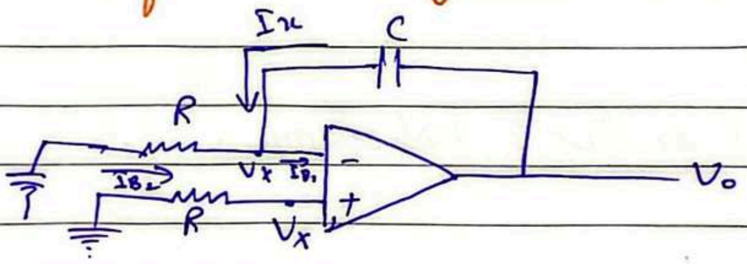


$$V_o = V_{os} + \frac{1}{RC} \left(V_{os} \times t\right)$$

$$V_o = V_{os} + \left(\frac{V_{os}}{RC}\right) t$$

$$y = mt + C$$

Impact of I_B on integrator's behaviour



$$0 - V_x = I_{B2} R$$

$$V_x = -R I_{B_2}$$

$$i = \frac{0 - (-I_{B_2} \cancel{R})}{\cancel{R}}$$

$$i = I_{B_2}$$

Date

$$i_x + I_{B_2} = I_{B_1}$$

$$i_x = I_{B_1} - I_{B_2}$$

$$V_C = \frac{1}{C}\int_0^t i_x(t)\,dt$$

$$= \frac{1}{C}\int_0^t I_{B_1} - I_{B_2}\,dt$$

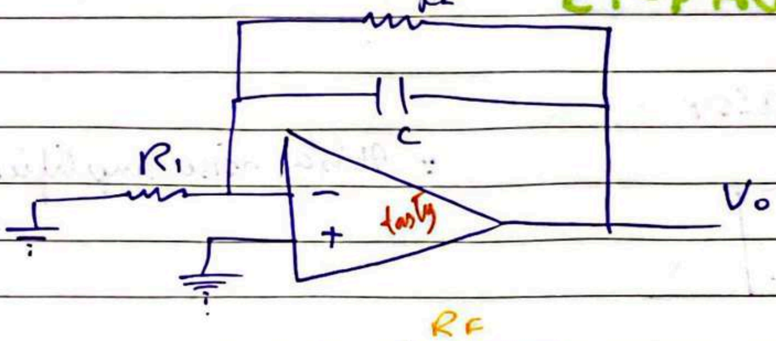
$$V_C = \left(\frac{I_{B_1} - I_{B_2}}{C}\right)t$$

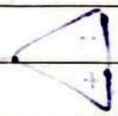
$$V_C = V_o - V_X$$

$$V_o = V_C + V_X$$

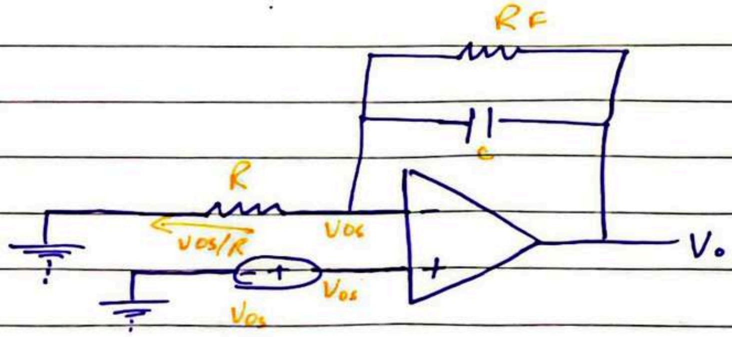
$$\boxed{V_o = -I_{B_2}R + \left(\frac{I_{B_1} - I_{B_2}}{C}\right)t}$$

DAY-13





∴ current will always choose resistor over capacitor.



$$V_o = V_{os} + \left(\frac{V_{os}}{R}\right)R_F$$

$$\boxed{V_o = \left(1 + \frac{R_F}{R}\right)V_{os}}$$

Scanned with CamScanner

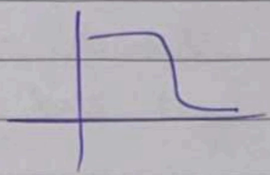
Date

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$

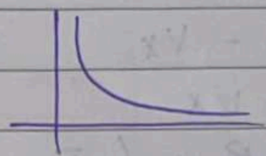
$$= -\frac{1/Cs \parallel R_F}{R_1}$$

$$G(s) = \frac{-R_F/R_1}{1 + s(CR_F)}$$

$$G(s) = \frac{-1}{(RC)s}$$



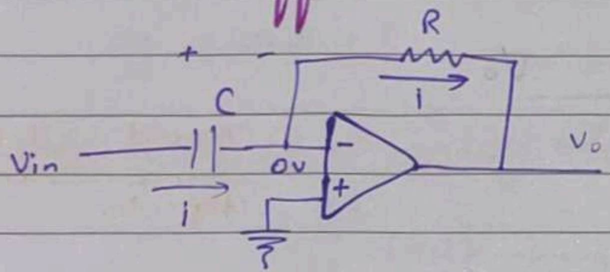
(low pass filter)



DAY-13

OP-AMP Differentiator :-

∵ It is a noise amplifier



$$i = C\frac{dv_c}{dt}$$

$$i = C\frac{dv_{in}}{dt} \longrightarrow (1)$$

$$0 - V_o = iR$$

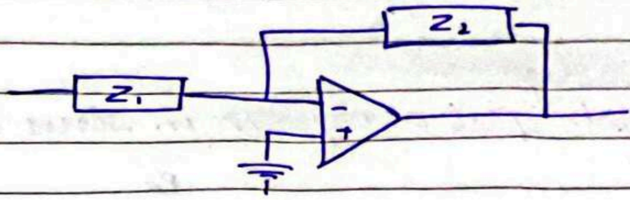
$$V_o = -iR \longrightarrow \text{put in } (1)$$

$$V_o = -RC\frac{dv_{in}}{dt}$$

Date

$RC = 1$

$V_o = -\dfrac{dV_i}{dt}$



$$\frac{V_o(s)}{V_i(s)} = \frac{-Z_2}{Z_1}$$

$$\frac{V_o}{V_i} = -\frac{R}{\frac{1}{Cs}}$$

$$\boxed{\frac{V_o}{V_i} = -RCs}$$

$$V_o(s) = -RC\left(V_i(s)\, s\right)$$

$$\int V(t)\,dt \xrightarrow{L} \frac{V(s)}{s}$$

$$\frac{dV(t)}{dt} \xrightarrow{L} sV(s)$$

Taking L^{-1}

$$\boxed{V_o(t) = -RC\left(\frac{dV_i}{dt}\right)}$$

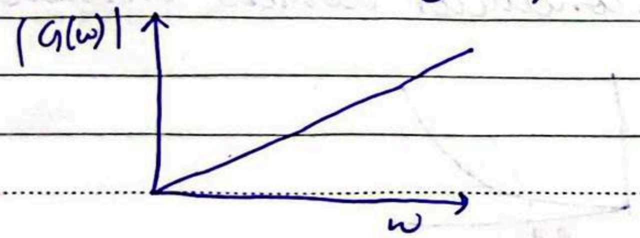
$$\frac{V_o(s)}{V_i(s)} = -RCs$$

$$s = j\omega$$

$$G(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = -RC(j\omega) = 0 - j(RC\omega)$$

$$|G(\omega)| = RC\omega$$

$$\angle G(\omega) = \tan^{-1}\left(\frac{-RC\omega}{0}\right) = -90^\circ$$

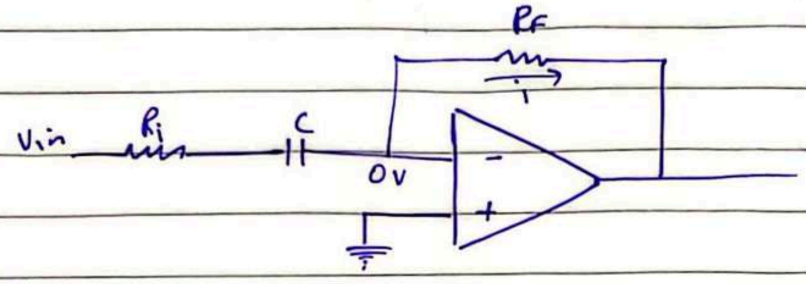


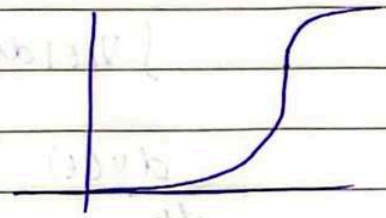
Scanned with CamScanner

Date

Remedy:-

We put a resistor in series with a capacitor





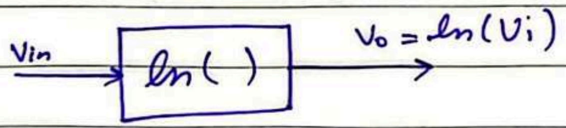
(high pass filter)

$$\frac{V_o}{V_i} = -\frac{R}{R_i + 1/Cs}$$

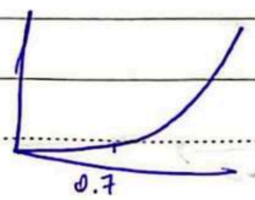
$$\boxed{\frac{V_o}{V_i} = -\frac{R/R_i}{1 + 1/RCs}}$$

Non-linear Operation Of Op-AMP :-

Logarithmic Amp:-



All the PN Junction / semiconductor devices exhibit an exp. behaviour

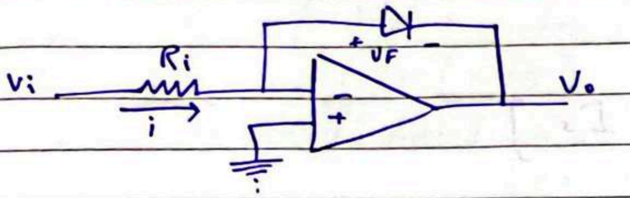


Scanned with CamScanner

Date

log amp. is an inverting amp. The feedback resistor is replaced with any PN Junction device (Diode)

Log Amplifier :- $V_o \propto \ln(V_i)$



$$i = \frac{V_i}{R_i}$$

Diode I-V Eq :-

$$I = I_s\left(e^{V_F/\eta V_T} - 1\right)$$

η = eta = $\begin{cases} 1 \text{ for silicon} \\ 2 \text{ for germanium} \end{cases}$

V_T = thermal voltage

I = forward current.
I_s = Reverse sat. current
V_F = Voltage across diode

Assume $V_F > \eta V_T \longrightarrow e^{V_F/\eta V_T} >> 1$

neglecting 1

$$\therefore I = I_s e^{V_F/\eta V_T}$$

taking ln() on both sides of eq

$$\ln(I) = \ln\left(I_s e^{V_F/\eta V_T}\right)$$

$$\ln I = \ln I_s + \ln e^{V_F/\eta V_T}$$

$$\ln I = \ln I_s + \frac{V_F}{\eta V_T}$$

Date

$$V_F = \eta V_T (\ln I - \ln I_s)$$

$$V_F = \eta V_T \left(\ln\left(\frac{V_i}{R}\right) - \ln I_s \right)$$

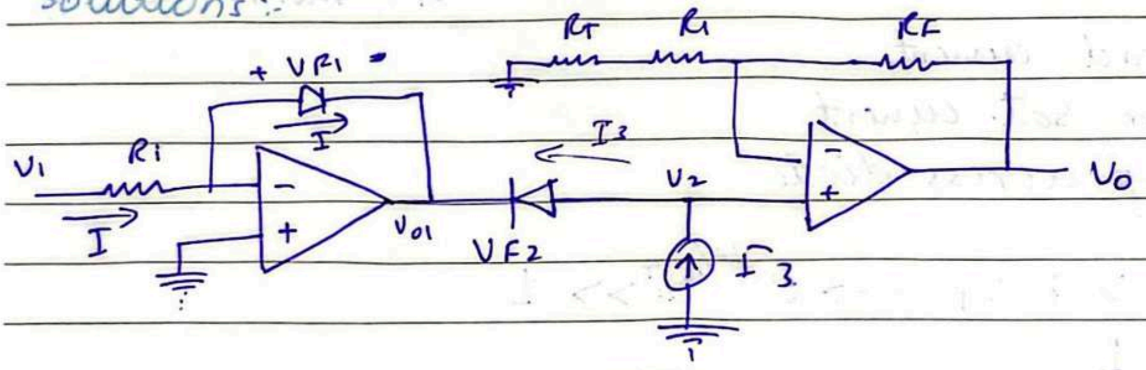
$$0 - V_0 = V_F$$
$$V_0 = - V_F$$

$$V_0 = - \eta V_T \left[\ln\left(\frac{V_i}{R} \right) - \ln I_s \right]$$

Problems:-

1) $\ln(I_s)$ needs to be removed. Put similar diode in opposite direction
2) $V_0 \propto \eta V_T$

Solutions:-



(Modified Log. Amp)

$$V_{01} = -\eta V_T \left(\ln\left(\frac{V_i}{R}\right) - \ln(I_s) \right)$$

$$V_{F2} = +\eta V_T \left(\ln I_3 - \ln(I_s) \right)$$

$$\therefore V_{F2} = V_2 - V_{01}$$

Scanned with CamScanner

Date

$V_2 = V_{F2} + V_{O1}$

$V_2 = \eta V_T [\ln(I_3) - \ln(I_s)] - \eta V_T [\ln(V_i/R) - \ln I_s]$

$= \eta V_T \ln I_3 - \eta \cancel{V_T \ln}(I_s) - \eta V_T \ln(V_i/R) + \eta \cancel{V_T \ln} I_s$

$V_2 = \eta V_T (\ln I_3 - \ln(V_i/R))$

$V_2 = -\eta V_T (\ln(V_i/R) - \ln(I_3))$

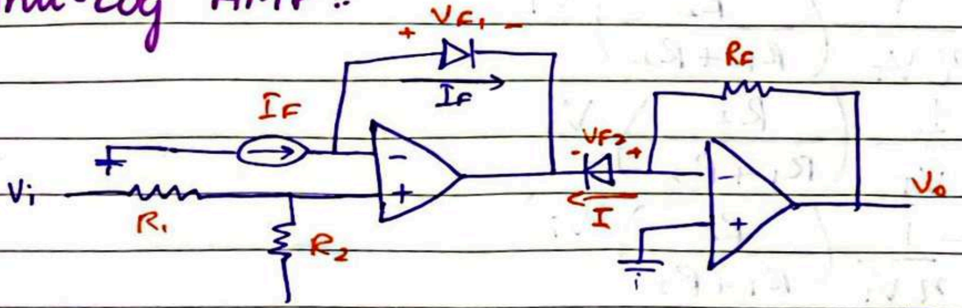
$V_2 = -\eta V_T \left(\ln\left(\frac{V_i}{I_3 R}\right)\right)$ (1st prob. solved)

$V_o = \left(1 + \frac{R_F}{R_1 + R_T}\right) V_2$ (as sec amp is non inverting)

$$\boxed{V_o = -\left(1 + \frac{R_F}{R_1 + R_T}\right)\left(\eta T \ln\left(\frac{V_i}{\underbrace{I_3 R_i}_{\text{const.}}}\right)\right)}$$

DAY-14

Anti-Log AMP :-



$I = \frac{V_o}{R}$

$V_x = \left(\frac{R_2}{R_1 + R_2}\right) V_i$

$V_{F1} = +\eta V_T (\ln I_F - \ln I_s)$

Date

$$V_x - V_{o1} = V_{F1}$$

$$V_{o1} = V_x - V_{F1}$$

$$V_{o1} = \left(\frac{R_2}{R_1 + R_2}\right) V_i - \eta V_T \left(\ln I_F - \ln I_s\right) \longrightarrow ①$$

$$0 - V_{o1} = V_{F2}$$

$$V_{o1} = -V_{F2} \longrightarrow ②$$

$$V_{F2} = +\eta V_T \left(\ln I - \ln I_s\right)$$

substitute ① in ②

$$-V_{F2} = V_x - V_{F1}$$

$$-\eta V_T \left(\ln I - \ln I_s\right) = \left(\frac{R_2}{R_1 + R_2}\right) V_i - \eta V_T \left(\ln I_F - \ln I_s\right)$$

$$-\eta V_T \ln I + \eta V_T \cancel{\ln I_s} + \eta V_T \ln I_F - \eta V_T \cancel{\ln I_s} = \left(\frac{R_2}{R_1 + R_2}\right) V_i$$

$$-\eta V_T \left(\ln I - I_F\right) = \left(\frac{R_2}{R_1 + R_2}\right) V_i$$

$$\left(\frac{R_2}{R_1 + R_2}\right) V_i = \eta V_T \left(\ln I_F - \ln I\right)$$ DAY-11

$$\ln\left(\frac{I_F}{I}\right) = \frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) V_i$$

$$\ln\left(\frac{I}{I_F}\right) = -\frac{1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) V_i$$

$$\frac{I}{I_F} = \ln^{-1}\left(\frac{-1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) V_i\right)$$

$$I = I_F \ln^{-1}\left(\frac{-1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) V_i\right)$$

$$\frac{V_o}{R_F} = I_F \ln^{-1}\left(\frac{-1}{\eta V_T} \left(\frac{R_2}{R_1 + R_2}\right) V_i\right)$$

Scanned with CamScanner

Date

$$V_o = I_F R_F \ln^{-1}\left(-\frac{1}{\eta V_T}\left(\frac{R_2}{R_1+R_2}\right)V_i\right)$$

We replace R_2 with R_T

$$\boxed{V_o = I_F R_F \ln^{-1}\left(-\frac{V_i}{R_{F1}}\right)}$$

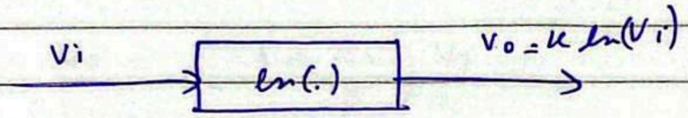
$R_2 = R_T$

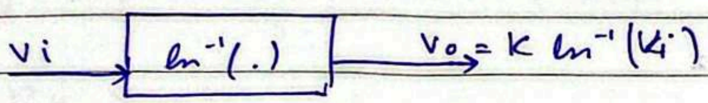
$\frac{1}{\eta V_T}\left(\frac{R_T}{R_1+R_T}\right)$

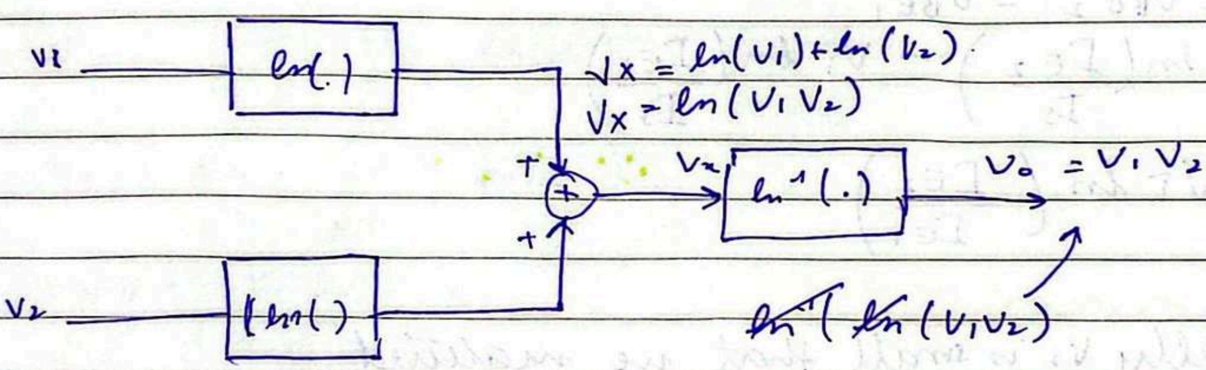
$R_1 >> R_T$

$\frac{1}{\eta V_T}\left(\frac{R_T}{R_1}\right)$

$\frac{1}{\eta R_1}\left(\cancel{\frac{R_T}{V_T}}\right)$

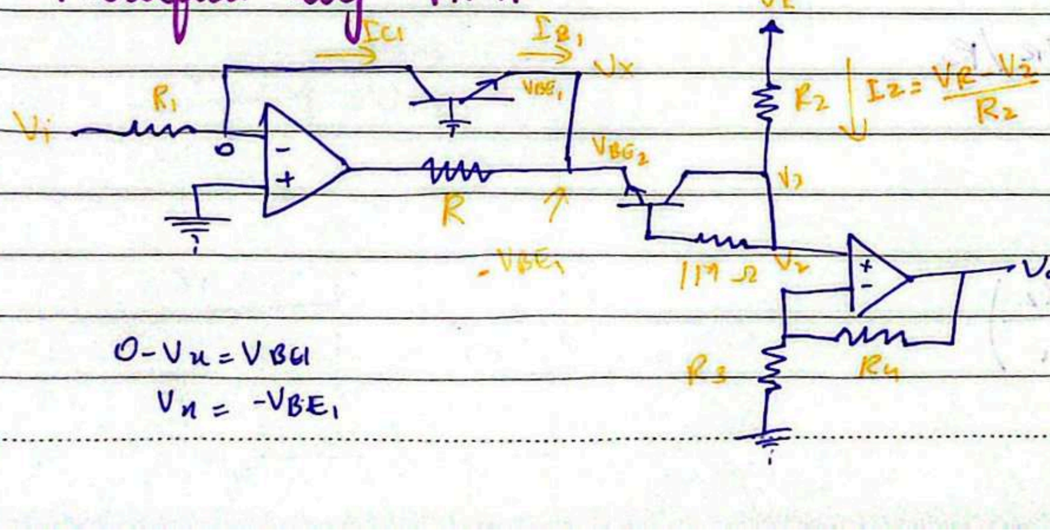






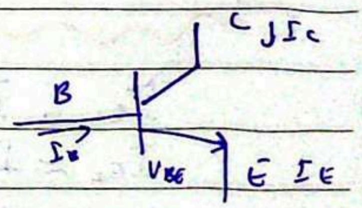
$\cancel{\ln^{-1}}(\cancel{\ln}(V_1 V_2))$

Modified 'log' AMP



$0 - V_x = V_{BE1}$

$V_x = -V_{BE_1}$



$\beta I_B = I_C$

$I_C + I_B = I_E$

$V_{BE} \geq 0.7$

Scanned with CamScanner

Date

BJT x-tics:

$$I_E = I_S e^{V_{BE}/V_T}$$

$$V_{BE} = V_T \ln\left(\frac{I_E}{I_S}\right)$$

$$I_{C1} = \frac{V_i}{R_1} \approx I_{E1}$$

$$V_{BE_1} = V_T \ln\left(\frac{I_B}{I_S}\right)$$

$$V_{BE2} = V_T \ln\left(\frac{I_{E2}}{I_S}\right)$$

$$V_2 - (-V_{BE_1}) = V_{BE2}$$

$$V_2 + V_{BE1} = V_{BE2}$$

$$V_2 = V_{BE2} - V_{BE1}$$

$$V_2 = V_T \ln\left(\frac{I_{E2}}{I_S}\right) - V_T \ln\left(\frac{I_{E1}}{I_S}\right)$$

$$V_2 = V_T \ln\left(\frac{I_{E2}}{I_{E1}}\right)$$

generally V_2 is small that we neglected

$V_2 << V_R$

$$I_{C2} = I_2 = V_R / R_2$$

$$I_{E2} \approx I_{C2}$$

$$I_{E2} = \frac{V_R}{R_2}$$

$$V_2 = V_T \ln\left(\frac{V_R/R_2}{V_i/R_1}\right)$$

Date

$$V_2 = V_T \ln\left(\frac{V_R\, R_1}{V_i R_2}\right)$$

$$V_2 = -V_T \ln\left(\frac{V_i\, R_2}{R_1 \cdot V_R}\right)$$

$$V_2 = V_T \ln(V_i)$$

suppose
$V_R = 10V$, $R_1 = 10K$
$R_2 = 100K$

From the last amp

$$V_o = \left(1 + \frac{R_4}{R_3}\right) V_2$$

$$V_o = -\left(1 + \frac{R_4}{R_3}\right) V_T \ln(V_i)$$

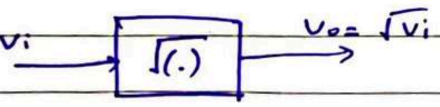
$$\frac{R_4}{R_3} >>> 1$$

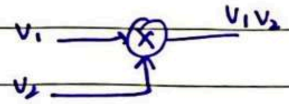
$$V_o = -\left(\frac{R_4}{R_3}\right) V_T \ln(V_i)$$

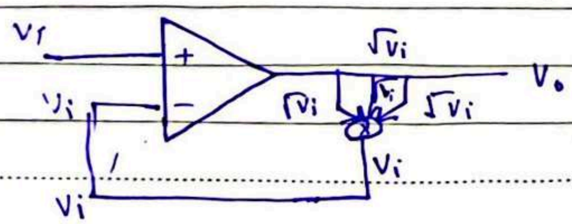
$$V_o = -R_4 \left(\frac{\cancel{V_T}}{\cancel{R_3}}\right) \ln(V_i)$$

$$V_o = -K \ln(V_i)$$

Square Root Extractor :





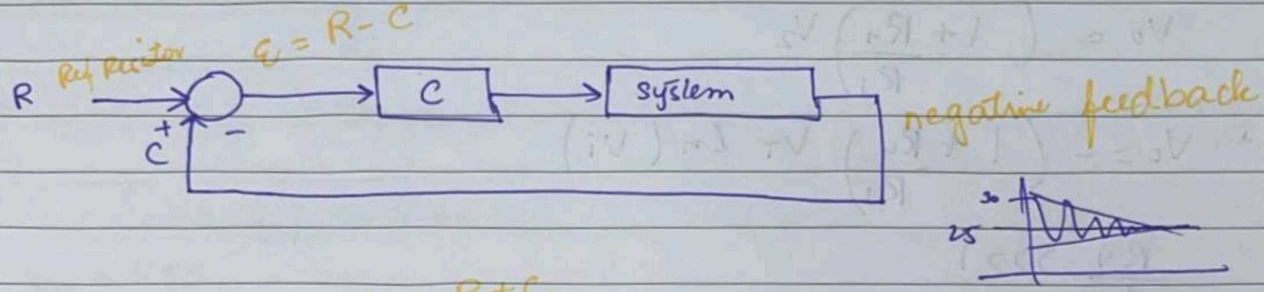


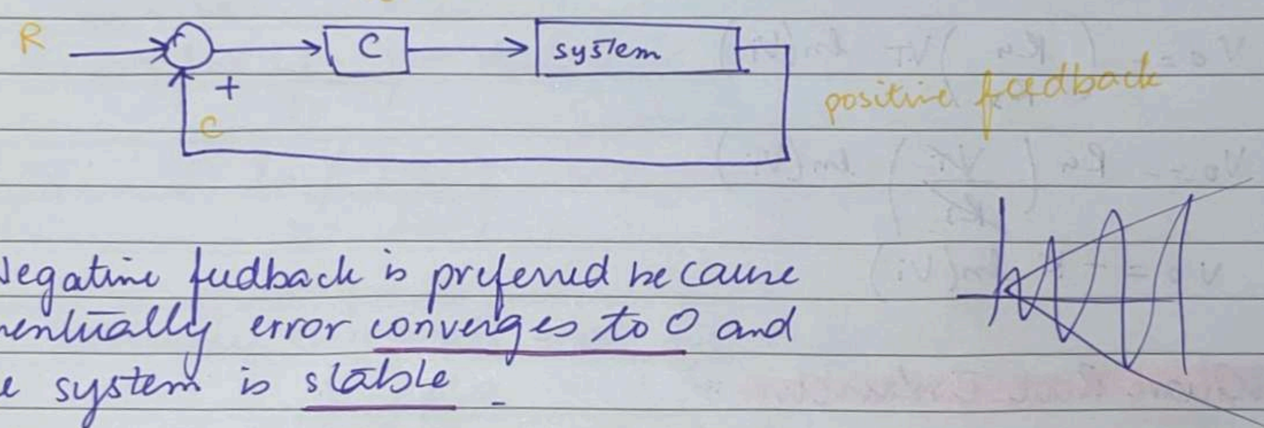
Date

DAY-15

Sine Wave Generator (oscillator):

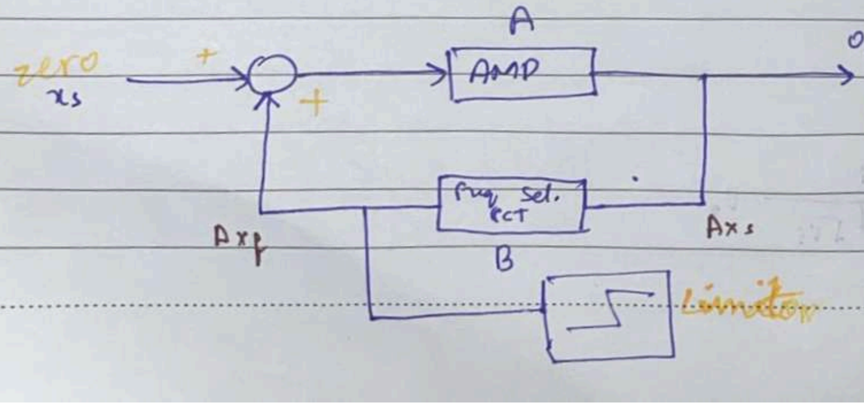
Osci-llators :- Oscillators generate a finite output signal by using an amplifier followed by a frequency selective circuit connected in positive feedback loop with NO input signal.





Negative feedback is prefered because eventually error converges to 0 and the system is stable.

Sinusoidal Oscillator:-



Date

Suppose:

$V_i = x_s$

x_s = reservoir of infinite sinusoid frequencies

$= M_1 \sin(wt) + M_2 \sin(2wt) + M_3 \sin(3wt) - \cdots + M_n \sin(nwt)$

Magnitude of $M_1, M_2, M_3 + \cdots M_n \rightarrow$ is V.V.V small of the order 10^{-30}

I

$x_s + 0 \xrightarrow{\text{sum}} A x_s \xrightarrow{\text{AMP}} A\beta x_F$ freq.

II

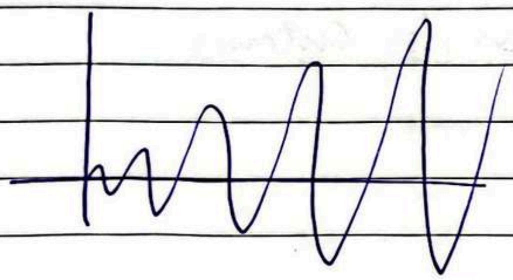
$x_s + A\beta x_F \rightarrow A x_s + A^2\beta^2 x_F \rightarrow A^2\beta^2 x_F$ $A = 10^5$ $\beta = 1$

III

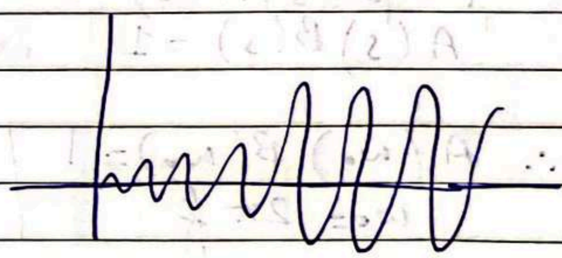
$x_s + A^2\beta^2 x_F \rightarrow A x_s + A^3\beta^3 x_F \rightarrow A^3\beta^3 x_F$

IV

$x_s + A^3\beta^3 x_F \rightarrow A x_s + A^4\beta^4 x_F \rightarrow A^4\beta^4 x_F$

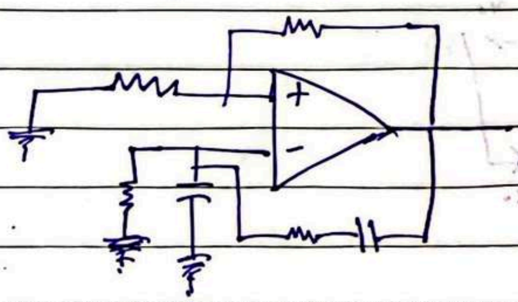


Before limitor



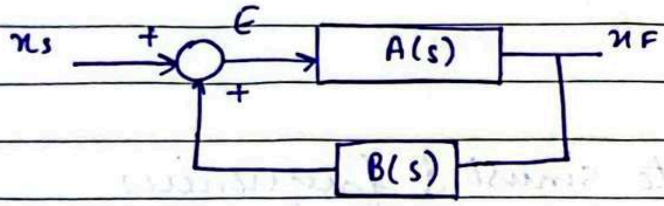
After limitor

We use thermal noise of the resistors



(Example)

Date



$E = x_s + \beta(s) x_F \longrightarrow 1$

$x_F = A(s) E \longrightarrow 2$ (put 1 in 2)

$x_F = A(s)\left(x_s + \beta(s)\, x_F\right)$

$x_F\left(1 - A(s)\beta(s)\, x_F\right) = A(s)\, x_s$

$$\frac{x_F}{x_s} = \frac{A_s}{1 - A(s)\beta(s)}$$

$$\frac{x_F}{x_s} = \frac{x_F}{0} = \infty \quad \text{(Acc. to def.)}$$

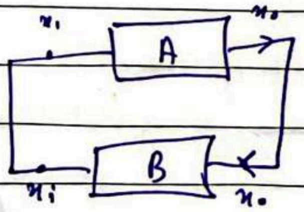
$$\frac{A(s)}{1 - \beta(s)A(s)} = \infty$$

$$1 - \beta(s)A(s) = 0$$

$\boxed{A(s)B(s) = 1}$ Bark Hausan's Criteria

$$\boxed{\begin{gathered} \therefore\ A(\omega_0)\,\beta(\omega_0) = 1 \\ \omega_0 = 2\pi f \end{gathered}} \longrightarrow A$$

If we apply zero input, the output would be a finite signal of frequency f_0 if and only if this condition (A) holds at the frequency f_0



$x_i = \beta x_0 \longrightarrow 1$

Date

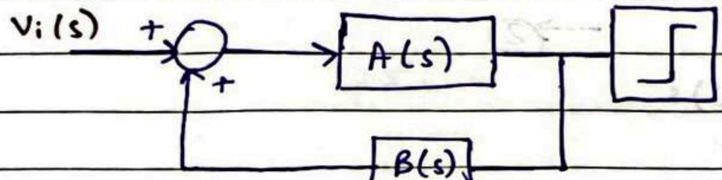
$x_0 = A x_i \rightarrow (2)$

$\cancel{x_0} = A B \cancel{x_0}$

$\boxed{AB = 1}$

DAY-16

Wein-Bridge oscillator

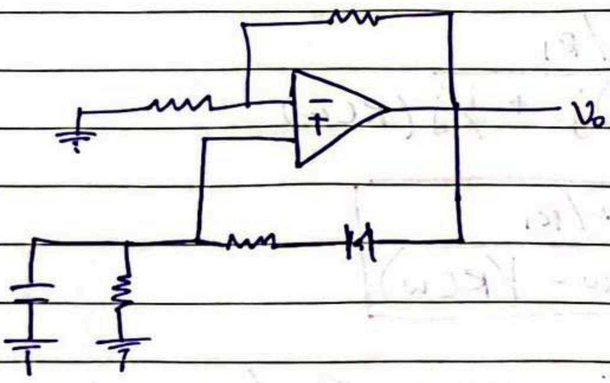


Bark-Housan's Criteria :

To get a finite sinu signal at $f = f_0$

$A(\omega_0)\,\beta(\omega_0) = 1$

$\omega_0 = 2\pi f_0$



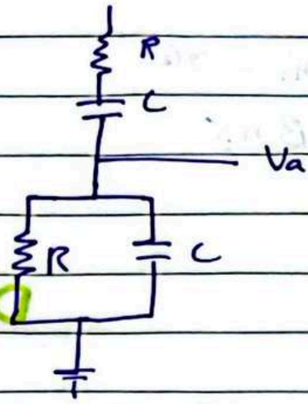
According to the def. of oscillators.

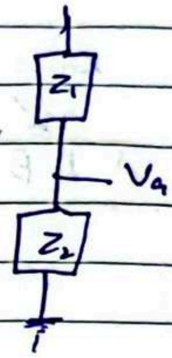
$$V_0(s) = \left(1 + \frac{R_2}{R_1}\right) V_a(s)$$

$$\frac{V_0(s)}{V_a(s)} = 1 + \frac{R_2}{R_1}$$

Scanned with CamScanner

$\therefore \; A(s) = 1 + \dfrac{R_2}{R_1} \longrightarrow ①$





Using VDR

$\beta(s) = \dfrac{Z_2}{Z_1 + Z_2}$

$= \dfrac{R \parallel 1/sC}{(R + 1/sC) + (R \parallel 1/sC)}$

$B(s) = \dfrac{1}{3 + (RC)s + 1/(RC)s} \longrightarrow ②$

According to Bark Hausen's criteria

$A(s) \cdot B(s) = 1$

$A(s)\,B(s) = \dfrac{1 + R_2/R_1}{3 + (RC)s + 1/RC(s)}$

put $s = j\omega$

$A(\omega)\,\beta(\omega) = \dfrac{1 + R_2/R_1}{3 + (RC\omega)j + 1/j(RC\omega)}$

$$\boxed{A(\omega)\,B(\omega) = \dfrac{1 + R_2/R_1}{3 + j\,(RC\omega - 1/RC\omega)}}$$

To get a real finite sine wave of freq. f_0,

$A(\omega_0)\,B(\omega_0) = 1$ at $f = f_0$

$A(\omega_0)\,B(\omega_0) = \dfrac{1 + R_2/R_1}{3 + \underbrace{j\,(RC\omega_0 - 1/RC\omega_0)}_{\text{zero}}}$

Date

If we make the imag. part = 0, then this whole thing becomes REAL

To make the system real:

$$RC\omega_0 - \frac{1}{RC\omega_0} = 0$$

$$RC\omega_0 = \frac{1}{RC\omega_0}$$

$$\omega_0^2 = \frac{1}{(RC)^2}$$

$$\omega_0 = \frac{1}{RC}$$ (Filter Design)

$$A(\omega_0)\,\beta(\omega_0) = 1$$

$$\frac{1 + R_2/R_1}{3 + j\left(RC\omega_0 - 1/RC\omega_0\right)} = 1$$

$$\frac{1 + R_2/R_1}{3} = 1$$

$$1 + \frac{R_2}{R_1} = 3$$

$$\frac{R_2}{R_1} = 2$$

$$\boxed{\frac{R_2}{R_1} \geq 2}$$

Filter:

$$\omega_0 = \frac{1}{RC}$$ → we need to find C.

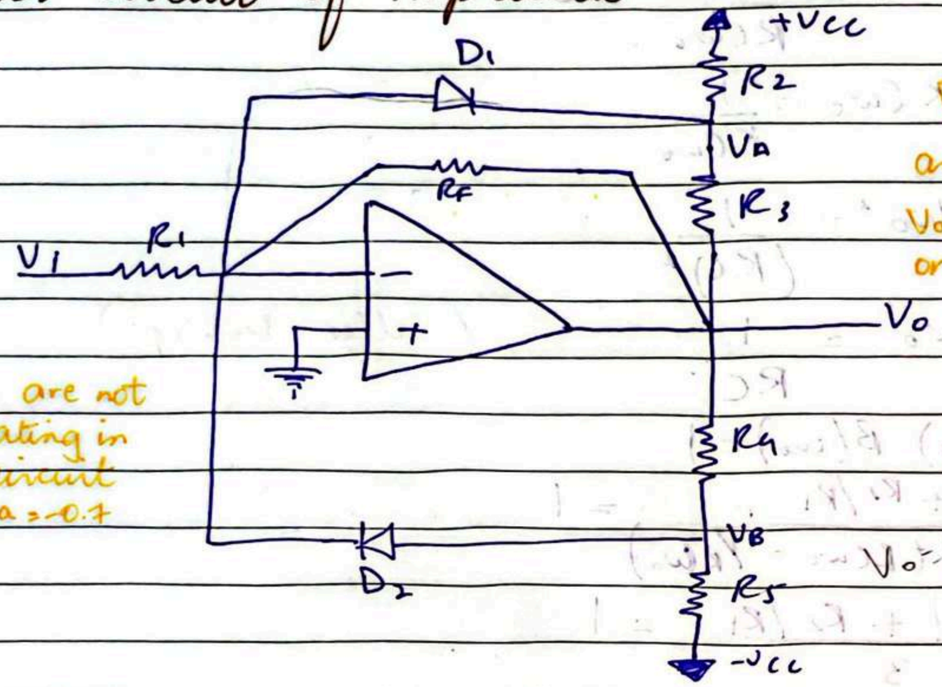
given   assume

Date

Amp. $\longrightarrow$ put R_2 just a little more that twice

$$\frac{R_2}{R_1} \geq 2$$

Limiter Circuit of Amplitude control :



D₁ depends on V_A and V_A depends on V_o and V_o depends on V_i.

diodes are not participating in this circuit unless $V_A = -0.7$

- V_i is very small (close to zero) but +ve
- $V_o = -\left(\frac{R_F}{R_i}\right) V_i$
- V_o ~~is~~ v. small but −ve ≈ -0.01
- V_i will gradually increase (+ve)
- V_o is will be more & more (−ve)
- V_A will continue to decrease in magnitude until $V_A = -0.7V$. At this point D_1 turn on.
- When diode turns on and becomes short circuited, it brings $R_3 \parallel R_F$

Scanned with CamScanner

Date

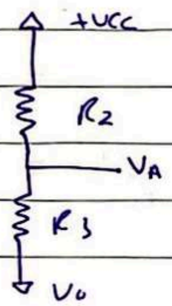
$$V_0 = -\left(\frac{R_3 \| R_F}{R_1}\right) V_i$$

$$\frac{R_F}{R_i} >> \frac{R_3 \| R_F}{R_i}$$

$$G_1 >> G_2$$

For negative half cycle
V_i is becoming more & -ve
V_o is becoming more +ve
V_B is becoming less -ve

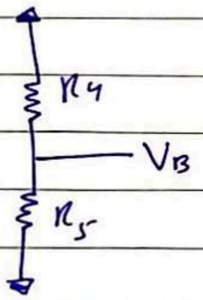
$$V_0 = -\left(\frac{R_F \| R_4}{R_i}\right) V_i$$



$$V_A = \left(\frac{R_3}{R_3+R_2}\right) V_{cc} + \left(\frac{R_2}{R_3+R_2}\right) V_o$$

$V_A = -V_D$ $\qquad$ $V_o = L_-$

$$\boxed{L_- = -V_{cc}\left(\frac{R_3}{R_2}\right) - V_D\left(1+\frac{R_3}{R_2}\right)}$$



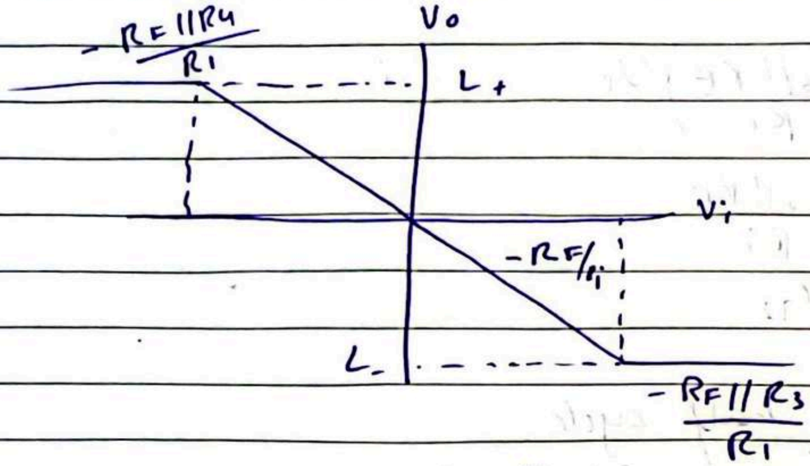
$$V_B = \left(\frac{R_4}{R_4+R_5}\right)(-V_{cc}) + \left(\frac{R_5}{R_4+R_5}\right) V_o$$

$V_B = +V_D$ $\qquad$ $V_o = L_+$

$$\boxed{L_+ = +V_{cc}\left(\frac{R_4}{R_5}\right) + V_D\left(1+\frac{R_4}{R_5}\right)}$$

Scanned with CamScanner

Date



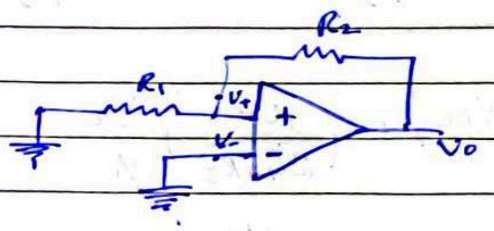
DAY-17

Multi-vibrator cct's for signal generation :-

1 BISTABLE MVB (2 stable states)
2 MONO STABLE MVB (1 stable state)
3 A-stable MVB (No stable state)

BISTABLE MVB

Bistability : DC Amplifier in +ve feedback loop.
Open loop gain : $A\beta > 1$
Feedback contain a voltage divider of R_1 & R_2



$V_- \rightarrow 0V$
No ext. input voltage
electrical noise in cct introduces a small +ve voltage at V_+

Date

Hence,

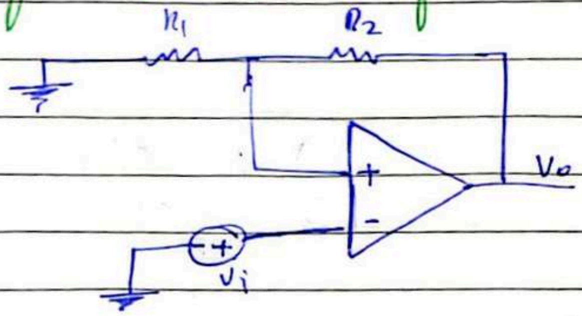
$$V_o' = A(\overset{+ve}{V_+} - \overset{0}{V_-})$$

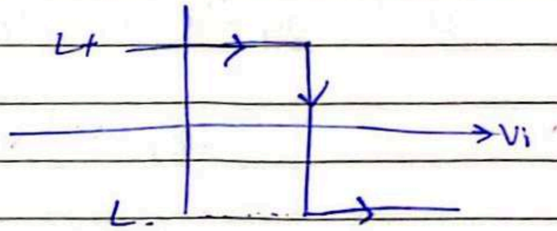
$$\Rightarrow V_o' > 0$$

Output is fed back at $V_+ = \left(\frac{R_1}{R_1+R_2}\right) V_o'$

$$V_o = \beta V_o'$$

Transfer Characteristics of Bistable MVB :-

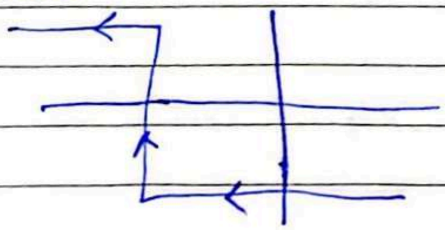




Suppose $V_o = L_+$; $V_+ = \left(\frac{R_1}{R_1+R_2}\right) L_+ = \beta L_+$

- $V_+ = \beta L_+$
- $V_- = V_i$
- Amp ; $V_o = A(\overset{\beta L_+}{V_+} - \overset{0}{V_-})$
- initially , $V_- = 0V$
- We start increasing V_i gradually

Now



Date

Suppose $v_o = L_-$, $V_+ = \left(\frac{R_1}{R_1 + R_2} \right) L_- = \beta L_-$

$V_+ = \beta L_- = -5V$

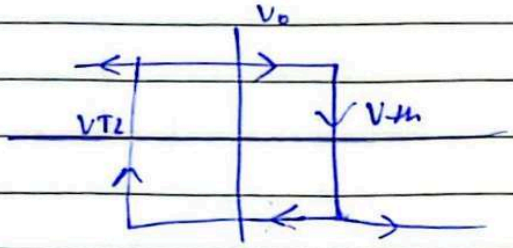
$V_- = V_i = 0$

Amp = $V_o = A(V_+ - V_-)$

initially $V_- = 0$

We start decreasing V_i gradually

Combining the two graphs



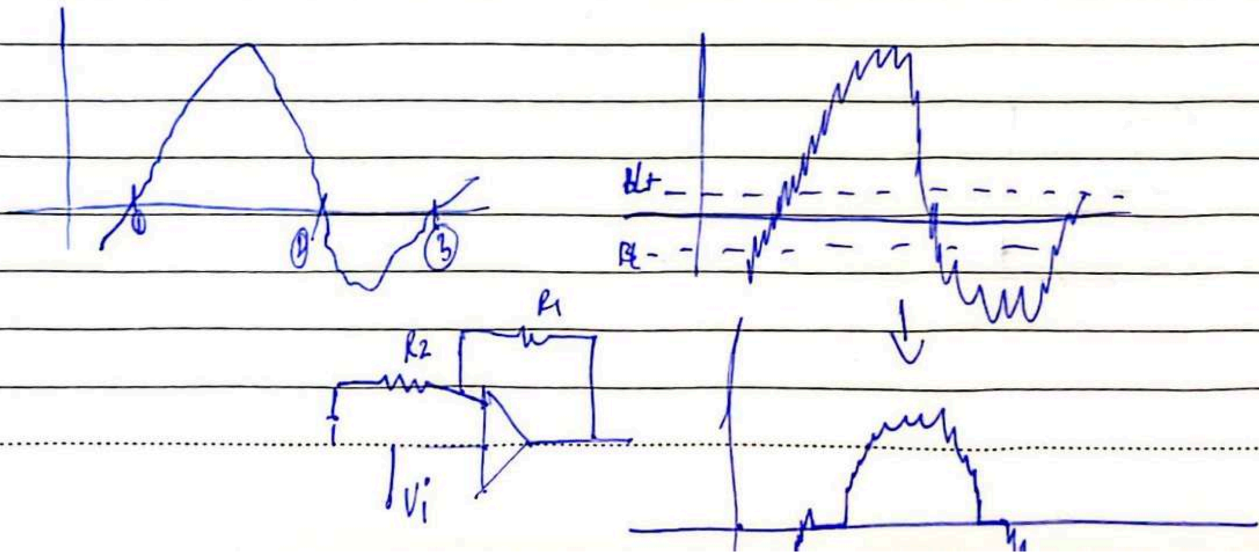
$V_{TH} = \beta L_+$

$V_{TL} = \beta L_-$

Hysteresis Comparator or Inverting Bistable MVB

If trigger (V_i) $> \beta L_+$ } V_o goes from L_+ to L_-

If trigger (V_i) $< \beta L_-$ } v_o goes from V_- to L_+

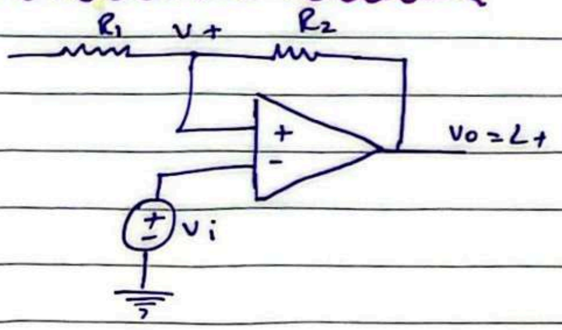


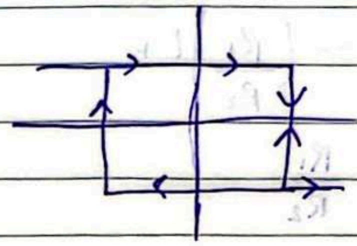
Date

When you have
Hysteresis comparator helps to eliminate the ripple near zero crossing of signal waveform

DAY- 18

Inverting Bistable MVB :



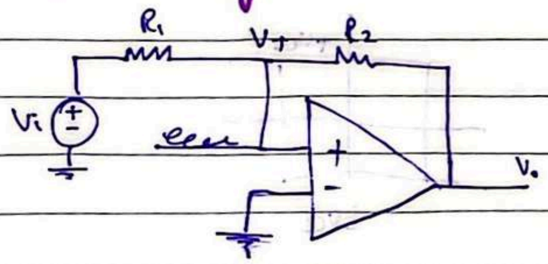


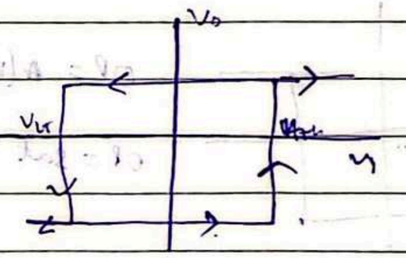
$V_{TH} = \beta L_+$

$V_{T2} = \beta L_-$

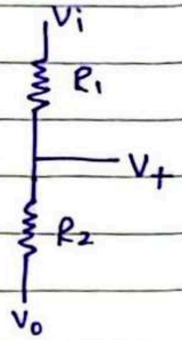
$\beta = \dfrac{R_1}{R_1 + R_2}$

Non-inverting Bistable MVB :





Date



$$V_+ = \left(\frac{R_1}{R_1+R_2}\right) V_o$$

$$V_+ = \left(\frac{R_2}{R_1+R_2}\right) V_i$$

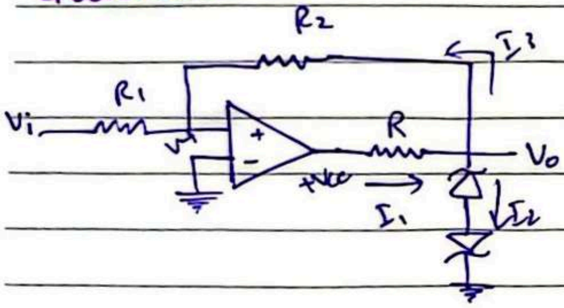
$$V_+ = \left(\frac{R_1}{R_1+R_2}\right) V_o + \left(\frac{R_2}{R_1+R_2}\right) V_i$$

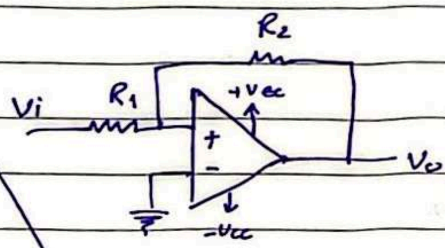
$$V_{th} = -\left(\frac{R_1}{R_2}\right) -$$

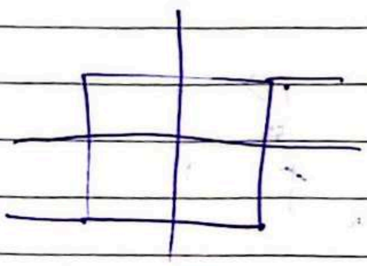
$$V_{T2} = -\left(\frac{R_1}{R_2}\right) L_+$$

$$\beta = \frac{R_1}{R_2}$$

Question :-

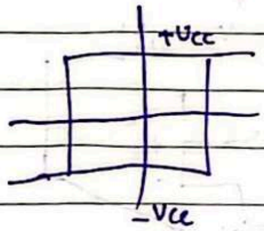






$OP = A(V_+ - V_-)$

$OP =$ sat at $+V_{cc}$



Date

P1 - VAD

$V_Z = 6.8V$

$V_D = 0.7V$

$V_o = V_Z + V_D$

$V_o = 7.5V$

If $V_+ < 0$,

$V_o = -7.5V$

$I_1 = \frac{V_{cc} - 7.5}{7.5k} = 10mA$

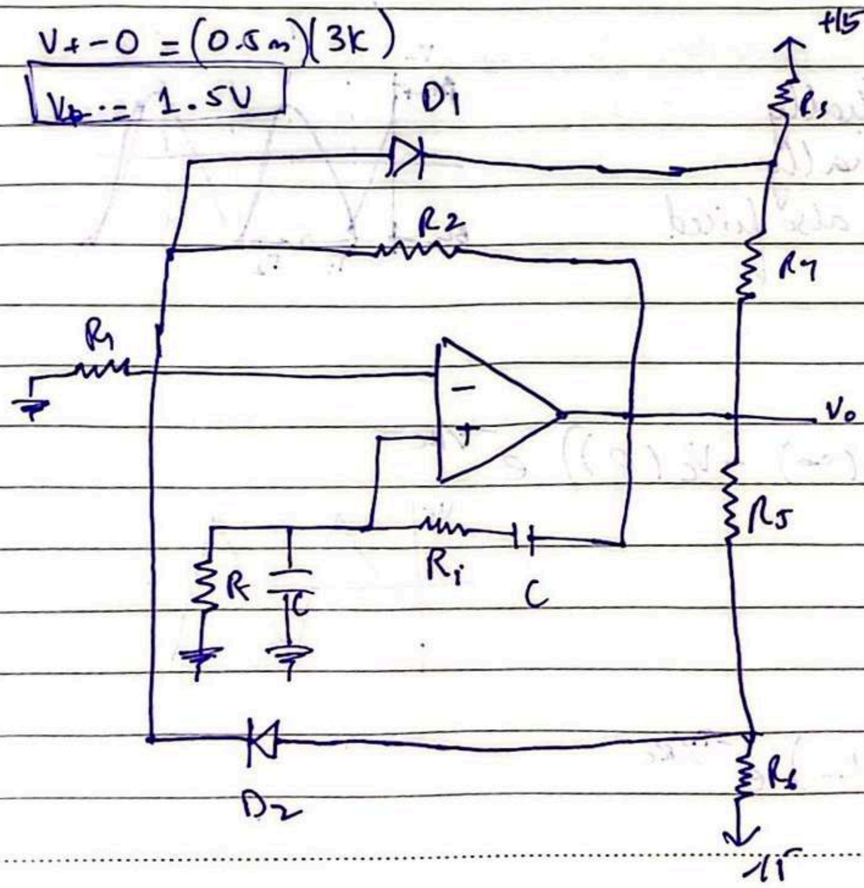
$I_3 = \frac{7.5 - 0}{12 + 3k} = 0.5mA$

$I_2 = I_3 - I_1$

$= 0.5mA$

$V_+ - 0 = (0.5m)(3k)$

$\boxed{V_+ = 1.5V}$

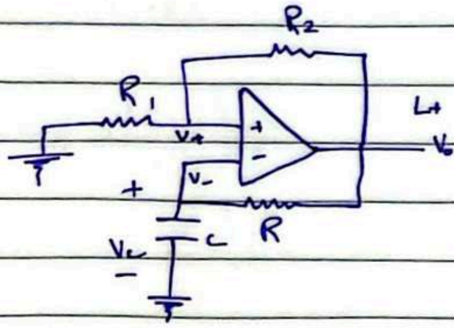


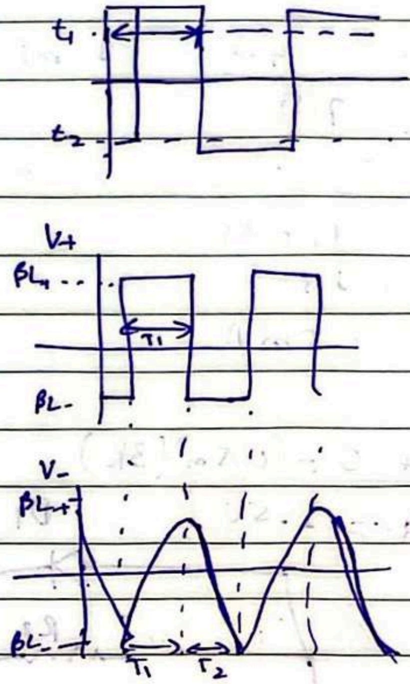
Date

DAY-19

ASTABLE MVB (using inv. bistable config.) :-

- No stable state
- keeps toggling b/w L_+ & L_-
- Used to generate square waves & rectangular waves.
- Uses BISTABLE MVB cct. with an RC cct. in -ve feedback.





$V_o = A(V_+ - V_-)$

$V_- = V_c$

initially $V_o = L_+$, $V_+ = \beta L_+$

V_c will start rising exponentially

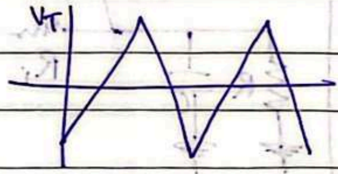
V_- will start rising exponentially

Because $V_o = L_+$ is fixed, V_+ is also fixed

Time Period :-

T_1 =

$$V_c(t) = V_c(\infty) - (V_c(\infty) - V_c(0))\, e^{-t/RC}$$



$V_c(0) = \beta L_-$

$V_c(\infty) = L_+$

$t = T_1$

at $t = T_1$, $V_c(t) = \beta L_+$

$$\beta L_+ = L_+ - (L_+ - \beta L_-)\, e^{-T_1/RC}$$

Date

$$T_1 = RC \ln\left(\frac{1 - \beta\left(L_-/L_+\right)}{1-\beta}\right)$$

For sym. power supplies ; $L_+ = -L_-$

$$\boxed{T_2 = RC \ln\left(\frac{1+\beta}{1-\beta}\right)}$$

Now for T_2:

T_2

$V_C(0) = \beta L_+$

$V_C(\infty) = L_-$

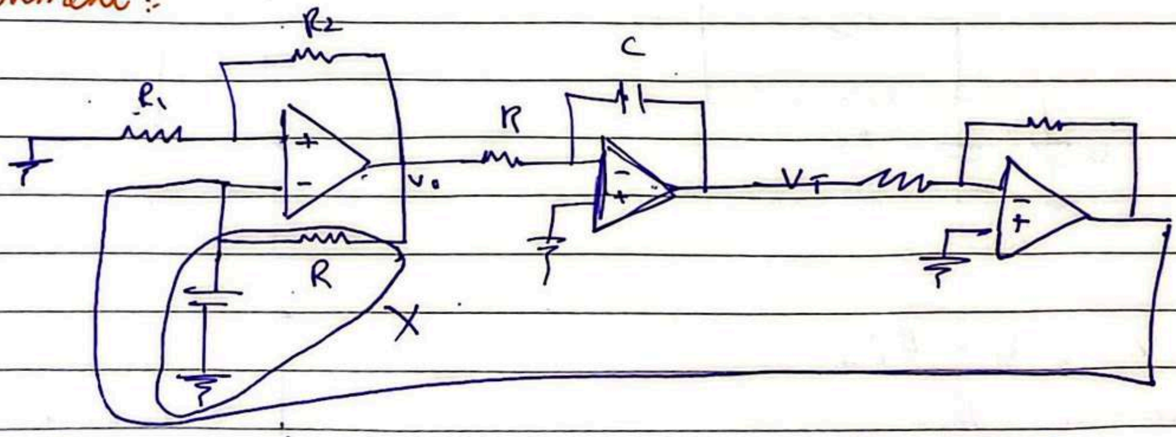
$t = T_2$

at $t = T_2$, $V_C(t) = \beta L_-$ OS-VAO

$$\boxed{T_2 = RC \ln\left(\frac{1+\beta}{1-\beta}\right)}$$

having $T_1 = T_2$ because of having sym. power supplies, we will have perfect square wave.

Improvement :-



$$V_T = \frac{-1}{RC}\int V_0\, dt$$

$\beta = \frac{R_1}{R_1+R_2}$

$T = T_1 + T_2$

Date

$$= -\frac{1}{RC} \int 10\, dt$$

$RC = 1$

$$T_1 = RC \left(\frac{V_{TH} - V_{TL}}{L_+} \right)$$

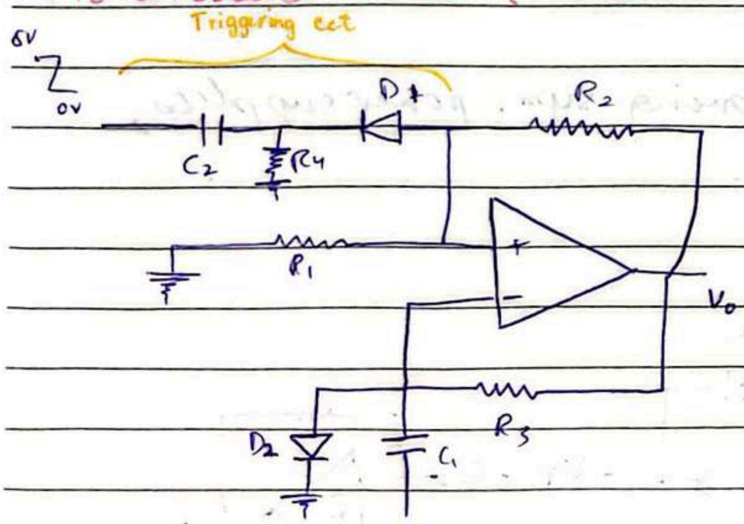
$$T_2 = RC \left(\frac{V_{TH} - V_{TL}}{-L_-} \right)$$

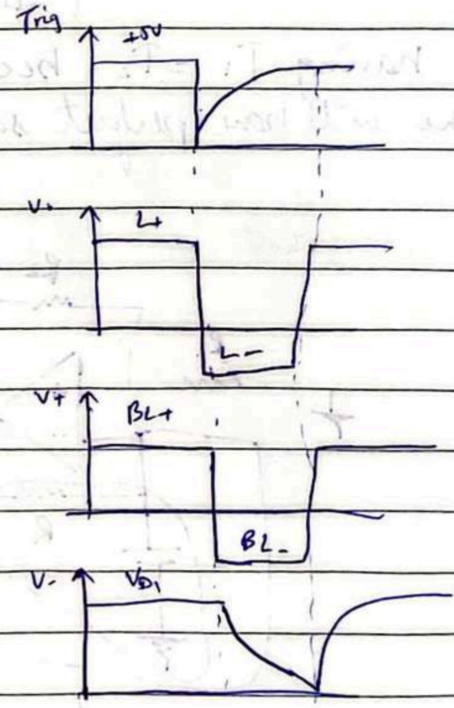
$V_{TH} = \beta L_+$

$V_{TL} = \beta L_-$

DAV-20

Monostable MVB (Pulse Generator):





- $V_o = L_+$, $V_+ = \beta L_+$
- Because $V_K > \beta L_+$
- D_1 is reversed biased.
- D_1 is open

Date

- For I_n flow, D_2 is forward biased
- D_2 is short cct.

Low pass filter acts as an integrator
High pass filter acts as a differentiator

- $V_o = A(V_+ - V_-)$
  $= A(\beta L_+ - 0.7)$

- When I apply a trigger ; $V_K = 0V$
- D_1 becomes forward biased (short cct)
- D_1 is short cct, 0V at V_K will travel ahead & flow itself at V_+.
- $V_o = A(V_+ - V_-)$
  $= A(0 - 0.7)$
- In this ~~dist~~ duration, V_K becomes +ve again. D_1 is open again.

$V_o = A(V_+ - V_-)$
$= A(\beta L_- - 0.7)$
$-12 - (-12.001)$

Mathematical Derivation :-

$V_c(t) = V_c(\infty) - (V_c(\infty) - V_c(0))e^{-t/R_3C_1}$

$V_c(0) = V_{D_1}$

$V_c(\infty) = L_-$

at $t = T$; $V_c(T) = \beta L_-$

$\beta L_- = L_- - (L_- - V_{D_1})e^{-T/R_3C_1}$

Scanned with CamScanner

Date

solving

$$T = R_3 C_1 \ln\left(\frac{V_{D_1} - L_-}{\beta L_- - L_-}\right)$$

Scanned with CamScanner

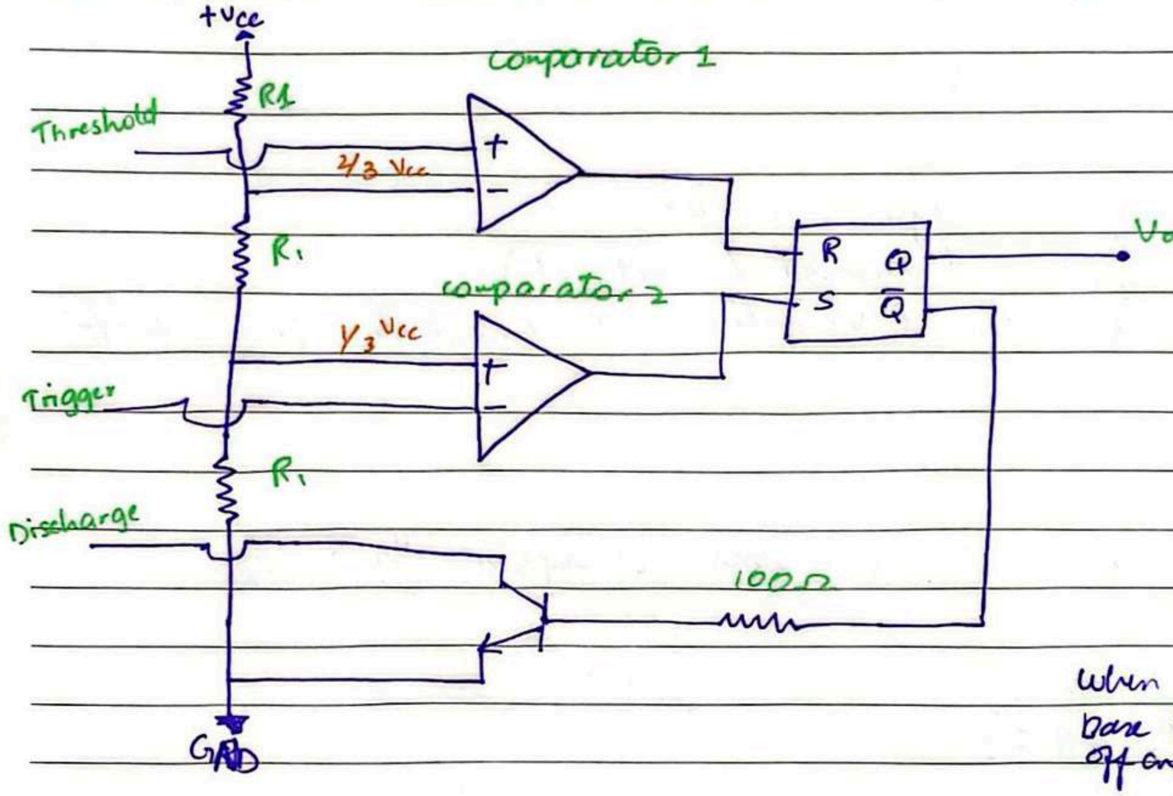
Date

solving

$$T = R_3 C_1 \ln\left(\frac{V_{D_1} - L_-}{\beta L_- - L_-}\right)$$

DAY - 21 (After Mid 2)

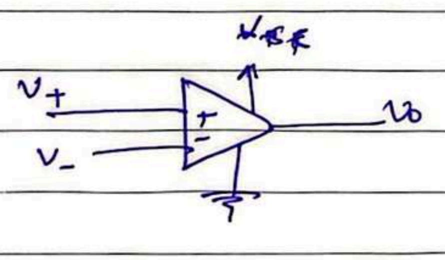
555 Timer Circuit:



R	S	Q	$\overline{Q}$
0	0	Prev. State	
0	1	1	0
1	0	0	1
1	1	undefined	

when we apply 0 at base NPN transistor turns off and become open circuit

General concept:-

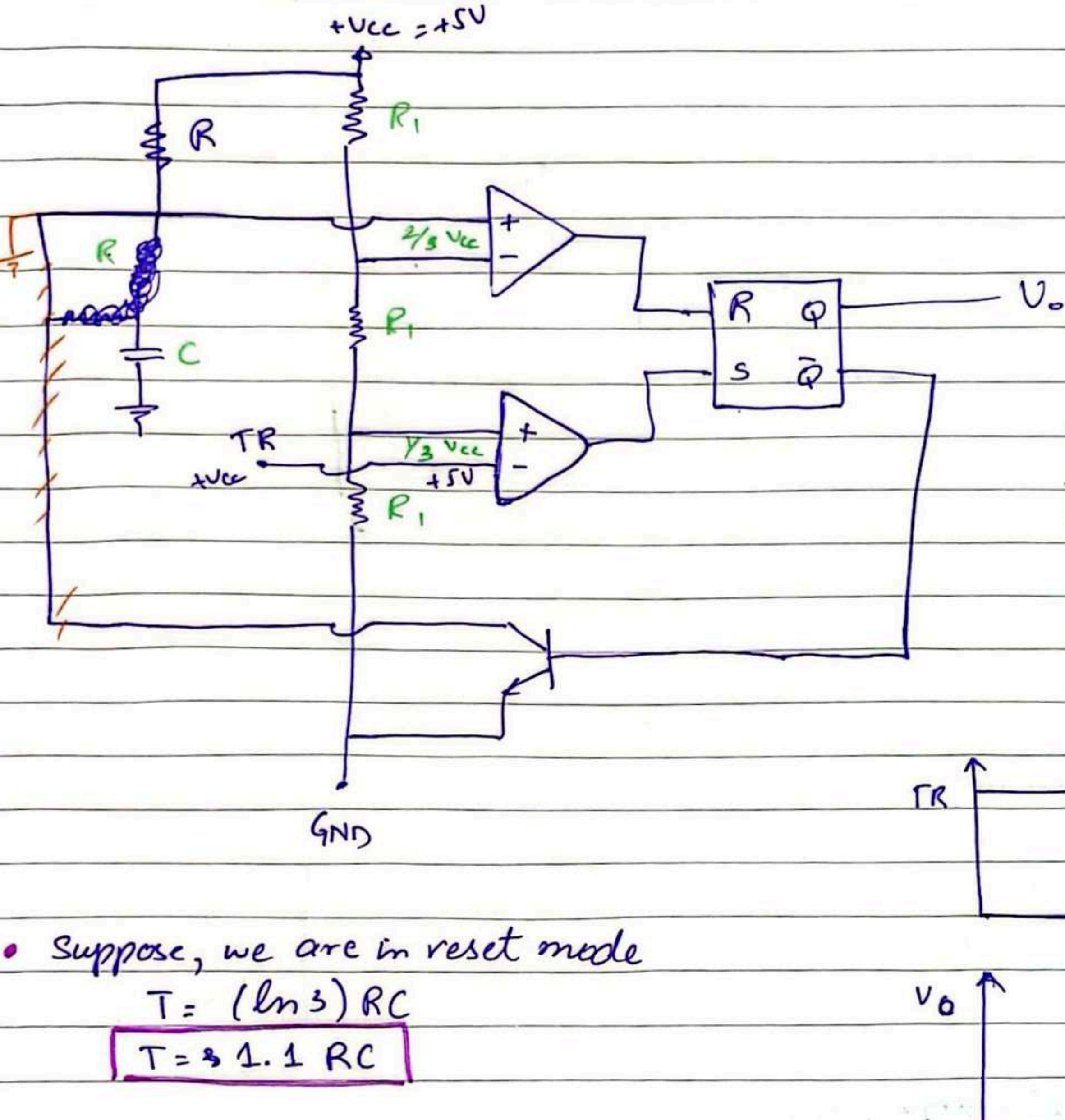


Date

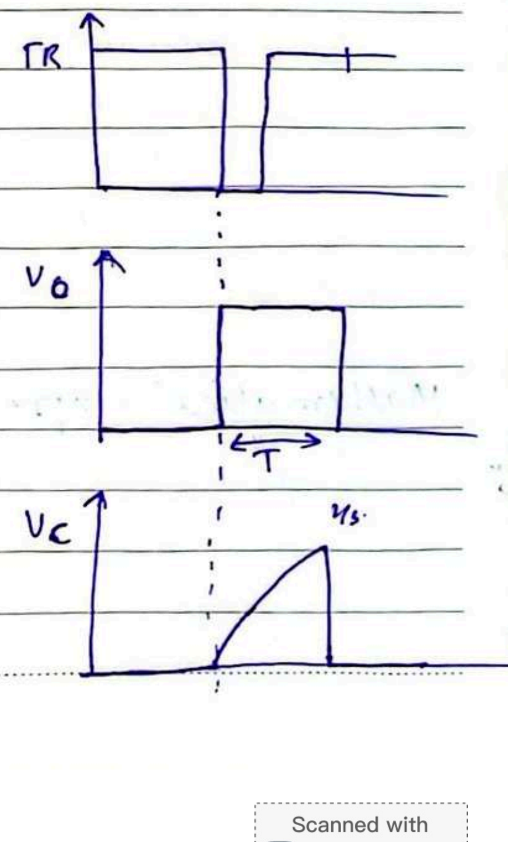
$V_+ > V_-$ $V_o = +V_{cc}$ '1'

$V_+ < V_-$ $V_o =$ GND or $-V_{cc}$ '0'

555 Timer Circuit as a monostable MVB:



R	S	Q
0	0	prev.
0	1	1
1	0	0
1	1	∞



- Suppose, we are in reset mode

$T = (\ln 3) RC$

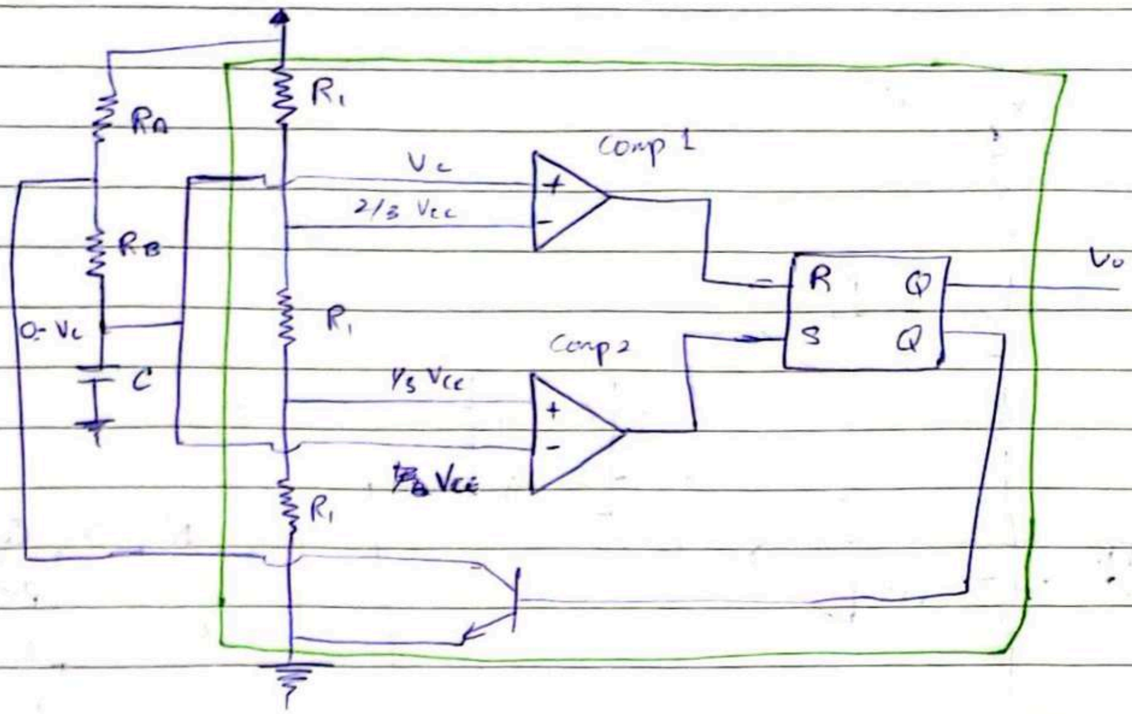
$\boxed{T \approx 1.1 RC}$

∵ At 1, transistor turns on (short ~~open~~ circuit)

Date

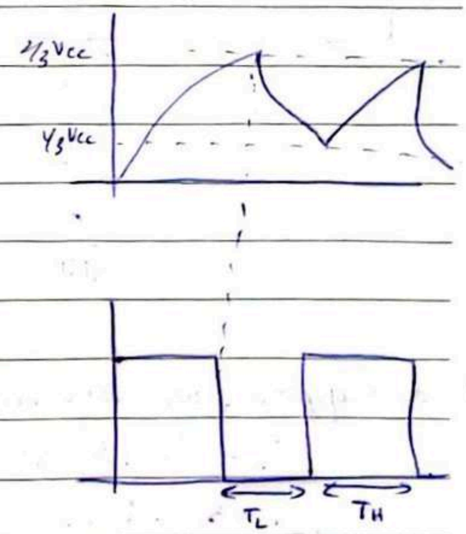
DAY-22

ASTABLE MVB using 555 Timer :-



R	S	Q
0	0	Prev
0	1	1
1	0	0
1	1	∞

Initially $V_c = 0$



When capacitor is charging, it is through R_A & R_B. So $\tau = (R_A + R_B)C$

When capacitor is discharging, it is through R_B only so $\tau = (R_B)C$

So T_H is larger than T_L

Mathematical Expression :-

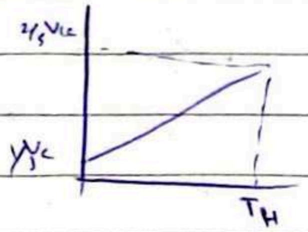
$$V_c(t) = V(\infty) - (V(\infty) - V(0))e^{-t/RC}$$

$$V(\infty) = V_{cc}$$

$$V(0) = \frac{1}{3}V_{cc}$$

Date

at $t = T_H$, $V_C = \frac{2}{3} V_{CC}$



$$\frac{2}{3} V_{CC} = V_{CC} - \left(V_{CC} - \frac{1}{3} V_{CC}\right) e^{-\frac{T_H}{(R_A + R_B)C}}$$

$$\boxed{T_H = \ln 2 \, (R_A + R_B) C}$$ → imp Formula

For discharging

$$V_C(t) = V(\infty) - \left(V(\infty) - V(0)\right) e^{-t/RC}$$

$$V(\infty) = 0$$

$$V(0) = \frac{2}{3} V_{CC}$$

at $t = T_L$, $V_C = \frac{1}{3} V_{CC}$

$$\frac{1}{3} V_{CC} = 0 - \left(0 - \frac{2}{3} V_{CC}\right) e^{-T_L/R_B C}$$

$$\boxed{T_L = \ln 2 \, (R_B C)}$$ → imp Formula

$$T = T_H + T_L$$

$$\boxed{T = (\ln 2)(R_A + 2R_B) C}$$

Duty cycle :-

"The percentage of the total time period for which the signal is logic high or logic 1."

$$\boxed{D = \frac{T_H}{T_H + T_L} \times 100}$$

$$\boxed{D = \frac{R_A + R_B}{R_A + 2R_B}}$$

} IMP. FORMULAS

Scanned with CamScanner

Date

One of the limitations of this circuit is that T_H is never equal to T_L in current setting of the circuit and therefore you can never achieve a 50% duty cycle and you can never achieve a square wave.

For 50% Duty cycle :-

$T_H = (\ln 2)(R_A)C$

$T_L = (\ln 2)(R_B)C$

$T = T_H + T_L$

$= (\ln 2)(R_A + R_B)C$

$D = \dfrac{T_H}{T_H + T_L}$

$= \dfrac{R_A}{R_A + R_B}$

$R_A = R_B$

DAY-23

555 Timer Circuit :-

For duty cycle >50%

$T_1 = (\ln 2)(R_A + R_B)C$

$T_2 = (\ln 2)(R_B)C$

$T = T_1 + T_2$

$= (\ln 2)(R_A + 2R_B)C$

$f = \dfrac{1}{T}$

$D = \dfrac{R_A + R_B}{R_A + 2R_B}$

Date

For duty cycle ≤ 50%

$T_1 = (\ln 2)(R_A)C$

$T_2 = (\ln 2)(R_B)C$

$T = (\ln 2)(R_A + R_B)C$

$f = \dfrac{1}{T}$

$D = \dfrac{R_A}{R_A + R_B}$

Question :-

Use a 1nF capacitor in a 555 timer circuit, find the value of R to generate a pulse of 10μS.

Sol

$T = (\ln 3)RC$

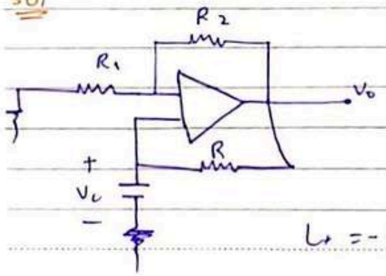
$10\mu s = (\ln 3) \times R \times 10^{-9}$

$R =$

Question :-

Design and generate 2 circuit diagram that generate a 50% (square cycle) MVB that can generate perfect square wave signal.

Sol



and

second circuit is a Astable MVB using 555 cct.

$T = RC \ln\left(\dfrac{1+\beta}{1-\beta}\right)$

$L_+ = -L_-$

Date

NOTE:

→ If we are talking about generating pulse, we are gonna draw (MVB monostable diagram)

→ If we talking about generating a signal or wane we gonna draw Astable MVB

Question ::

Use a 680pF capacitor, design a circuit with 555 timer that generate a rect. signal of 75% duty cycle & 50 KHz frequency.

Sol

$$\frac{R_A + R_B}{R_A + 2R_B} = 0.75 = \frac{3}{4}$$

$$4R_A + 4R_B + 3R_A + 6R_B$$

$$R_A = 2R_B$$

$$f = 50K = \frac{1}{f} = 20\mu s = T$$

$$T = (\ln 2)(R_A + 2R_B)C$$

$$20\mu s = (\ln 2)(R_A + 2R_B)(680p)$$

$$R_A + 2R_B = 42.432K$$

$$R_B = 10.5K\Omega$$

$$R_A = 21K\Omega$$

Scanned with CamScanner

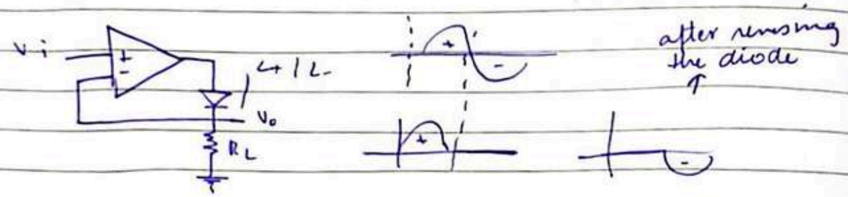
Date

DAY-24

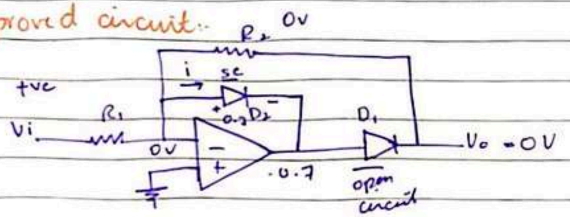
Cons :

This circuit does not provide any for signal amplification.

Inflexible - it doesnot offer the feature as to which half cycle need to be passed.

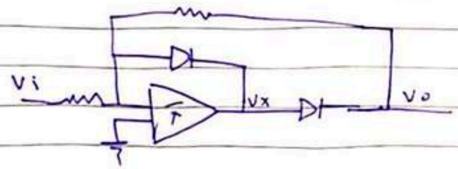


Improved circuit :



For +ve half cycle

- D_2 is forward biased → short circuit
- 'i' passes through D_2
- D_1 will be reversed biased → open circuit



Date

For -ve half cycle

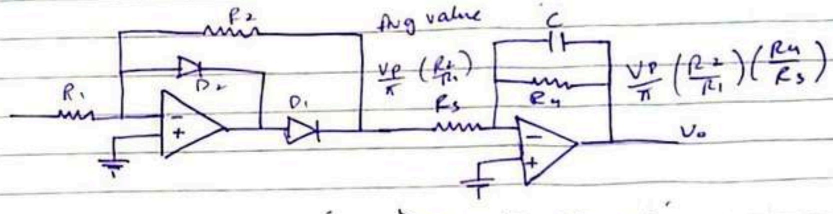
D_2 is reversed biased → OC

D_1 is forward biased → SC

It becomes an inverting OP-amp

$$V_o = -\left(\frac{R_2}{R_1}\right) V_i$$

AC Voltmeter : (V.V.V. IMP).



Precision Rectifier

DC gain = R_2/R_1

Low Pass Filter

DC gain = R_4/R_3

$$\text{Avg value} = \frac{V_p}{\pi}$$

Filter design

cutoff freq

$$\omega = \frac{1}{CR_4}$$

$$f = \frac{1}{2\pi CR_4}$$

$$V_{o,avg} = \frac{V_p}{\pi}\left(\frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3}\right)$$

Date

$$W_c = \frac{1}{C R_4}$$

Question:

Design an AC voltmeter for a 10Hz input signal. It is calibrated for sine wave signal to provide +10V o/p for i/p of 1.0 V_{rms}. The R_{in} must as high as possible. Keep the gain of AC rectifier part to unity. The largest resistor available is 1.0 MΩ.

sol.

$$\frac{R_2}{R_1} = 1.0$$

Set R_1 & R_2 = 1MΩ

convert RMS to peak

$$V_{RMS} = \frac{V_P}{\sqrt{2}}$$

putting in formula

$$V_0 = \frac{V_P}{\pi}\left(\frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3}\right)$$

let R_4 = 1M Ω

R_3 = 45 KΩ

$$w = \frac{1}{C R_4}$$

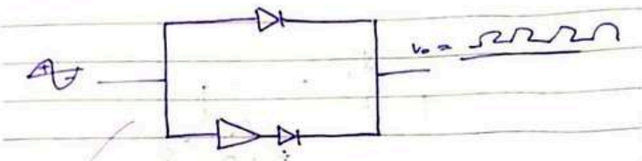
$$f = \frac{1}{2\pi C R_4}$$

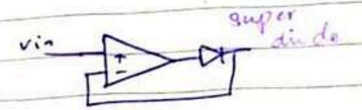
C = 15.9nF

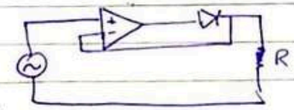
Date

DAY-25

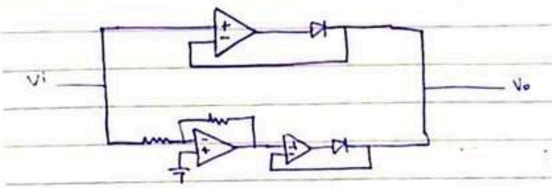
Precision Full Wave Rectifier :

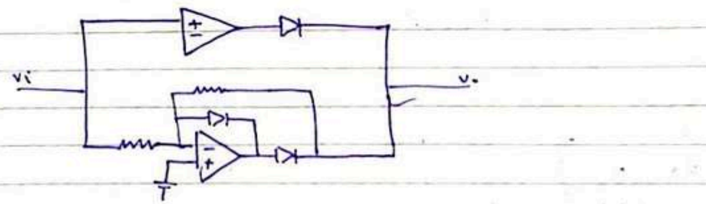




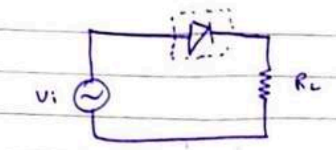


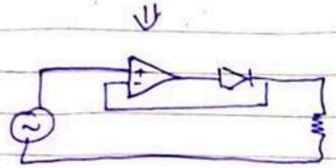
For precision we replace diode with super diode.

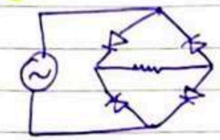


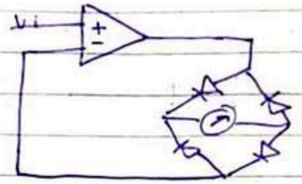


Date



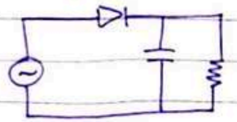


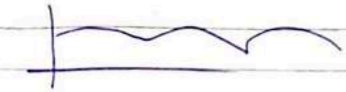




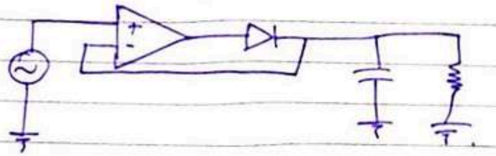
We can also use 0.1 Ω resist

Peak Rectifier:

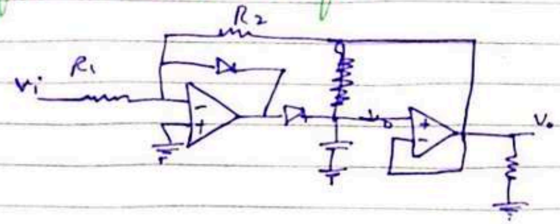




Precision Peak Rectifier:

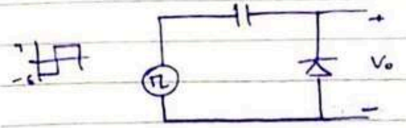


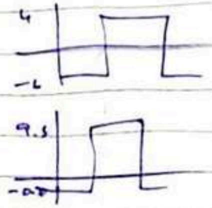
Buffered Precision Rectifier

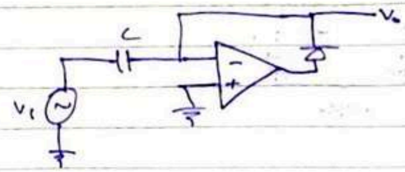


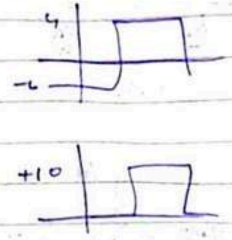
Date

CLAMPER:-





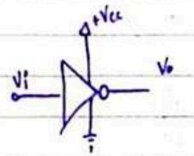




DAY-26

Digital logic Inverter:

Function of inverter:-

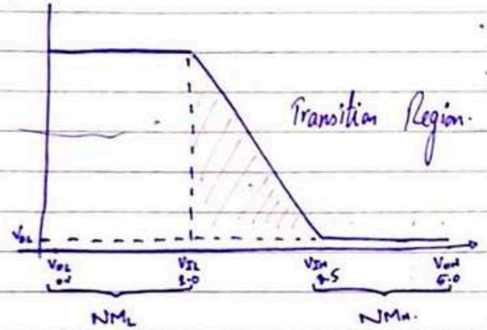


[0.1 - 2.0V] → '0'
[3.5 - 5.0V] → '1'
[2.0 - 3.5V] →

'1' → $V_{OH} = 5.0V$.
'0' → $V_{OL} = 0.1V$.
$V = 2.2V$ (Undefined)

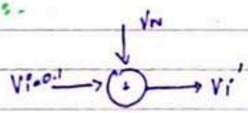
Date

Voltage transfer ratio:

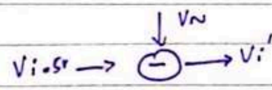


Noise Margin :-

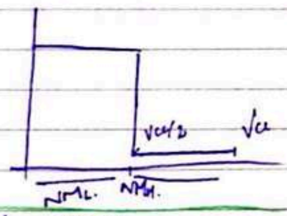
For NM_L.



For NM_H.



ideal noise margin graph can be like



$NM_H = NM_L = \frac{V_{CC}}{2}$

$V_{IL} = V_{IH} = \frac{V_{CC}}{2}$

when $V_{OL} < V_i < V_{IL}$; $V_o = V_{OH}$.

when $V_{IH} < V_i < V_{OH}$; $V_o = V_{OL}$.

V_{IL} -> Max. V_i of an inverter that can be interpreted as logic low (0).

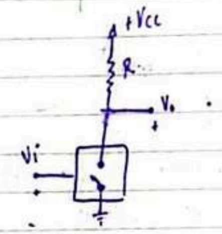
Date

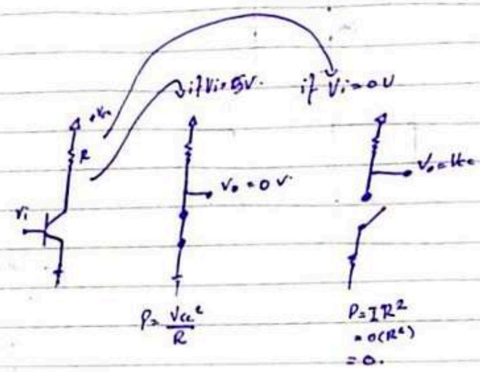
V_{IH} => Min V_i of an inverter that can be interporated as logic high (1).

$NM_L = V_{IL} - V_{OL}$

$NM_H = V_{OH} - V_{IH}$

Inverter Implementation:





Using Complementary switches.

n-type :

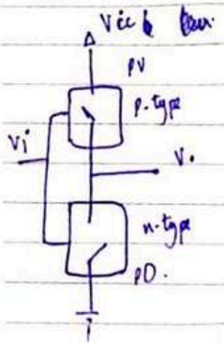
logic 1 -> SW close.

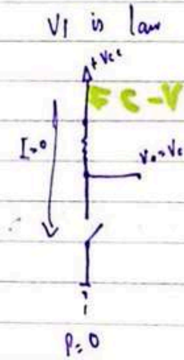
logic 0 -> SW open.

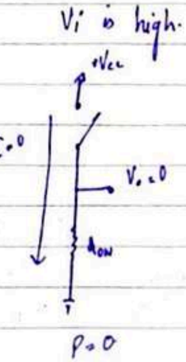
p-type:

logic 1 -> SW open.

logic 0 -> SW close.

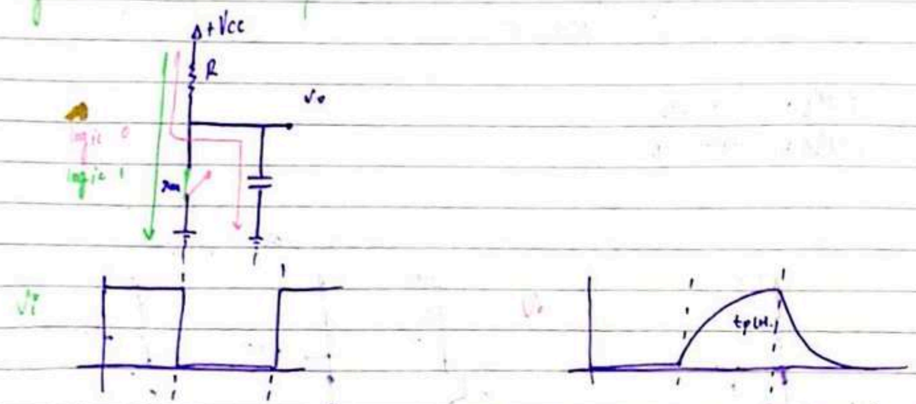






Date

Dynamic Power Dissipation:-



- Once the capacitor fully charged up current will stop flowing.
- Capacitor will discharge through short ron resistor.

Through charging & discharging power dissipation will be formed.
So

$$P_{dynamic} = f\,C V_{cc}^2$$

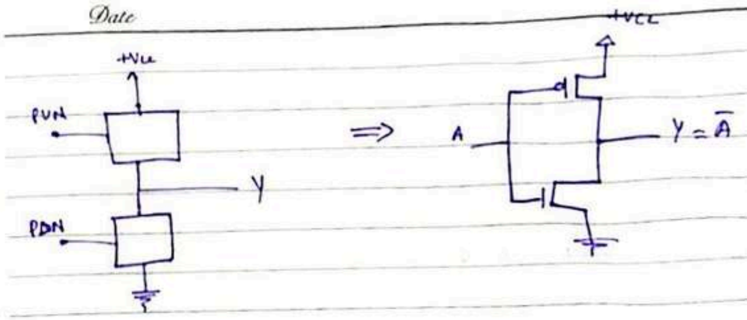
This topic was about CMOS. bcz of it's low power dissipation.

DAY-27

CMOS Logic Gate Circuits:

1. Logic - level Inverter contain a PUN & PDN.
2. NMOS Transistor → PDN
   PMOS transistor → PUN

Date



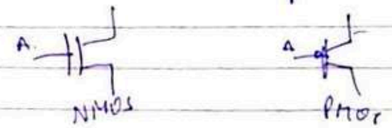
- Operated in in complimentary function
- A = '1'. NMOS is ON, PMOS is off. Y = '0'
- A = '0'. PMOS is ON, NMOS is off. Y = '1'

For other logic gates:
- PDN contain NMOS transistor ~~only~~ only — conducts when gate input is high
- PUN contain PMOS transistor only — conducts when gate input is low.

Two NMOS transistors in PDN or
Two PMOS transistors in PUN
1) Perform OR function when connected in parallel.
Perform AND function when connected in series

PDN is exact opposite of PUN.
i.e. two series NMOS transistor PDN can interpreted as two parallel PMOS " PUN

PMOS is not gate followed by NMOS.



Date

Two i/p NOR Gate:

$Y = \overline{A+B}$

PUN : $Y = \overline{A+B}$

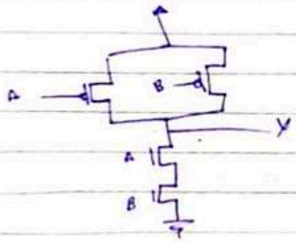
$\left.\begin{aligned} Y &= \overline{A+B} \\ Y &= A+B \end{aligned}\right\}$ complement of PUN

implement PDN first!

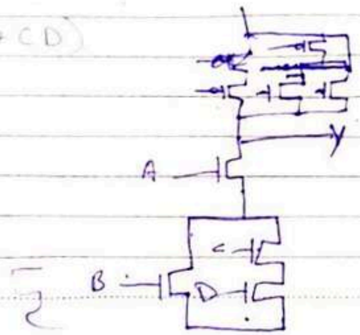
Complement PDN to get PUN!

Two i/p NAND Gate:

$Y = \overline{AB}$

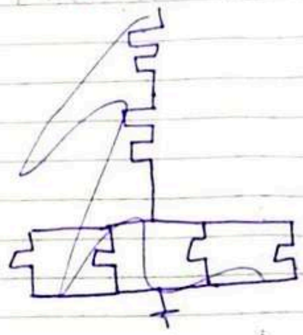


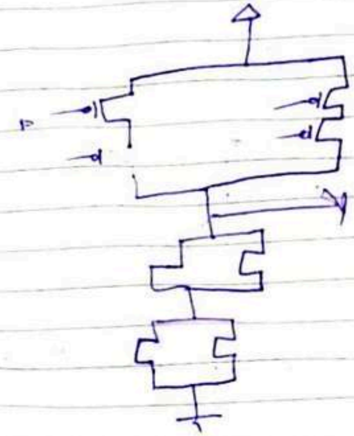
$Y = A(B+CD)$



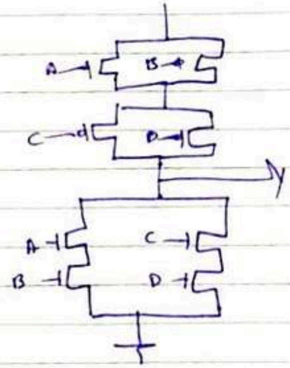
Date

$Y = \overline{(A+B)(C+D)}$





$F = \overline{AB+CD}$

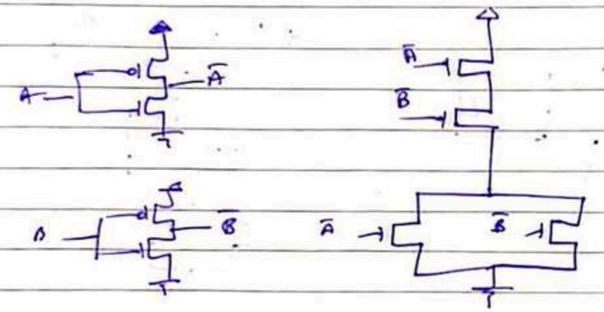


This was the first method of CMOS implementation

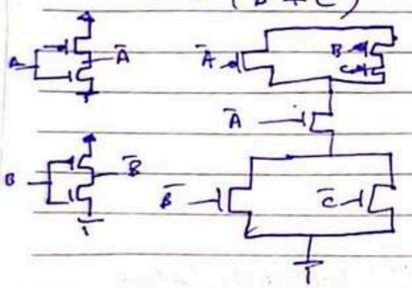
Date

2^{nd} Method of CMOS implementation

$$Y = AB$$
$$\overline{Y} = \overline{AB} = \overline{A} + \overline{B} \Rightarrow PDN$$



$$Y = A + BC$$
$$\overline{Y} = \overline{A + BC}$$
$$= \overline{A}\,(\overline{B} + \overline{C})$$

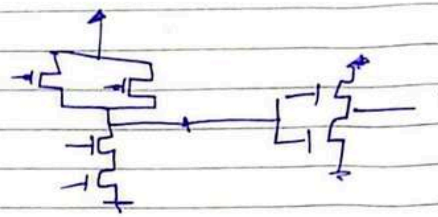


3^{rd} Method of CMOS implementation (Post Inversion)

$$Y = AB$$

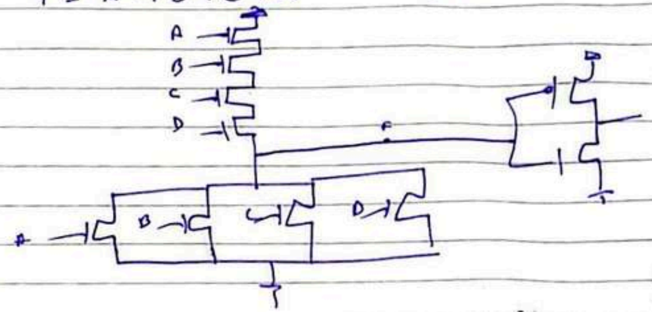
Date

$$F = \overline{AB}$$



POST INVERTION

$$Y = A + B + C + D$$



In first method, the equation is PUN. In second, we invert and have PUN. In third, we need a fudge.

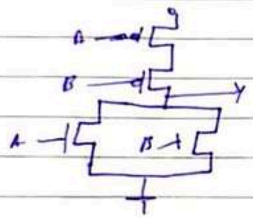
DAY-28

2nd Method

$$Y = AB$$
$$PUN = Y = AB$$

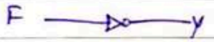
Date

PDN = $\overline{AB} = \bar{A} + \bar{B}$



3rd Method

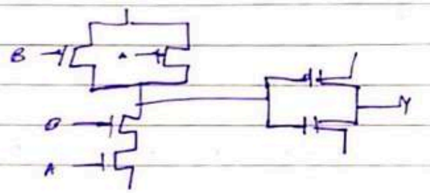
Y = AB



Y = $\bar{F}$

PUN = F = $\overline{AB}$

PDN = $\bar{F}$ = AB

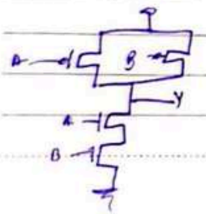


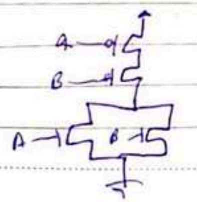
4th Method

8S-VAKI

Y = $\overline{AB}$ NAND

Y = $\overline{A+B}$ NOR

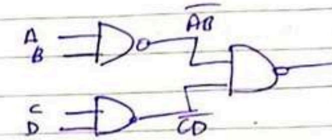




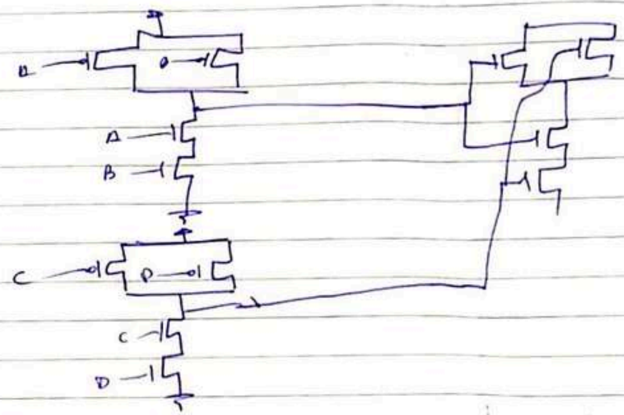
Date

Y = AB + CD

Make the CMOS circuit using gates only



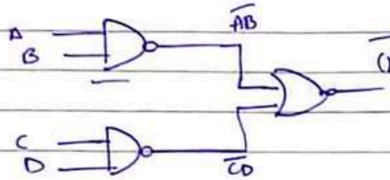
$\overline{(\overline{AB})(\overline{CD})}$ = AB + CD



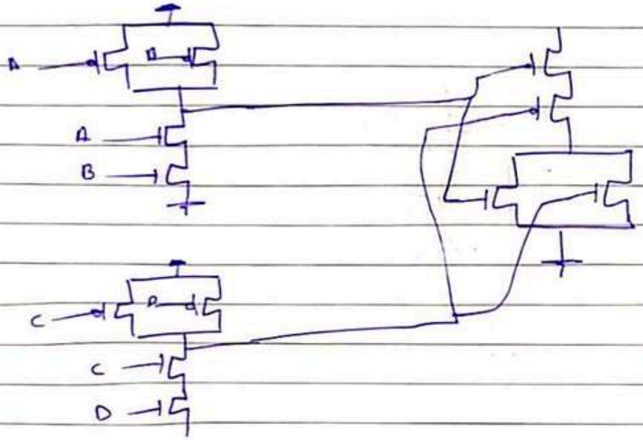
Scanned with CamScanner

Date

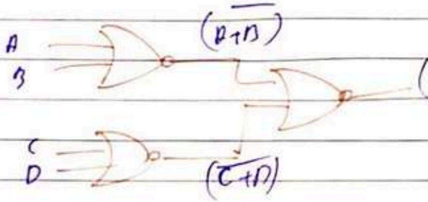
$Y = ABCD$



$\overline{(\overline{AB}) + (\overline{CD})} \Rightarrow ABCD$



$Y = (A+B)(C+D)$

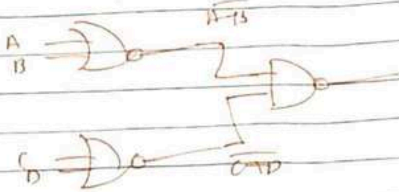


$\overline{\overline{(A+B)} + \overline{(C+D)}}$

$= (A+B)(C+D)$

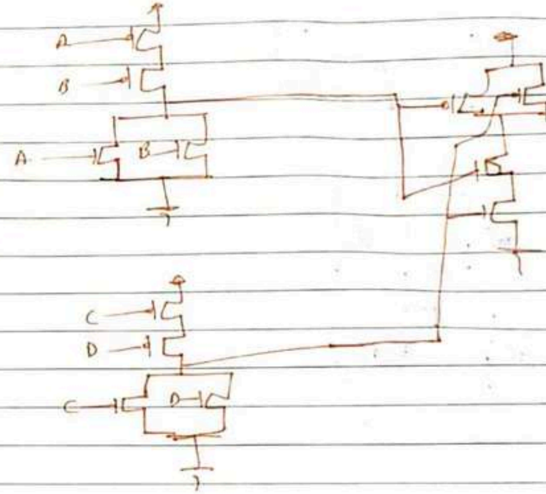
Date

$A+B+C+D$



$\overline{(\overline{A+B})(\overline{C+D})}$

$A+B+C+D$



$Y = AB + C+D$

$Y = AB(C+D)$

Date

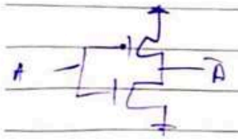
XOR Gate

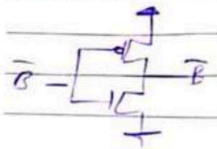
① $(A+B)(\bar{A}+\bar{B})$

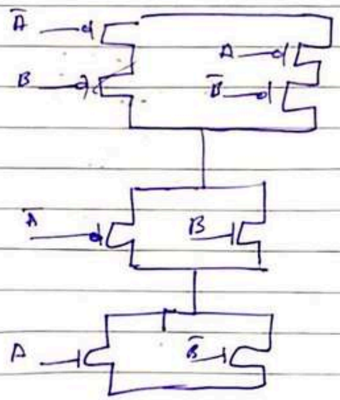
PUN $Y = A\bar{B} + \bar{A}B$

PDN $= \bar{Y} = \overline{A\bar{B} + \bar{A}B}$

$= (\bar{A}+B)(A+\bar{B})$







②

PUN $= Y = A\bar{B} + \bar{A}B$

PDN $= \bar{Y} = \bar{A}\bar{B} + AB$

Date